ENGLISH FOR WORK / EVIDENCE / DEVELOPMENT / QUALITY / COMMUNICATION

Pharmaceutical English

State the evidence. Qualify the claim. Close the communication gap.



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Eight practical lessons. Eleven extended conversations.
Vocabulary, phrase banks, guided speaking, and explained answers.

Revised 2026 edition / Upper-intermediate to advanced

Your route through the book

01 Drug Development Strategy, Unmet Need, TPP, and Evidence Logic, page 4

Replace an overstated development claim with a precise finding and a testable next question.

02 Regulatory Pathways, IND Readiness, NDA/BLA Strategy, and Agency Interaction, page 17

Frame a focused regulatory question and accurately record the scope of the response.

03 Clinical Trial Design, Protocols, GCP, and Operations, page 31

Resolve conflicting trial instructions through an explicit, documented clarification rather than choosing a convenient version.

04 Endpoints, Estimands, Statistics, Data Readouts, and Clinical Meaning, page 44

Explain a trial readout without replacing the prespecified question or overstating statistical certainty.

05 Pharmacovigilance, Safety Signals, Risk-Benefit, and Label Updates, page 58

Preserve reported facts, avoid premature causality judgments, and route a safety report without waiting for perfect information.

06 CMC, CGMP, Quality Events, Manufacturing, and Supply, page 72

Report a quality-documentation gap precisely and separate investigation status from batch disposition.

07 Labeling, Medical Affairs, Promotion Review, and Compliance Boundaries, page 85

Identify an unsupported promotional implication and return a precisely scoped claim through the required review route.

08 Market Access, RWE, Launch Readiness, Biosimilars, and Lifecycle Management, page 97

Explain differences between study designs and keep evidence, economic assumptions, and access decisions distinct.

Professional communication notes, page 3

Three additional workplace cases, page 112

Answers and explanations, page 123

Phrase finder, page 180

Vocabulary index, page 182

References and publication notes, page 187

Suggested study range: B2-C1. This is an editorial study range, not a certified proficiency assessment.

Precision protects the meaning.

Pharmaceutical conversations often connect specialists who use similar words for different purposes. A result, a hypothesis, a submission, and an approval are not interchangeable. These fictional cases practice concise clarification, careful evidence language, and explicit handoffs without turning English exercises into technical operating instructions.

Name the evidence level

Say whether a finding comes from a laboratory model, a clinical trial, or routine-care data. Keep the population, comparator, outcome, and limitations attached to the claim.
"The laboratory finding supports a hypothesis; it does not establish patient benefit."

Verify the controlling document

A familiar attachment may be outdated or not applicable to a particular site. Ask for the document identifier, version, approval status, and effective status before treating conflicting wording as settled.
"Please confirm the version authorized for this site and the exact section that governs the visit."

Report uncertainty without losing urgency

Incomplete information is not the same as evidence that nothing happened. Follow the applicable safety or quality process and preserve what was reported, what is missing, and who owns follow-up.
"The report is incomplete; I am routing the available information through the required channel now."

Keep review and approval distinct

Receiving a submission, reviewing a draft, and authorizing use are different events. State the actual status and next decision instead of using reassuring language that implies more than the record supports.
"Comments are resolved, but final approval for this version is still pending."

Professional boundaries

Original fictional language practice, not medical advice, clinical training, regulatory guidance, or authorization to perform regulated work. US terminology is used where identified; requirements vary by jurisdiction, product, and context. Follow current approved procedures, qualified professional judgment, and applicable requirements.

A Read the situation

LESSON 01 / DRUG DEVELOPMENT STRATEGY, UNMET NEED, TPP, AND EVIDENCE LOGIC

Replace an overstated development claim with a precise finding and a testable next question.

A laboratory signal becomes a patient-benefit claim

An internal slide for fictional candidate PX-17 says it improves patient mobility. The available experiment used a cell-based assay, not patients. Across three experimental runs, the team observed a change in a disease-related marker at one tested concentration. Human exposure, tolerability, and clinical benefit have not been established. Scientist Leila and development lead Owen are preparing a portfolio discussion. The proposed target product profile describes a desired future benefit, not an achieved product characteristic. No clinical study is authorized by this discussion.

People: Leila, Translational scientist; Owen, Development lead.

Read the record

Choose one answer per question using the facts above.

1. Where was the observed result obtained?

A. In a cell-based assay across three experimental runs
B. In a completed patient mobility trial
C. In routine treatment at multiple hospitals
D. In an approved product label

2. What does the target product profile represent here?

A. Desired future characteristics to guide development
B. Proof that patient mobility has improved
C. Authorization to begin a clinical study
D. A final regulatory approval

3. Which claim exceeds the evidence?

A. A disease-related marker changed in the assay.
B. Human tolerability remains unestablished.
C. Further evidence is required.
D. PX-17 improves patient mobility.

Answers and explanations, page 123

B Vocabulary in use

CORE TERMS

unmet medical need
A health-related need not adequately addressed by available options.

characterize the unmet medical need

development candidate
A selected investigational substance being evaluated for further development.

nominate a development candidate

target engagement
Evidence that an intervention interacts with its intended biological target.

assess target engagement

cell-based assay
A laboratory test using cells to assess a specified response.

qualify the cell-based assay

surrogate endpoint
A substitute outcome used to predict a clinical outcome, with context-dependent support.

justify a surrogate endpoint

translational evidence
Information connecting findings across experimental and human contexts.

strengthen translational evidence

target product profile
A planning description of desired future product characteristics.

refine the target product profile

mechanism of action
The biological process through which an intervention produces an effect.

investigate the mechanism of action

proof of concept
Evidence supporting the feasibility of a defined scientific or clinical idea.

specify the proof-of-concept question

biomarker
A measured characteristic indicating a biological process or response.

evaluate a candidate biomarker

clinical benefit
A meaningful favorable effect on how patients feel, function, or survive.

establish clinical benefit

biological plausibility
A reasoned basis for believing a biological explanation could be credible.

assess biological plausibility

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

unmet medical need UNMET MEDICAL NEED

characterize the unmet medical need

Keep the meaning precise: A health-related need not adequately addressed by available options.

proof of concept PROOF OF CONCEPT

specify the proof-of-concept question

Keep the meaning precise: Evidence supporting the feasibility of a defined scientific or clinical idea.

B Precision and professional shorthand

EXTEND YOUR RANGE

dose-response relationship

How a measured effect varies with administered dose.

characterize the dose-response relationship

exposure-response relationship

How an effect varies with drug exposure in the body.

examine exposure-response relationships

pharmacokinetics

Study of the time course of drug exposure through absorption, distribution, metabolism, and elimination.

characterize pharmacokinetics

pharmacodynamics

Study of a drug's biological effects and their relation to exposure.

measure pharmacodynamic effects

nonclinical study

Research conducted outside human clinical studies.

interpret nonclinical findings

tolerability

The extent to which unwanted effects can be endured in a defined context.

evaluate tolerability

therapeutic window

The exposure range associated with desired effects and acceptable toxicity in context.

investigate the therapeutic window

repeatability

Agreement between repeated results under the same specified measurement conditions.

assess repeatability

assay limitation

A restriction affecting what a test can establish.

disclose assay limitations

go/no-go criterion

A defined condition informing whether development proceeds or stops.

set go/no-go criteria

B Precision and professional shorthand / continued

evidence gap

Missing information needed to support a proposed conclusion.

prioritize evidence gaps

development milestone

A defined point of progress or review in a development program.

plan the next development milestone

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

biological plausibility BIOLOGICAL PLAUSIBILITY

assess biological plausibility

Keep the meaning precise: A reasoned basis for believing a biological explanation could be credible.

nonclinical study NONCLINICAL STUDY

interpret nonclinical findings

Keep the meaning precise: Research conducted outside human clinical studies.

C Phrases that move the work forward

Locate the evidence

The observation comes from a cell-based assay, not a patient study.

Protect the distinction

Clinical benefit has not been established.

Ask for the bridge

What evidence connects this marker to the intended patient outcome?

Name the limitation

We tested one concentration, so this does not characterize a dose-response relationship.

Avoid implied safety

The current experiment does not establish human tolerability.

Separate stages

An early signal supports investigation, not a treatment recommendation.

Clarify the measure

Are we discussing target engagement, a biomarker, or a patient outcome?

Mark authority

This portfolio discussion does not authorize a clinical study.

Narrow the claim

We observed a marker change under the tested conditions.

Clarify the aspiration

The target product profile describes what we aim to demonstrate.

Qualify repetition

Three runs support repeatability within these conditions, not general clinical effectiveness.

State the next question

We need to test whether the finding persists under relevant follow-up conditions.

Check the population

Which patients would the proposed product eventually be intended to serve?

Request a criterion

What result would support the next development decision?

Keep the goal conditional

The desired benefit remains a hypothesis to be evaluated.

Close accurately

Retain the finding, its conditions, and the unresolved evidence gap.

Precision in real exchanges

Purpose: Clarify the aspiration
Useful wording: The target product profile describes what we aim to demonstrate.

Purpose: Ask for the bridge
Useful wording: What evidence connects this marker to the intended patient outcome?

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which replacement is best supported?

A. PX-17 is safe at every human dose.
B. PX-17 has an approved mobility indication.
C. PX-17 changed the marker in the tested cell assay; patient benefit remains unestablished.
D. PX-17 restores mobility in patients because the marker changed.

2. What does testing one concentration fail to characterize?

A. How the effect changes across different concentrations
B. Whether the tested assay recorded any response
C. Whether the experiment used cells
D. The number of experimental runs stated in the brief

3. A colleague needs the term for this meaning: The biological process through which an intervention produces an effect. Which term fits?

A. assay limitation
B. pharmacokinetics
C. mechanism of action
D. surrogate endpoint

D Choose the response that fits / continued

4. Rebuild the exchange from "A laboratory signal becomes a patient-benefit claim". Owen: "The commercial planning page also lists improved mobility. That is in the target profile, so perhaps someone copied it into the result summary." Leila: [missing response] Owen: "Can we call this proof of concept? The phrase is familiar, but different colleagues may hear very different claims when they read it." Which response belongs in the gap?

A. The target product profile is aspirational at this stage. We should label the desired outcome separately from the finding so readers do not confuse the development goal with achieved performance.

B. Start with the evidence gap between the assay response and the intended outcome. The scientific team should define the follow-up question and its limits through the appropriate development process.

C. The runs used the same analyst, method, and equipment over a short interval. They inform repeatability. Show all three results and their variation, not just the strongest one.

D. Good. The next development milestone remains subject to the required review. This discussion does not authorize a clinical study, and the final slide should say no more than the evidence supports.

Answers and explanations, page 123

One useful distinction

A biomarker change is not automatically clinical benefit. Its meaning depends on the marker, context, and supporting evidence. Here, one concentration and three laboratory runs do not establish human exposure, tolerability, dose response, or improved mobility.

E Conversation

A laboratory signal becomes a patient-benefit claim

A laboratory signal becomes a patient-benefit claim

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

cell-based assay / dose-response relationship / development milestone / repeatability / go/no-go criterion / biomarker / tolerability / clinical benefit / target product profile / evidence gap

01 **Owen:** The portfolio slide says PX-17 improves patient mobility. I thought the latest package contained laboratory work only. Have I missed a clinical result?

02 **Leila:** No. The result comes from a [1] _____. We saw the marker change across three runs at one tested concentration. Nobody measured mobility in patients.

03 **Owen:** Then the headline has moved beyond the evidence. The team wanted a clear link to the disease, but the wording makes that link sound demonstrated.

04 **Leila:** We can describe the [2] _____ response precisely and explain why it interests us. We still need evidence connecting that response to the outcome we hope to influence.

05 **Owen:** The commercial planning page also lists improved mobility. That is in the target profile, so perhaps someone copied it into the result summary.

06 **Leila:** The [3] _____ is aspirational at this stage. We should label the desired outcome separately from the finding so readers do not confuse the development goal with achieved performance.

E Conversation / continued

07 **Owen:** Can we call this proof of concept? The phrase is familiar, but different colleagues may hear very different claims when they read it.

08 **Leila:** Specify the concept and setting. It is not evidence of [4] _____. A broad proof-of-concept label could conceal exactly the distinction we are trying to restore.

09 **Owen:** Why is there a rising dose-response arrow if we tested only one concentration? The graphic seems to be claiming something the experiment did not measure.

10 **Leila:** Remove that implication. A [5] _____ has not been characterized here. The visual should not make a stronger statement than the accompanying words or the experiment itself.

Conversation continues on the next page.

A laboratory signal becomes a patient-benefit claim

CONVERSATION / CONTINUED

11 **Owen:** What can we say about the three runs? I do not want to discard useful repetition merely because the result is still early.

12 **Leila:** The runs used the same analyst, method, and equipment over a short interval. They inform [6] _____. Show all three results and their variation, not just the strongest one.

13 **Owen:** Someone has added well tolerated to the summary. There were no human participants, so I cannot see what that statement is based on.

14 **Leila:** Human [7] _____ is unestablished. We can report the actual observations, but we should not translate the absence of a particular laboratory finding into a general human safety claim.

15 **Owen:** Let us make the next decision concrete. What missing information matters most before the program can support a stronger development proposal?

16 **Leila:** Start with the [8] _____ between the assay response and the intended outcome. The scientific team should define the follow-up question and its limits through the appropriate development process.

17 **Owen:** The committee will ask what would count as enough progress to continue. A broad instruction to produce more data will not help them compare programs.

18 **Leila:** Propose a defined [9] _____ for qualified review. It should match the next decision, without pretending that an early laboratory threshold alone establishes a product suitable for patients.

A laboratory signal becomes a patient-benefit claim / continued

19 **Owen:** I will replace the mobility headline with the assay finding. Send me the limits and proposed next question so the committee sees what this result supports.

20 **Leila:** Good. The next [10] _____ remains subject to the required review. This discussion does not authorize a clinical study, and the final slide should say no more than the evidence supports.

CHECK THE GAPS

1. cell-based assay / **2.** biomarker / **3.** target product profile / **4.** clinical benefit / **5.** dose-response relationship / **6.** repeatability / **7.** tolerability / **8.** evidence gap / **9.** go/no-go criterion / **10.** development milestone

Why these answers fit, page 124

F Check accuracy and scope

A laboratory signal becomes a patient-benefit claim

Return to the case

An internal slide for fictional candidate PX-17 says it improves patient mobility. The available experiment used a cell-based assay, not patients. Across three experimental runs, the team observed a change in a disease-related marker at one tested concentration. Human exposure, tolerability, and clinical benefit have not been established. Scientist Leila and development lead Owen are preparing a portfolio discussion. The proposed target product profile describes a desired future benefit, not an achieved product characteristic. No clinical study is authorized by this discussion.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. Which additional question most directly tests the proposed marker-to-mobility claim?

A. What evidence links changes in this marker to patient mobility in the intended context?

B. How can three runs at one concentration be relabeled as a dose-response study?

C. Which patient population should be added to the slide although no patients were studied?

D. Can the desired benefit in the target product profile substitute for clinical measurement?

2. What can the portfolio discussion decide on the supplied authority?

A. Final marketing authorization

B. A universal human dosing recommendation

C. A proposed next investigation for appropriate review, not automatic clinical-study authorization

D. Immediate patient treatment with the candidate

Answers and explanations, page 126

Listen for the distinction

Switch roles for turns 11-20. Distinguish repeatability under the stated conditions from human tolerability and the desired mobility benefit.

A narrower claim can be more useful

Do not soften an unsupported assertion merely by adding promising. State what was measured, where it was measured, and what remains unknown. Then connect the result to the next question. This lets a colleague support further investigation without endorsing a conclusion the evidence cannot carry.

F Say it / transfer to a new case

A goal is not a result

A planning profile targets once-daily use. The candidate has no human dosing data. A laboratory test measured target engagement but did not measure patient symptoms.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

unestablished / goal / target engagement / improvement

01 **Scientist:** Once-daily use is currently a [1] _____.

02 **Reviewer:** The measured laboratory finding concerns [2] _____.

03 **Scientist:** Human dosing remains [3] _____.

04 **Reviewer:** We cannot claim symptom [4] _____.

CHECK THE GAPS

1. goal / **2.** target engagement / **3.** unestablished / **4.** improvement

Why these answers fit, page 126

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "A laboratory signal becomes a patient-benefit claim". Owen: "Someone has added well tolerated to the summary. There were no human participants, so I cannot see what that statement is based on." Leila: [missing response] Owen: "Let us make the next decision concrete. What missing information matters most before the program can support a stronger development proposal?" Which response belongs in the gap?

A. Propose a defined go/no-go criterion for qualified review. It should match the next decision, without pretending that an early laboratory threshold alone establishes a product suitable for patients.

B. We can describe the biomarker response precisely and explain why it interests us. We still need evidence connecting that response to the outcome we hope to influence.

C. Human tolerability is unestablished. We can report the actual observations, but we should not translate the absence of a particular laboratory finding into a general human safety claim.

D. The runs used the same analyst, method, and equipment over a short interval. They inform repeatability. Show all three results and their variation, not just the strongest one.

Answers and explanations, page 127

A Read the situation

LESSON 02 / REGULATORY PATHWAYS, IND READINESS, NDA/BLA STRATEGY, AND AGENCY INTERACTION

Frame a focused regulatory question and accurately record the scope of the response.

> **A meeting question that asks for blanket agreement**
>
> A US development team is preparing an agency meeting package for fictional investigational product RV-8. Its draft asks whether the entire development plan is acceptable. The immediate unresolved question concerns the proposed patient population and the rationale for a planned study endpoint. Supporting summaries exist, but two supporting reports are unfinished. Regulatory lead Priya and clinical lead Evan must distinguish the sponsor proposal, available evidence, and specific advice requested. No agency response has been received, and meeting preparation does not constitute authorization to begin the study.

People: Priya, Regulatory lead; Evan, Clinical lead.

Read the record

Choose one answer per question using the facts above.

1. What is wrong with the current question?

A. It identifies too narrow a product name.
B. It asks for information already supplied as final approval.
C. It asks broadly about the entire plan instead of the specific unresolved proposal.
D. It contains a confirmed agency decision.

2. What is the status of two supporting reports?

A. Irrelevant because the meeting is scheduled
B. Unfinished
C. Approved by the agency
D. Replaced by a marketing authorization

3. Which statement is accurate now?

A. An NDA has been approved for RV-8.
B. No agency response has been received.
C. The study is authorized because a package is being prepared.
D. The entire program has formal agreement.

Answers and explanations, page 128

B Vocabulary in use

CORE TERMS

regulatory strategy
A plan for addressing applicable regulatory requirements and interactions.

align the regulatory strategy

regulatory pathway
The applicable route for development, review, or authorization.

confirm the regulatory pathway

investigational new drug application
A US submission supporting proposed clinical investigation of a drug.

prepare an IND submission

new drug application
A US application seeking approval to market a new drug.

plan the NDA content

biologics license application
A US application seeking licensure of a biological product.

prepare the BLA strategy

agency interaction
A formal or informal exchange with a regulatory authority.

document the agency interaction

meeting package
Materials submitted to support a scheduled regulatory discussion.

assemble the meeting package

briefing document
A structured account of background, evidence, proposals, and questions.

finalize the briefing document

sponsor position
The development sponsor's proposed approach and rationale.

state the sponsor position

focused question
A question limited to a defined issue that can receive a useful answer.

formulate a focused question

supporting rationale
The reasoning and evidence behind a proposal.

substantiate the supporting rationale

information gap
Missing material relevant to assessing a proposal.

disclose an information gap

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

regulatory strategy REGULATORY STRATEGY

align the regulatory strategy

Keep the meaning precise: A plan for addressing applicable regulatory requirements and interactions.

agency interaction AGENCY INTERACTION

document the agency interaction

Keep the meaning precise: A formal or informal exchange with a regulatory authority.

B Precision and professional shorthand

EXTEND YOUR RANGE

submission readiness

The state of being sufficiently prepared for an intended submission.

assess submission readiness

dossier

An organized collection of documents supporting regulatory review.

maintain a coherent dossier

cross-reference

A pointer connecting a statement to supporting material.

verify the cross-reference

version control

Management of document revisions and their status.

maintain version control

regulatory commitment

An undertaking made in the relevant regulatory context.

track regulatory commitments

meeting minutes

The documented account of a meeting's discussion and outcomes.

reconcile meeting minutes

written response

A documented reply to a submitted question or issue.

review the written response

conditional advice

Advice whose applicability depends on stated circumstances.

preserve conditional advice

clinical hold

A regulatory order delaying or suspending a clinical investigation in the US context.

address a clinical hold

approval status

The actual state of a specified authorization decision.

verify approval status

indication

The disease, condition, or use for which a product is intended or authorized in context.

define the proposed indication

regulatory affairs

The function coordinating regulatory strategy, submissions, and interactions.

consult regulatory affairs

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

information gap INFORMATION GAP

disclose an information gap

Keep the meaning precise: Missing material relevant to assessing a proposal.

regulatory commitment REGULATORY COMMITMENT

track regulatory commitments

Keep the meaning precise: An undertaking made in the relevant regulatory context.

C Phrases that move the work forward

Focus the question

We seek advice on the proposed population and endpoint rationale.

State the sponsor view

Our proposed approach is set out in section three.

Supply the reason

The rationale rests on the following evidence and assumptions.

Disclose incompleteness

Two supporting reports are not yet final.

Avoid blanket agreement

This question does not ask for approval of the entire program.

Check the reference

Please verify that the cited section supports this statement.

Distinguish status

The package is in preparation; no agency response has been received.

Preserve a condition

Record the qualification alongside the advice, not in a separate summary.

Confirm the pathway

Regulatory affairs should confirm the applicable route for this product.

Separate submissions

An IND and a marketing application serve different purposes.

Resolve a discrepancy

The summary and supporting report currently describe different populations.

Control the version

Use the agreed document version for the submitted package.

Track an undertaking

Identify the owner and due date for each recorded commitment.

Limit the conclusion

Advice on this question does not settle every development issue.

Mark study authority

Meeting preparation is not permission to begin the study.

Close the record

Distinguish answered questions, conditions, and issues still open.

Precision in real exchanges

Purpose: Disclose incompleteness
Useful wording: Two supporting reports are not yet final.

Purpose: Avoid blanket agreement
Useful wording: This question does not ask for approval of the entire program.

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which agency question is focused on the immediate unresolved proposal?

A. Can unfinished supporting reports be treated as final if their summaries are included?

B. Will acceptance of this endpoint establish the eventual marketing indication?

C. Does the proposed population and endpoint rationale in section three address the intended study objective?

D. Does the agency endorse the entire development program on the basis of this meeting package?

2. Which package statement is accurate?

A. All supporting reports are final because the meeting is planned.

B. No evidence needs to support a sponsor position.

C. A scheduled meeting constitutes marketing authorization.

D. Two reports remain unfinished and their status is identified.

3. A colleague needs the term for this meaning: A US application seeking approval to market a new drug. Which term fits?

A. sponsor position

B. new drug application

C. cross-reference

D. clinical hold

D Choose the response that fits / continued

4. Rebuild the exchange from "A meeting question that asks for blanket agreement". Evan: "The supporting summaries are available, although two underlying reports are unfinished. I do not want the package to imply that those reports are final." Priya: [missing response] Evan: "Section three names a narrower population than the summary. Which one is our proposal? We need that settled before the package goes out." Which response belongs in the gap?

A. And keep approval status explicit. Advice on one issue is not blanket program approval, and preparation for a meeting is not permission to start the study.

B. Identify that information gap and explain its relevance. We should not conceal unfinished work or claim that a summary has the status of a completed report.

C. We need a focused question. The immediate issue is the proposed patient population and endpoint rationale, not every aspect of the program from development through launch.

D. State the sponsor position clearly, then give the rationale and relevant alternatives. Reviewers should not have to infer our proposal from a collection of background slides.

Answers and explanations, page 128

One useful distinction

IND means investigational new drug application; NDA means new drug application; BLA means biologics license application in US usage. These terms name different regulatory submissions. An IND is not a marketing approval, and preparing a meeting package does not authorize a study.

E Conversation

A meeting question that asks for blanket agreement

A meeting question that asks for blanket agreement

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

regulatory pathway / focused question / information gap / investigational new drug application / cross-reference / briefing document / conditional advice / meeting minutes / sponsor position / approval status

01 **Evan:** Our lead question asks whether the whole development plan is acceptable. It is concise, but I suspect it will not give us the advice we need.

02 **Priya:** We need a [1] _____. The immediate issue is the proposed patient population and endpoint rationale, not every aspect of the program from development through launch.

03 **Evan:** The team has a preferred approach. Should we present alternatives equally, or make clear which approach we are proposing and why we favor it?

04 **Priya:** State the [2] _____ clearly, then give the rationale and relevant alternatives. Reviewers should not have to infer our proposal from a collection of background slides.

05 **Evan:** The supporting summaries are available, although two underlying reports are unfinished. I do not want the package to imply that those reports are final.

06 **Priya:** Identify that [3] _____ and explain its relevance. We should not conceal unfinished work or claim that a summary has the status of a completed report.

E Conversation / continued

07 **Evan:** Section three names a narrower population than the summary. Which one is our proposal? We need that settled before the package goes out.

08 **Priya:** Reconcile the wording in the [4] _____. A precise question is not enough if the supporting text describes a different proposal or leaves the study population ambiguous.

09 **Evan:** I will also check the references. One footnote points to an earlier table, and the current table no longer contains the information cited.

10 **Priya:** Verify each [5] _____ after the content is settled. A broken evidence trail forces reviewers to guess which material we intended and can weaken an otherwise useful discussion.

Conversation continues on the next page.

A meeting question that asks for blanket agreement

CONVERSATION / CONTINUED

11 **Evan:** Some colleagues are calling the meeting package our IND approval package. That seems to mix the meeting purpose with the status of a regulatory submission.

12 **Priya:** Use the terms carefully. An [6] _____ concerns proposed clinical investigation in the US context. Preparing this meeting package does not authorize the study or establish a marketing approval.

13 **Evan:** The long-range plan also mentions an NDA or BLA. We should explain the applicable route without suggesting that either application is already ready.

14 **Priya:** Exactly. The [7] _____ depends on the product and context. Regulatory affairs should confirm the route, while the current discussion remains focused on the specific development question.

15 **Evan:** After the meeting, the executive team will want a short update. How do we keep it concise without losing qualifications that affect the advice?

16 **Priya:** Preserve any [8] _____ in the main summary. If a response depends on the proposed population or additional evidence, that condition belongs beside the conclusion, not buried elsewhere.

17 **Evan:** We should compare our notes with the official record through the applicable process. A colleague's enthusiastic recollection should not become the only account.

18 **Priya:** Yes, reconcile the [9] _____ and track unresolved differences. Assign owners to follow-up items and commitments rather than treating a successful conversation as completion of all required work.

A meeting question that asks for blanket agreement / continued

19 **Evan:** I will revise the question and flag the unfinished reports. Please check the executive summary too; it must not suggest that we already have agency agreement.

20 **Priya:** And keep [10] _____ explicit. Advice on one issue is not blanket program approval, and preparation for a meeting is not permission to start the study.

CHECK THE GAPS

1. focused question / **2.** sponsor position / **3.** information gap / **4.** briefing document / **5.** cross-reference / **6.** investigational new drug application / **7.** regulatory pathway / **8.** conditional advice / **9.** meeting minutes / **10.** approval status

Why these answers fit, page 129

F Check accuracy and scope

A meeting question that asks for blanket agreement

Return to the case

A US development team is preparing an agency meeting package for fictional investigational product RV-8. Its draft asks whether the entire development plan is acceptable. The immediate unresolved question concerns the proposed patient population and the rationale for a planned study endpoint. Supporting summaries exist, but two supporting reports are unfinished. Regulatory lead Priya and clinical lead Evan must distinguish the sponsor proposal, available evidence, and specific advice requested. No agency response has been received, and meeting preparation does not constitute authorization to begin the study.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. Which distinction is correct in US terminology?

A. An IND is a guarantee that an NDA will be approved.

B. A meeting package replaces every required application.

C. A BLA is merely a meeting calendar invitation.

D. An IND supports clinical investigation; an NDA seeks marketing approval for a new drug.

2. How should limited advice be summarized?

A. Expand it into approval of every proposed study.

B. Delete qualifications to make the update easier to read.

C. Treat the sponsor's preferred wording as the agency's conclusion.

D. Preserve its subject, assumptions, conditions, and unresolved issues.

Answers and explanations, page 131

Listen for the distinction

Switch roles for the document discussion. Preserve the two unfinished reports and the absence of any agency response or study authorization.

Make the question answerable

A focused question gives reviewers a proposal, supporting rationale, and a specific point on which advice is sought. Avoid treating a useful discussion as blanket agreement with every part of a program. Meeting records should preserve conditions, unresolved issues, and any distinction between advice and formal authorization.

F Say it / transfer to a new case

The response answers one question

An invented meeting note supports an endpoint approach only for the population described in the package. It requests further justification of follow-up duration. It grants no marketing authorization.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

conditional / population / authorization / unresolved

01 **Regulatory lead:** The endpoint advice is [1] _____.

02 **Clinical lead:** Follow-up duration remains [2] _____.

03 **Regulatory lead:** Preserve the stated [3] _____.

04 **Clinical lead:** This is not marketing [4] _____.

CHECK THE GAPS

1. conditional / **2.** unresolved / **3.** population / **4.** authorization

Why these answers fit, page 131

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "A meeting question that asks for blanket agreement". Evan: "The long-range plan also mentions an NDA or BLA. We should explain the applicable route without suggesting that either application is already ready." Priya: [missing response] Evan: "After the meeting, the executive team will want a short update. How do we keep it concise without losing qualifications that affect the advice?" Which response belongs in the gap?

A. We need a focused question. The immediate issue is the proposed patient population and endpoint rationale, not every aspect of the program from development through launch.

B. Reconcile the wording in the briefing document. A precise question is not enough if the supporting text describes a different proposal or leaves the study population ambiguous.

C. Exactly. The regulatory pathway depends on the product and context. Regulatory affairs should confirm the route, while the current discussion remains focused on the specific development question.

D. Verify each cross-reference after the content is settled. A broken evidence trail forces reviewers to guess which material we intended and can weaken an otherwise useful discussion.

Answers and explanations, page 132

Review 1 / Meaning and context

Use language from the lessons you have completed. Choose one answer per question.

> **Case file**
>
> Revisit "Drug Development Strategy, Unmet Need, TPP, and Evidence Logic" and "Regulatory Pathways, IND Readiness, NDA/BLA Strategy, and Agency Interaction". Choose precise terminology, then reconstruct the missing responses using both neighboring turns. The final two questions also revisit earlier conversations.

1. A colleague needs the term for this meaning: A selected investigational substance being evaluated for further development. Which term fits?

A. development candidate

B. repeatability

C. exposure-response relationship

D. biomarker

2. A colleague needs the term for this meaning: A structured account of background, evidence, proposals, and questions. Which term fits?

A. briefing document

B. written response

C. regulatory strategy

D. submission readiness

3. Rebuild the exchange from "A laboratory signal becomes a patient-benefit claim". Owen: "Why is there a rising dose-response arrow if we tested only one concentration? The graphic seems to be claiming something the experiment did not measure." Leila: [missing response] Owen: "What can we say about the three runs? I do not want to discard useful repetition merely because the result is still early." Which response belongs in the gap?

A. The runs used the same analyst, method, and equipment over a short interval. They inform repeatability. Show all three results and their variation, not just the strongest one.

B. We can describe the biomarker response precisely and explain why it interests us. We still need evidence connecting that response to the outcome we hope to influence.

C. Specify the concept and setting. It is not evidence of clinical benefit. A broad proof-of-concept label could conceal exactly the distinction we are trying to restore.

D. Remove that implication. A dose-response relationship has not been characterized here. The visual should not make a stronger statement than the accompanying words or the experiment itself.

Review 1 / Meaning and context / continued

4. Rebuild the exchange from "A meeting question that asks for blanket agreement". Evan: "We should compare our notes with the official record through the applicable process. A colleague's enthusiastic recollection should not become the only account." Priya: [missing response] Evan: "I will revise the question and flag the unfinished reports. Please check the executive summary too; it must not suggest that we already have agency agreement." Which response belongs in the gap?

A. Reconcile the wording in the briefing document. A precise question is not enough if the supporting text describes a different proposal or leaves the study population ambiguous.

B. Verify each cross-reference after the content is settled. A broken evidence trail forces reviewers to guess which material we intended and can weaken an otherwise useful discussion.

C. Preserve any conditional advice in the main summary. If a response depends on the proposed population or additional evidence, that condition belongs beside the conclusion, not buried elsewhere.

D. Yes, reconcile the meeting minutes and track unresolved differences. Assign owners to follow-up items and commitments rather than treating a successful conversation as completion of all required work.

Answers and explanations, page 163

A Read the situation

LESSON 03 / CLINICAL TRIAL DESIGN, PROTOCOLS, GCP, AND OPERATIONS

Resolve conflicting trial instructions through an explicit, documented clarification rather than choosing a convenient version.

> **Two visit windows and no confirmed controlling version**
>
> Coordinator Maya is scheduling next week. A protocol attachment gives a plus-or-minus-three-day visit window; a site quick-reference sheet gives seven. The controlling version and site implementation status are unconfirmed. No participant awaits an urgent care decision. Clinical research associate Ben must obtain clarification through the investigator and sponsor process. Neither may invent a window, silently change records, or assume the newest attachment governs the site.

People: Maya, Site coordinator; Ben, Clinical research associate.

Read the record

Choose one answer per question using the facts above.

1. What is the documented conflict?

A. Two confirmed identical protocol versions
B. A participant refusing all care
C. A completed decision approving a new window
D. Three-day versus seven-day visit windows in two sources

2. What has not been confirmed?

A. Whether a quick-reference sheet exists
B. Whether the two wordings differ
C. The controlling version and its implementation status at the site
D. Whether appointments are being prepared for next week

3. Which response avoids an unauthorized assumption?

A. Rewrite historical visit records to match the preferred version.
B. Escalate the exact discrepancy for documented clarification through the applicable process.
C. Select seven days because it is more convenient.
D. Treat the newest email attachment as automatically controlling.

Answers and explanations, page 133

B Vocabulary in use

CORE TERMS

protocol
The document specifying a clinical study's objectives, design, and procedures.

follow the applicable protocol

protocol amendment
A documented change to a study protocol.

implement an authorized protocol amendment

visit window
The allowed timing range for a specified study visit or assessment.

confirm the visit window

schedule of assessments
The planned timing of study procedures and measurements.

reconcile the schedule of assessments

controlling version
The authorized document version applicable to the work in question.

identify the controlling version

effective date
The date from which a document or change applies in context.

verify the effective date

site activation
Confirmation that a site meets the applicable conditions to begin specified study activity.

confirm site activation status

principal investigator
The investigator responsible for leading the study at a site.

consult the principal investigator

clinical research associate
A professional supporting study oversight and monitoring within assigned responsibilities.

contact the clinical research associate

study coordinator
A site professional coordinating assigned trial activities.

brief the study coordinator

good clinical practice
Principles and standards for ethical, reliable clinical trial conduct.

apply good clinical practice

informed consent
The voluntary agreement process based on adequate study information and understanding.

document the informed-consent process

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

protocol PROTOCOL

follow the applicable protocol

Keep the meaning precise: The document specifying a clinical study's objectives, design, and procedures.

effective date EFFECTIVE DATE

verify the effective date

Keep the meaning precise: The date from which a document or change applies in context.

B Precision and professional shorthand

EXTEND YOUR RANGE

institutional review board

A body reviewing research involving human participants in the US context.

confirm institutional review board review

eligibility criterion

A condition determining whether someone may enter a study.

verify eligibility criteria

randomization

Allocation to study groups using a chance-based method.

protect the randomization process

blinding

Concealment of treatment allocation from specified parties.

maintain blinding

source record

The original record or certified copy containing relevant study observations.

preserve the source record

electronic case report form

An electronic tool for collecting protocol-required participant data.

complete the electronic case report form

data query

A documented request to clarify or resolve a data issue.

resolve a data query

protocol deviation

A departure from the study protocol.

assess a potential protocol deviation

monitoring finding

An issue identified during study monitoring.

document a monitoring finding

audit trail

A traceable record of changes and associated information.

preserve the audit trail

B Precision and professional shorthand / continued

training record

Documentation of completed instruction or qualification.

verify the training record

essential record

A record supporting evaluation of trial conduct and result reliability.

maintain essential records

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

informed consent INFORMED CONSENT

document the informed-consent process

Keep the meaning precise: The voluntary agreement process based on adequate study information and understanding.

source record SOURCE RECORD

preserve the source record

Keep the meaning precise: The original record or certified copy containing relevant study observations.

C Phrases that move the work forward

State the discrepancy

The protocol attachment says three days; the quick-reference sheet says seven.

Identify the source

I will include the document identifiers and exact sections.

Escalate precisely

Please resolve the timing discrepancy through the investigator and sponsor process.

Separate status

Received by email does not necessarily mean approved for implementation.

Limit the response

I am confirming the governing instruction, not proposing a new window.

Assess impact

After confirmation, determine whether any completed visits require review.

Keep care distinct

Administrative clarification must not replace appropriate clinical judgment.

Control obsolete copies

Handle superseded quick-reference material through the document process.

Request the authority

Which approved version is effective for this site?

Avoid a guess

I cannot confirm the visit window from these conflicting files.

Protect the record

We must preserve the original entry and any documented correction history.

Check training

Has the relevant site team completed the required update?

Close the loop

Please acknowledge the clarification and identify any affected appointments.

Avoid blame

The two sources conflict; let us establish the applicable instruction.

Document the outcome

Record the confirmed version, effective status, and source of clarification.

Confirm understanding

Please read back the confirmed window and the document it comes from.

Precision in real exchanges

Purpose: Avoid a guess
Useful wording: I cannot confirm the visit window from these conflicting files.

Purpose: Escalate precisely
Useful wording: Please resolve the timing discrepancy through the investigator and sponsor process.

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which message best resolves the three-day versus seven-day discrepancy?

A. Please revise the calendar to five days while the document owners reconcile the difference.

B. Please identify the approved site-effective protocol version and section governing this visit window.

C. Please confirm that the latest email attachment supersedes the site record by its file date.

D. Please authorize the seven-day reference-sheet window because it fits next week's appointments.

2. What does a newer file date prove by itself?

A. That all required training is complete

B. That older source records may be overwritten

C. Only that the file carries a later date, not that it governs the site

D. That every site has implemented the document

3. A colleague needs the term for this meaning: The planned timing of study procedures and measurements. Which term fits?

A. schedule of assessments

B. randomization

C. clinical research associate

D. monitoring finding

D Choose the response that fits / continued

4. Rebuild the exchange from "Two visit windows and no confirmed controlling version". Maya: "I will quote both wordings in the request. Calling it a scheduling question without the numbers would make the reviewer reconstruct the discrepancy." Ben: [missing response] Maya: "Nobody is waiting for an urgent care decision. This concerns next week's appointments, but I want to resolve it before we send instructions to participants." Which response belongs in the gap?

A. Please send the identifiers and exact sections for both. We need the controlling version before confirming the schedule. I cannot choose a window from the filenames alone.

B. And verify the relevant training record. Even a correct document may not resolve the operational problem if the people using it have not received the required update.

C. Preserve the audit trail. Any necessary correction must follow the applicable process and retain traceability. We must not rewrite the history to make the schedule appear consistent.

D. Include the relevant schedule of assessments references as well. The team needs enough context to identify whether the quick-reference material reflects an authorized change or an error.

Answers and explanations, page 133

> **One useful distinction**
>
> A later file date does not establish that a document is approved and effective for a particular site. Confirm identity, version, authorization, and applicability. A quick-reference sheet should not silently replace the controlling protocol or become an unofficial amendment.

E Conversation

Two visit windows and no confirmed controlling version

Two visit windows and no confirmed controlling version

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

controlling version / training record / essential records / principal investigator / effective date / read-back / audit trail / protocol amendment / protocol deviation / schedule of assessments

01 **Maya:** I am scheduling next week's visits and have two different windows. The protocol attachment says plus or minus three days; the quick-reference sheet says seven.

02 **Ben:** Please send the identifiers and exact sections for both. We need the [1] _____ before confirming the schedule. I cannot choose a window from the filenames alone.

03 **Maya:** The seven-day sheet arrived in a newer email. The coordinator who forwarded it assumed that made it the current instruction for every site.

04 **Ben:** A later email does not establish the [2] _____ or implementation status. We should confirm which approved document applies here through the investigator and sponsor process.

05 **Maya:** I will quote both wordings in the request. Calling it a scheduling question without the numbers would make the reviewer reconstruct the discrepancy.

06 **Ben:** Include the relevant [3] _____ references as well. The team needs enough context to identify whether the quick-reference material reflects an authorized change or an error.

E Conversation / continued

07 **Maya:** Nobody is waiting for an urgent care decision. This concerns next week's appointments, but I want to resolve it before we send instructions to participants.

08 **Ben:** Ask the [4] _____ to handle the site issue through the appropriate route. We should not invent clinical instructions or let an administrative document question substitute for medical judgment.

09 **Maya:** If the seven-day wording came from an amendment, we need more than a copy of the amended page. We need to know its status here.

10 **Ben:** Correct. A [5] _____ has an applicable approval and implementation process. Confirm those facts rather than treating circulation of the file as evidence that every required step occurred.

Conversation continues on the next page.

Two visit windows and no confirmed controlling version

CONVERSATION / CONTINUED

11 **Maya:** The quick-reference sheet is useful at the desk, but it cannot create a separate set of rules. We should reconcile it after the answer is confirmed.

12 **Ben:** And verify the relevant [6] _____. Even a correct document may not resolve the operational problem if the people using it have not received the required update.

13 **Maya:** Should we review earlier visits as well? I need the version that applied then, not just today's confirmed window, before labeling any visit outside it.

14 **Ben:** Exactly. Assess any potential [7] _____ through the appropriate process after establishing the facts. Do not assume that today's clarification automatically governs every historical visit.

15 **Maya:** One colleague suggested changing the old dates in the tracker to remove the apparent conflict. That would conceal what happened rather than resolve it.

16 **Ben:** Preserve the [8] _____. Any necessary correction must follow the applicable process and retain traceability. We must not rewrite the history to make the schedule appear consistent.

17 **Maya:** Once we receive the response, I will send a short site update with the confirmed version and the appointments that need attention.

18 **Ben:** Use a [9] _____ for the critical timing instruction. Ask the relevant colleague to repeat the window and its source so we can detect a misunderstanding before acting on it.

Two visit windows and no confirmed controlling version / continued

19 **Maya:** Send me the authorized clarification for the study record. I will reconcile the desk sheet and route affected data questions so the answer is not lost in email.

20 **Ben:** Good. The [10] _____ should show the discrepancy, the authorized resolution, and the follow-up. That closes the communication loop without turning a convenient reference sheet into an unofficial protocol.

CHECK THE GAPS

1. controlling version / **2.** effective date / **3.** schedule of assessments / **4.** principal investigator / **5.** protocol amendment / **6.** training record / **7.** protocol deviation / **8.** audit trail / **9.** read-back / **10.** essential records

Why these answers fit, page 135

F Check accuracy and scope

Two visit windows and no confirmed controlling version

Return to the case

Coordinator Maya is scheduling next week. A protocol attachment gives a plus-or-minus-three-day visit window; a site quick-reference sheet gives seven. The controlling version and site implementation status are unconfirmed. No participant awaits an urgent care decision. Clinical research associate Ben must obtain clarification through the investigator and sponsor process. Neither may invent a window, silently change records, or assume the newest attachment governs the site.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. How should a necessary record correction be handled?

A. Through the applicable traceable correction process, preserving the original history
B. By silently replacing the original value
C. By deleting the discrepancy from the record
D. By changing all records before checking the protocol

2. What should follow an authorized clarification?

A. Declare every prior visit compliant without review.
B. Treat clarification as permission to change any unrelated procedure.
C. Confirm understanding and assess affected appointments or records through the appropriate process.
D. Assume no further communication is needed.

Answers and explanations, page 136

Listen for the distinction

Switch roles for the clarification and read-back. Repeat the document identity, site-effective status, and exact window only when the script supplies confirmation.

Read back the conflict exactly

A useful escalation states both wordings, identifies their sources, and asks a specific question about the applicable version. Avoid describing the issue merely as confusion at the site. Precise language helps colleagues resolve the document problem without assigning blame or improvising clinical instructions.

F Say it / transfer to a new case

Receipt does not establish applicability

A coordinator receives version 5 by email. The site record still identifies version 4 as effective. No implementation confirmation for version 5 has been supplied.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

recency / confirmation / received / unconfirmed

01 **Coordinator:** Version 5 has been [1] _____.

02 **Monitor:** Its site applicability remains [2] _____.

03 **Coordinator:** I will request implementation [3] _____.

04 **Monitor:** Do not infer authority from file [4] _____.

CHECK THE GAPS

1. received / **2.** unconfirmed / **3.** confirmation / **4.** recency

Why these answers fit, page 136

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "Two visit windows and no confirmed controlling version". Maya: "Should we review earlier visits as well? I need the version that applied then, not just today's confirmed window, before labeling any visit outside it." Ben: [missing response] Maya: "One colleague suggested changing the old dates in the tracker to remove the apparent conflict. That would conceal what happened rather than resolve it." Which response belongs in the gap?

A. Exactly. Assess any potential protocol deviation through the appropriate process after establishing the facts. Do not assume that today's clarification automatically governs every historical visit.

B. Include the relevant schedule of assessments references as well. The team needs enough context to identify whether the quick-reference material reflects an authorized change or an error.

C. Please send the identifiers and exact sections for both. We need the controlling version before confirming the schedule. I cannot choose a window from the filenames alone.

D. A later email does not establish the effective date or implementation status. We should confirm which approved document applies here through the investigator and sponsor process.

Answers and explanations, page 137

A Read the situation

LESSON 04 / ENDPOINTS, ESTIMANDS, STATISTICS, DATA READOUTS, AND CLINICAL MEANING

Explain a trial readout without replacing the prespecified question or overstating statistical certainty.

A favorable secondary result takes over the headline

A fictional trial measures week-12 symptoms; lower scores are better. The primary treatment-minus-control estimate is -2 points (95% confidence interval: -5 to +1; p=0.19). The prespecified two-sided threshold is 0.05. A secondary endpoint has nominal p=0.01, but the testing plan requires primary success before a confirmatory secondary claim. Statistician Anika and medical lead Luis must correct the headline: the trial proved efficacy.

People: Anika, Statistician; Luis, Medical lead.

Read the record

Choose one answer per question using the facts above.

1. Did the primary analysis meet the specified significance threshold?

A. No; p=0.19 is above 0.05.
B. Yes; every negative point estimate is significant.
C. Yes; the secondary p-value replaces it.
D. No; because the confidence interval contains only negative values.

2. What is the secondary endpoint's status under this plan?

A. A guarantee of clinical benefit in every patient
B. A nominal result that cannot support the stated confirmatory claim after primary failure
C. Automatic proof that the primary endpoint succeeded
D. A replacement primary endpoint chosen after the readout

3. What does the negative point estimate indicate?

A. The treatment is worse because negative numbers are always unfavorable.
B. The confidence interval excludes no difference.
C. The estimated difference favors treatment on this lower-is-better scale.
D. The trial has proved no possible benefit.

Answers and explanations, page 138

B Vocabulary in use

CORE TERMS

primary endpoint
The outcome designated to address the study's principal objective.

report the primary endpoint

secondary endpoint
An additional prespecified outcome supporting other study objectives.

interpret secondary endpoints

exploratory endpoint
An outcome analyzed to investigate questions without the same confirmatory status.

label exploratory endpoints

estimand
The precise treatment-effect target defined for a clinical question.

specify the estimand

intercurrent event
An event after treatment starts that affects interpretation or existence of relevant measurements.

address intercurrent events

treatment-policy strategy
An estimand approach targeting outcomes regardless of specified intercurrent events.

define a treatment-policy strategy

hypothetical strategy
An estimand approach targeting a scenario in which specified intercurrent events would not occur.

justify a hypothetical strategy

analysis population
The defined group included in an analysis.

specify the analysis population

statistical analysis plan
The detailed prespecified methods for study analyses.

follow the statistical analysis plan

point estimate
A single numerical estimate of a population quantity.

present the point estimate

confidence interval
An interval from a procedure designed to cover the target parameter at a stated long-run rate.

report the confidence interval

p-value
The probability, under the null model, of a result at least as incompatible with it as observed.

interpret the p-value

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

primary endpoint PRIMARY ENDPOINT

report the primary endpoint

Keep the meaning precise: The outcome designated to address the study's principal objective.

treatment-policy strategy TREATMENT-POLICY STRATEGY

define a treatment-policy strategy

Keep the meaning precise: An estimand approach targeting outcomes regardless of specified intercurrent events.

B Precision and professional shorthand

EXTEND YOUR RANGE

significance threshold

The prespecified criterion used for a statistical test decision.

apply the significance threshold

testing hierarchy

A prespecified order and conditions for confirmatory tests.

respect the testing hierarchy

effect size

The magnitude of a difference or relationship on a stated scale.

interpret the effect size

missing data

Required or relevant observations not available for analysis.

assess missing data

subgroup analysis

Examination of results within defined subsets of participants.

qualify a subgroup analysis

multiplicity

The issue of repeated testing increasing opportunities for false positive conclusions.

control multiplicity

nominal p-value

A p-value reported without the relevant multiple-testing adjustment or confirmatory protection.

label a nominal p-value

clinical importance

The relevance of an effect to patients or clinical decisions.

assess clinical importance

sensitivity analysis

An analysis examining robustness to assumptions for the same target question.

perform sensitivity analysis

post hoc analysis

An analysis developed after the relevant data were examined.

label a post hoc analysis

B Precision and professional shorthand / continued

data cutoff

The date or point defining the data included in a readout.

state the data cutoff

topline readout

An initial summary of key study results.

prepare a balanced topline readout

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

p-value P-VALUE

interpret the p-value

Keep the meaning precise: The probability, under the null model, of a result at least as incompatible with it as observed.

effect size EFFECT SIZE

interpret the effect size

Keep the meaning precise: The magnitude of a difference or relationship on a stated scale.

C Phrases that move the work forward

Lead with the primary

The primary analysis did not meet the prespecified significance threshold.

Preserve uncertainty

The confidence interval includes no difference.

Respect the plan

Primary success was required before a confirmatory secondary claim.

Separate importance

Statistical significance and clinical importance are different questions.

Check the population

Confirm which participants are included in this analysis.

Ask about missingness

What assumptions support the handling of missing outcomes?

Clarify the scale

The difference is treatment minus control, measured in score points.

Avoid a rescue headline

A favorable secondary result does not replace the primary outcome.

State the direction

The point estimate favors treatment on this lower-is-better scale.

Qualify the secondary

The secondary p-value is nominal under this testing plan.

Avoid a false negative claim

Failure to meet the threshold does not prove there is no effect.

Define the target

Which treatment effect does the estimand specify?

Handle events explicitly

State how treatment discontinuation affects the target question.

Label later work

This post hoc analysis was not the prespecified primary analysis.

Keep dates visible

Identify the data cutoff for this readout.

Close with the full result

Present the estimate, uncertainty, testing status, and clinical context together.

Precision in real exchanges

Purpose: Qualify the secondary
Useful wording: The secondary p-value is nominal under this testing plan.

Purpose: Respect the plan
Useful wording: Primary success was required before a confirmatory secondary claim.

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which topline preserves the testing plan and uncertainty?

A. Primary threshold not met; the favorable secondary result remains nominal under the hierarchy.

B. Primary effect absent; the secondary result establishes efficacy at p=0.01.

C. Primary estimate favors treatment; therefore the primary significance requirement is satisfied.

D. Secondary efficacy confirmed; the primary result can be retained as supportive evidence.

2. What does the interval from -5 to +1 show?

A. It includes values favoring treatment, zero, and values favoring control on this scale.

B. It excludes every possibility of no difference.

C. It contains only unfavorable treatment differences.

D. It proves that 95% of patients improved by two points.

3. A colleague needs the term for this meaning: The precise treatment-effect target defined for a clinical question. Which term fits?

A. estimand

B. subgroup analysis

C. statistical analysis plan

D. testing hierarchy

D Choose the response that fits / continued

4. Rebuild the exchange from "A favorable secondary result takes over the headline". Luis: "The interval is minus five to plus one. Someone suggested calling that no effect because it crosses zero." Anika: [missing response] Luis: "What about the secondary result? Its p-value is smaller. Could that become the main efficacy message instead?" Which response belongs in the gap?

A. I will review the topline readout against the plan. We can report uncertainty clearly without claiming proven efficacy or turning an unsuccessful test into proof of no effect.

B. Yes. The primary endpoint did not meet its prespecified threshold: point one nine is above point zero five. We should lead with that result.

C. The confidence interval includes zero and values in both directions. It does not establish an effect of exactly zero or prove that every patient had the same response.

D. Report the nominal p-value with its limitation. The observation remains available for discussion, but it does not have the confirmatory status implied by proved efficacy.

Answers and explanations, page 138

> **One useful distinction**
>
> The estimate favors treatment, but the interval includes zero and the primary p-value exceeds 0.05. This is not proof of no effect. It is also not a successful primary test. The secondary finding retains its actual status under the prespecified plan.

E Conversation

A favorable secondary result takes over the headline

A favorable secondary result takes over the headline

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

testing hierarchy / intercurrent event / estimand / confidence interval / topline readout / Clinical importance / point estimate / sensitivity analyses / nominal p-value / primary endpoint

01 **Luis:** The headline says the trial proved efficacy. It leads with the secondary p-value of point zero one. Is that too strong?

02 **Anika:** Yes. The [1] _____ did not meet its prespecified threshold: point one nine is above point zero five. We should lead with that result.

03 **Luis:** But the treatment-minus-control difference is minus two points. Lower symptoms are better, so the direction is favorable. Can we still say that?

04 **Anika:** Certainly. The [2] _____ favors treatment. Keep it beside the uncertainty rather than dropping the favorable direction or presenting it as definitive efficacy.

05 **Luis:** The interval is minus five to plus one. Someone suggested calling that no effect because it crosses zero.

06 **Anika:** The [3] _____ includes zero and values in both directions. It does not establish an effect of exactly zero or prove that every patient had the same response.

E Conversation / continued

07 **Luis:** What about the secondary result? Its p-value is smaller. Could that become the main efficacy message instead?

08 **Anika:** Not under this [4] _____. Confirmatory secondary testing required primary success. Changing the headline cannot change the rule after we have seen the data.

09 **Luis:** I do not want to hide the secondary observation either. How should it appear in the readout?

10 **Anika:** Report the [5] _____ with its limitation. The observation remains available for discussion, but it does not have the confirmatory status implied by proved efficacy.

Conversation continues on the next page.

A favorable secondary result takes over the headline

CONVERSATION / CONTINUED

11 **Luis:** The clinicians will ask whether two points matter to patients. We have not included any explanation of the symptom scale.

12 **Anika:** [6] _____ needs that context and supporting evidence. A p-value alone cannot tell them whether the magnitude would be meaningful.

13 **Luis:** There is another inconsistency. One slide says all assigned patients; another talks only about people who stayed on treatment. Are those equivalent?

14 **Anika:** We need to check the [7] _____. It defines the treatment effect we intended to estimate, including population, outcome, treatments, event handling, and summary measure.

15 **Luis:** So stopping treatment is not automatically a reason to remove someone from the question or to assume every later observation is missing?

16 **Anika:** Correct. Discontinuation can be an [8] _____. Its treatment depends on the defined question; missing observations are a related but different issue.

17 **Luis:** Would alternative assumptions help show how robust the result is? I do not want the team shopping for the most favorable answer.

18 **Anika:** Use the planned [9] _____ for the same target question. Label exploratory work separately, especially when it changes the population or treatment-effect question.

19 **Luis:** I will revise the headline, retain the estimate and interval, and label the secondary finding nominal. Please check the population wording before circulation.

20 **Anika:** I will review the [10] _____ against the plan. We can report uncertainty clearly without claiming proven efficacy or turning an unsuccessful test into proof of no effect.

CHECK THE GAPS

1. primary endpoint / **2.** point estimate / **3.** confidence interval / **4.** testing hierarchy / **5.** nominal p-value / **6.** Clinical importance / **7.** estimand / **8.** intercurrent event / **9.** sensitivity analyses / **10.** topline readout

Why these answers fit, page 139

F Check accuracy and scope

A favorable secondary result takes over the headline

Return to the case

A fictional trial measures week-12 symptoms; lower scores are better. The primary treatment-minus-control estimate is -2 points (95% confidence interval: -5 to +1; p=0.19). The prespecified two-sided threshold is 0.05. A secondary endpoint has nominal p=0.01, but the testing plan requires primary success before a confirmatory secondary claim. Statistician Anika and medical lead Luis must correct the headline: the trial proved efficacy.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. Why does clinical importance need separate discussion?

A. A negative sign always proves a meaningful improvement.
B. Clinical importance is identical to the number of endpoints.
C. The practical meaning of a score difference is not established by a p-value alone.
D. Any p-value below one guarantees patient benefit.

2. Which statement about later subgroup work is appropriate?

A. Assume smaller groups cannot produce chance findings.
B. Label it according to its actual prespecification and multiplicity status.
C. Present it as the original primary analysis regardless of timing.
D. Hide the overall trial result whenever a subgroup looks favorable.

Answers and explanations, page 141

Listen for the distinction

Switch roles for the secondary-endpoint discussion. Stress nominal and the prespecified hierarchy; do not relabel primary failure as proof of no effect.

Do not repair a result by changing the question

Lead with the prespecified primary result and explain uncertainty before highlighting other findings. A disappointing primary test does not make every observation worthless, but a favorable secondary result does not erase the testing plan. Keep statistical evidence and clinical importance separate in the discussion.

F Say it / transfer to a new case

Relative and absolute reductions

In a separate fictional comparison, events occur in 200 of 1,000 control participants and 150 of 1,000 treatment participants. Use only these counts; no significance result is supplied.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

five percentage points / 15% / 25% / 20%

01 **Analyst:** The control event rate is [1] _____.

02 **Reviewer:** The treatment event rate is [2] _____.

03 **Analyst:** The absolute reduction is [3] _____.

04 **Reviewer:** Relative to control, the reduction is [4] _____.

CHECK THE GAPS

1. 20% / **2.** 15% / **3.** five percentage points / **4.** 25%

Why these answers fit, page 141

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "A favorable secondary result takes over the headline". Luis: "There is another inconsistency. One slide says all assigned patients; another talks only about people who stayed on treatment. Are those equivalent?" Anika: [missing response] Luis: "So stopping treatment is not automatically a reason to remove someone from the question or to assume every later observation is missing?" Which response belongs in the gap?

A. Not under this testing hierarchy. Confirmatory secondary testing required primary success. Changing the headline cannot change the rule after we have seen the data.

B. Yes. The primary endpoint did not meet its prespecified threshold: point one nine is above point zero five. We should lead with that result.

C. We need to check the estimand. It defines the treatment effect we intended to estimate, including population, outcome, treatments, event handling, and summary measure.

D. Certainly. The point estimate favors treatment. Keep it beside the uncertainty rather than dropping the favorable direction or presenting it as definitive efficacy.

Answers and explanations, page 142

Review 2 / Meaning and context

Use language from the lessons you have completed. Choose one answer per question.

> **Case file**
>
> Revisit "Clinical Trial Design, Protocols, GCP, and Operations" and "Endpoints, Estimands, Statistics, Data Readouts, and Clinical Meaning". Choose precise terminology, then reconstruct the missing responses using both neighboring turns. The final two questions also revisit earlier conversations.

1. A colleague needs the term for this meaning: The allowed timing range for a specified study visit or assessment. Which term fits?

A. visit window

B. eligibility criterion

C. principal investigator

D. protocol deviation

2. A colleague needs the term for this meaning: The defined group included in an analysis. Which term fits?

A. primary endpoint

B. missing data

C. analysis population

D. significance threshold

3. Rebuild the exchange from "A meeting question that asks for blanket agreement". Evan: "I will also check the references. One footnote points to an earlier table, and the current table no longer contains the information cited." Priya: [missing response] Evan: "Some colleagues are calling the meeting package our IND approval package. That seems to mix the meeting purpose with the status of a regulatory submission." Which response belongs in the gap?

A. Exactly. The regulatory pathway depends on the product and context. Regulatory affairs should confirm the route, while the current discussion remains focused on the specific development question.

B. Verify each cross-reference after the content is settled. A broken evidence trail forces reviewers to guess which material we intended and can weaken an otherwise useful discussion.

C. We need a focused question. The immediate issue is the proposed patient population and endpoint rationale, not every aspect of the program from development through launch.

D. Reconcile the wording in the briefing document. A precise question is not enough if the supporting text describes a different proposal or leaves the study population ambiguous.

Review 2 / Meaning and context / continued

4. Rebuild the exchange from "A favorable secondary result takes over the headline". Luis: "Would alternative assumptions help show how robust the result is? I do not want the team shopping for the most favorable answer." Anika: [missing response] Luis: "I will revise the headline, retain the estimate and interval, and label the secondary finding nominal. Please check the population wording before circulation." Which response belongs in the gap?

A. Not under this testing hierarchy. Confirmatory secondary testing required primary success. Changing the headline cannot change the rule after we have seen the data.

B. Yes. The primary endpoint did not meet its prespecified threshold: point one nine is above point zero five. We should lead with that result.

C. Use the planned sensitivity analyses for the same target question. Label exploratory work separately, especially when it changes the population or treatment-effect question.

D. Certainly. The point estimate favors treatment. Keep it beside the uncertainty rather than dropping the favorable direction or presenting it as definitive efficacy.

Answers and explanations, page 166

A Read the situation

LESSON 05 / PHARMACOVIGILANCE, SAFETY SIGNALS, RISK-BENEFIT, AND LABEL UPDATES

Preserve reported facts, avoid premature causality judgments, and route a safety report without waiting for perfect information.

An incomplete safety report

Medical-information associate Nora receives a message from a clinician reporting dizziness after a patient used fictional product LM-4. The clinician calls the dizziness severe, but the message does not state whether hospitalization, a life-threatening condition, or another seriousness criterion applies. Dates, dose, outcome, and other medicines are missing. The company procedure requires prompt internal routing of available safety information and documented follow-up. Safety colleague Rafael must distinguish reported severity, unconfirmed seriousness, and unassessed causality. This conversation is an internal handoff, not advice to the patient.

People: Nora, Medical-information associate; Rafael, Safety colleague.

Read the record

Choose one answer per question using the facts above.

1. What has the clinician actually reported?

A. Dizziness described as severe after product use
B. Confirmed product-caused hospitalization
C. A completed causality assessment
D. A verified outcome and dosing history

2. Can severe automatically be treated as serious in the regulatory sense?

A. Yes; the words are always interchangeable.
B. Yes; any adjective establishes hospitalization.
C. No; severe events must never be routed.
D. No; intensity and seriousness criteria are different concepts.

3. What does the stated company procedure require?

A. Waiting until causality is proved before any handoff
B. Deleting reports with missing dates
C. Giving the patient a dosing recommendation during the handoff
D. Prompt internal routing of available information with documented follow-up

Answers and explanations, page 143

B Vocabulary in use

CORE TERMS

pharmacovigilance

Activities concerned with detecting, assessing, understanding, and preventing medicine-related problems.

support pharmacovigilance

adverse reaction

An undesirable response for which a causal relationship is at least reasonably suspected in context.

assess a suspected adverse reaction

severity

The intensity of an event or symptom.

record reported severity

expectedness

Whether the nature or severity of an event is consistent with the applicable reference safety information.

evaluate expectedness

safety signal

Information suggesting a possible new or changed risk that warrants further evaluation.

evaluate a safety signal

adverse event

An unfavorable medical occurrence reported during product use, without necessarily established causation.

capture an adverse event

seriousness

Classification based on applicable outcomes or medical criteria, not intensity alone.

assess seriousness

causality assessment

Evaluation of whether and how a product may be related to an event.

document causality assessment

reference safety information

The designated information used for defined safety assessments in context.

identify reference safety information

case intake

Initial receipt and recording of a safety report.

complete case intake

B Vocabulary in use / continued

individual case safety report

A structured account of an individual suspected safety case.

prepare an individual case safety report

reporter

The person or source providing event information.

preserve reporter details

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

pharmacovigilance PHARMACOVIGILANCE

support pharmacovigilance

Keep the meaning precise: Activities concerned with detecting, assessing, understanding, and preventing medicine-related problems.

causality assessment CAUSALITY ASSESSMENT

document causality assessment

Keep the meaning precise: Evaluation of whether and how a product may be related to an event.

B Precision and professional shorthand

EXTEND YOUR RANGE

verbatim term

The exact wording used by the original reporter.

retain the verbatim term

temporal association

A relationship in timing without necessarily proving causation.

distinguish temporal association from causation

dechallenge

What happens after a suspected product is withdrawn or reduced, when this occurs clinically.

record available dechallenge information

follow-up request

A request for additional information about an existing case.

issue a focused follow-up request

duplicate report

Another report concerning a case already recorded.

investigate a possible duplicate report

event onset

The time an event began.

confirm event onset

concomitant medication

Another medicine used during the relevant period.

document concomitant medications

rechallenge

What happens after re-exposure to a product, when this has occurred clinically.

record reported rechallenge information

case reconciliation

Checking that related records are consistent and accounted for across sources.

perform case reconciliation

MedDRA

Medical Dictionary for Regulatory Activities, a standardized terminology used for regulatory medical information.

apply appropriate MedDRA coding

B Precision and professional shorthand / continued

benefit-risk assessment

Evaluation of favorable and unfavorable effects in the relevant context.

update the benefit-risk assessment

aggregate safety review

Evaluation of safety information across cases or datasets.

conduct an aggregate safety review

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

reporter REPORTER

preserve reporter details

Keep the meaning precise: The person or source providing event information.

dechallenge DECHALLENGE

record available dechallenge information

Keep the meaning precise: What happens after a suspected product is withdrawn or reduced, when this occurs clinically.

C Phrases that move the work forward

Open the handoff

I have an incomplete report that needs safety intake.

Separate the classification

Seriousness has not yet been confirmed.

Name the gaps

Dates, dose, outcome, and concomitant medicines are missing.

Keep follow-up focused

Please clarify whether any applicable seriousness criterion was met.

Clarify timing

When did the event begin relative to product use?

Assign ownership

Who will make the follow-up request and track the response?

Avoid dismissal

Missing information does not justify discarding the report.

Separate care

Clinical treatment questions belong with the qualified treating professional.

Preserve the wording

The clinician described the dizziness as severe.

Avoid causality overreach

The event followed use; causality has not been assessed.

Route promptly

I will send the available information through the required internal channel now.

Protect the original

Retain the reporter's exact words alongside any coded term.

Mark an unknown

The outcome is unknown, not recovered.

Confirm receipt

Please acknowledge intake so we know the handoff reached the safety team.

Distinguish a signal

One report does not by itself establish a confirmed causal risk.

Close accurately

Record what was reported, what remains unknown, and the next responsible owner.

Precision in real exchanges

Purpose: Avoid causality overreach
Useful wording: The event followed use; causality has not been assessed.

Purpose: Name the gaps
Useful wording: Dates, dose, outcome, and concomitant medicines are missing.

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which intake summary accurately separates the three safety concepts?

A. Serious dizziness confirmed after use; severity and causality unassessed.
B. Severe dizziness caused by the product; seriousness awaiting hospitalization details.
C. Nonserious dizziness after use; no hospitalization or unresolved outcome reported.
D. Severe dizziness reported after use; seriousness unconfirmed and causality unassessed.

2. What should happen while missing details are sought?

A. Delete the original wording after choosing a code.
B. Give a medication-change instruction to improve the report.
C. Route available information promptly under the stated procedure and document follow-up.
D. Keep the report outside the safety process until every field is complete.

3. A colleague needs the term for this meaning: Classification based on applicable outcomes or medical criteria, not intensity alone. Which term fits?

A. seriousness
B. temporal association
C. safety signal
D. duplicate report

D Choose the response that fits / continued

4. Rebuild the exchange from "An incomplete safety report". Nora: "The event occurred after product use, but the message gives no exact dates. I cannot tell how close the timing was or whether other explanations were considered." Rafael: [missing response] Nora: "We are missing dose, outcome, and other medicines. I will list those gaps so they remain visible in the structured record." Which response belongs in the gap?

A. Correct. A safety signal requires evaluation in context. This individual report contributes information, but we should not announce a confirmed new risk or dismiss it without assessment.

B. Describe a reported temporal association, not proven causation. The safety assessment will consider the available evidence. Do not add a causal conclusion that the source did not establish.

C. Any benefit-risk assessment or labeling action belongs in the qualified review process. Our immediate task is accurate capture, appropriate routing, and follow-up, not deciding the product's overall profile.

D. Preserve the verbatim term. Appropriate coding can be added through the process, but the original wording remains important. Standardization should not erase what the reporter actually said.

Answers and explanations, page 143

One useful distinction

Severe describes intensity. Serious uses applicable criteria, including death, life-threatening events, inpatient hospitalization or its prolongation, persistent or significant disability, congenital anomalies, and other medically important events. Clarify missing facts; do not infer seriousness from intensity alone.

E Conversation

An incomplete safety report

An incomplete safety report

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

case intake / verbatim term / benefit-risk assessment / temporal association / concomitant medications / severity / follow-up request / case reconciliation / receipt confirmation / safety signal

01 **Nora:** I received a clinician's message about dizziness after LM-4 use. The clinician called it severe, but several details are missing. I want to route it correctly.

02 **Rafael:** Send the available information through [1] _____ now, as our procedure requires. We can document what is unknown and arrange follow-up without waiting for you to complete a safety assessment yourself.

03 **Nora:** I have sent the message through intake now, keeping the clinician's wording. It does not state hospitalization, a life-threatening condition, or another seriousness criterion.

04 **Rafael:** I can confirm receipt. Keep [2] _____ separate from seriousness. Severe describes intensity; it does not settle the regulatory classification. We need to clarify the relevant facts rather than translate one word directly into the other.

05 **Nora:** The event occurred after product use, but the message gives no exact dates. I cannot tell how close the timing was or whether other explanations were considered.

06 **Rafael:** Describe a reported [3] _____, not proven causation. The safety assessment will consider the available evidence. Do not add a causal conclusion that the source did not establish.

E Conversation / continued

07 **Nora:** We are missing dose, outcome, and other medicines. I will list those gaps so they remain visible in the structured record.

08 **Rafael:** Include [4] _____ in the follow-up request. Record unknown fields as unknown; absence of information must not become an assertion that no other medicine was used.

09 **Nora:** The message says severe dizziness. For the structured follow-up, should I standardize it or keep the clinician's exact phrase alongside the coded fields?

10 **Rafael:** Preserve the [5] _____. Appropriate coding can be added through the process, but the original wording remains important. Standardization should not erase what the reporter actually said.

Conversation continues on the next page.

An incomplete safety report

CONVERSATION / CONTINUED

11 **Nora:** I can ask when the dizziness started, whether it resolved, and whether hospitalization occurred. I should avoid asking the clinician to confirm a conclusion we prefer.

12 **Rafael:** Make the [6] _____ neutral and specific. Ask for the facts needed to assess the case, and document the request and response through the required channel.

13 **Nora:** Who owns that contact? I do not want both teams to approach the reporter independently and create two incomplete versions of the same conversation.

14 **Rafael:** We will assign an owner and use [7] _____ to keep the records aligned. If another report arrives, assess whether it is connected rather than assuming similar wording means a duplicate.

15 **Nora:** The commercial team may ask whether this is a new safety signal. I do not think this handoff alone answers that broader question.

16 **Rafael:** Correct. A [8] _____ requires evaluation in context. This individual report contributes information, but we should not announce a confirmed new risk or dismiss it without assessment.

17 **Nora:** I also want the internal update to avoid implying that a label change has been decided. We are much earlier in the information process.

18 **Rafael:** Any [9] _____ or labeling action belongs in the qualified review process. Our immediate task is accurate capture, appropriate routing, and follow-up, not deciding the product's overall profile.

An incomplete safety report / continued

19 **Nora:** The message is routed. I will add the agreed follow-up owner and outstanding questions to the record. Treatment questions belong with the qualified treating professional.

20 **Rafael:** Keep the [10] _____ with the handoff record. You have confirmed the information reached safety; now we need to track the missing facts and follow-up response.

CHECK THE GAPS

1. case intake / **2.** severity / **3.** temporal association / **4.** concomitant medications / **5.** verbatim term / **6.** follow-up request / **7.** case reconciliation / **8.** safety signal / **9.** benefit-risk assessment / **10.** receipt confirmation

Why these answers fit, page 144

F Check accuracy and scope

An incomplete safety report

Return to the case

Medical-information associate Nora receives a message from a clinician reporting dizziness after a patient used fictional product LM-4. The clinician calls the dizziness severe, but the message does not state whether hospitalization, a life-threatening condition, or another seriousness criterion applies. Dates, dose, outcome, and other medicines are missing. The company procedure requires prompt internal routing of available safety information and documented follow-up. Safety colleague Rafael must distinguish reported severity, unconfirmed seriousness, and unassessed causality. This conversation is an internal handoff, not advice to the patient.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. Which follow-up asks for a missing fact without leading the reporter?

A. Please confirm the product was definitely responsible.
B. Can you change severe to mild so the case is simpler?
C. Since the patient recovered, can we close the case?
D. Was the patient hospitalized, and what was the outcome?

2. What does a temporal association establish?

A. That the event and product use have a stated timing relationship, not necessarily causation
B. That the product caused the event in every similar patient
C. That a label update is automatically required
D. That further assessment is unnecessary

Answers and explanations, page 146

Listen for the distinction

Switch roles for the follow-up discussion. Preserve severe as the reported intensity, unknown seriousness, unassessed causality, and the clinician's original words.

Keep the report and the judgment separate

Use reported for the clinician's words and unconfirmed for missing facts. Do not delay the required internal handoff while trying to prove that a product caused the event. Equally, do not dismiss the report because causality is uncertain. Qualified safety assessment and clinical care have their own appropriate routes.

F Say it / transfer to a new case

Unknown is not negative

A report says a rash occurred after product use. It does not state hospitalization, outcome, or a causal assessment. The applicable internal process requires routing available information now.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

unknown / now / causation / follow-up

01 **Associate:** Hospitalization status is [1] _____.

02 **Reviewer:** The timing does not establish [2] _____.

03 **Associate:** I will route the available report [3] _____.

04 **Reviewer:** Record the missing details for [4] _____.

CHECK THE GAPS

1. unknown / **2.** causation / **3.** now / **4.** follow-up

Why these answers fit, page 146

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "An incomplete safety report". Nora: "Who owns that contact? I do not want both teams to approach the reporter independently and create two incomplete versions of the same conversation." Rafael: [missing response] Nora: "The commercial team may ask whether this is a new safety signal. I do not think this handoff alone answers that broader question." Which response belongs in the gap?

A. We will assign an owner and use case reconciliation to keep the records aligned. If another report arrives, assess whether it is connected rather than assuming similar wording means a duplicate.

B. Describe a reported temporal association, not proven causation. The safety assessment will consider the available evidence. Do not add a causal conclusion that the source did not establish.

C. Any benefit-risk assessment or labeling action belongs in the qualified review process. Our immediate task is accurate capture, appropriate routing, and follow-up, not deciding the product's overall profile.

D. Send the available information through case intake now, as our procedure requires. We can document what is unknown and arrange follow-up without waiting for you to complete a safety assessment yourself.

Answers and explanations, page 147

A Read the situation

LESSON 06 / CMC, CGMP, QUALITY EVENTS, MANUFACTURING, AND SUPPLY

Report a quality-documentation gap precisely and separate investigation status from batch disposition.

A gap in the batch record

Batch Q-214 lacks a verification entry; whether the check occurred is unknown. The batch remains under quality review, not released. Production lead Diego asks quality specialist Helen whether available laboratory results justify saying fully reviewed. Site procedure requires documented investigation and authorized disposition. Neither may backdate an entry, assume the check occurred, or promise release to meet a shipment date.

People: Diego, Production lead; Helen, Quality specialist.

Read the record

Choose one answer per question using the facts above.

1. What is established about batch Q-214?

A. The product definitely failed every laboratory test.
B. A verification entry is missing and quality review remains open.
C. The batch has been released.
D. The verification definitely occurred.

2. Why is fully reviewed inaccurate?

A. A shipment date automatically closes review.
B. A production request supplies quality authorization.
C. The required investigation and authorized disposition are not complete.
D. Available laboratory results make all records irrelevant.

3. Which action is excluded by the case?

A. Backdating an entry to make the record appear complete
B. Preserving the record
C. Investigating the gap
D. Giving supply an accurate status update

Answers and explanations, page 148

B Vocabulary in use

CORE TERMS

chemistry, manufacturing, and controls

Product and process information addressing composition, manufacture, and quality control.

assess CMC readiness

batch record

Documentation of the production and control activities for a specific batch.

review the batch record

verification entry

A record showing a required check and its associated information.

complete the verification entry

data integrity

The completeness, consistency, and accuracy of data through its lifecycle.

protect data integrity

traceability

The ability to follow an activity, material, or result through its records.

preserve traceability

root cause

The underlying cause supported by investigation.

establish the root cause

current good manufacturing practice

Applicable requirements and practices supporting consistent pharmaceutical quality.

maintain CGMP compliance

master production instruction

The authorized instruction defining how a product is to be manufactured.

follow master production instructions

documentation gap

Missing or incomplete recorded information.

investigate a documentation gap

contemporaneous record

A record made when the relevant activity occurs.

maintain contemporaneous records

deviation investigation

A structured examination of a departure from an applicable requirement or procedure.

conduct a deviation investigation

corrective action

Action addressing a detected problem and its cause in the relevant system.

implement corrective action

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

chemistry, manufacturing, and controls CHEMISTRY, MANUFACTURING, AND CONTROLS

assess CMC readiness

Keep the meaning precise: Product and process information addressing composition, manufacture, and quality control.

documentation gap DOCUMENTATION GAP

investigate a documentation gap

Keep the meaning precise: Missing or incomplete recorded information.

B Precision and professional shorthand

EXTEND YOUR RANGE

preventive action

Action intended to prevent a potential problem from occurring.

define preventive action

CAPA

Corrective and preventive action, the system for addressing actual and potential quality problems.

verify CAPA effectiveness

effectiveness check

A review of whether an action achieved its intended result.

plan an effectiveness check

out-of-specification result

A test result outside established acceptance specifications.

investigate an out-of-specification result

out-of-trend result

A result inconsistent with an expected pattern, even when it may meet specifications.

evaluate an out-of-trend result

specification

Defined tests, procedures, and acceptance criteria for a material or product.

confirm the applicable specification

certificate of analysis

A document reporting specified test results for a material or batch.

verify the certificate of analysis

batch disposition

The authorized decision on how a batch may be used or handled.

document batch disposition

release authorization

Formal permission for a batch to proceed to the defined use or distribution.

verify release authorization

change control

The managed evaluation and authorization of proposed changes.

route a change through change control

B Precision and professional shorthand / continued

stability program

Planned assessment of product quality over time under specified conditions.

maintain the stability program

supply constraint

A limit affecting availability or delivery.

communicate the supply constraint

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

corrective action CORRECTIVE ACTION

implement corrective action

Keep the meaning precise: Action addressing a detected problem and its cause in the relevant system.

out-of-trend result OUT-OF-TREND RESULT

evaluate an out-of-trend result

Keep the meaning precise: A result inconsistent with an expected pattern, even when it may meet specifications.

C Phrases that move the work forward

State the observed gap

The verification entry is missing from the batch record.

Report the status

The batch remains under quality review and is not released.

Protect integrity

Do not backdate or reconstruct an entry as though it were contemporaneous.

Name the owner

Quality will coordinate the documented investigation under the site process.

Preserve traceability

Any permitted correction must retain its source and history.

Check effectiveness

An action is not complete merely because a training session was scheduled.

Avoid blanket blame

Establish the cause before assigning an unsupported explanation.

Confirm authorization

Who has authority to make the batch-disposition decision?

Avoid an assumed cause

We have not established whether the check occurred.

Separate evidence

Available laboratory results do not complete the documentation review.

Request the investigation

Please identify the evidence needed to resolve this discrepancy.

Limit the supply update

Release timing is unconfirmed pending authorized disposition.

Distinguish results

This documentation issue is not itself an out-of-specification test result.

Escalate pressure

The shipment deadline does not change the evidence or release authority.

Keep actions separate

Correction of the record and prevention of recurrence are distinct tasks.

Close with an update

We will provide the next status update without promising a release date.

Precision in real exchanges

Purpose: Separate evidence
Useful wording: Available laboratory results do not complete the documentation review.

Purpose: Protect integrity
Useful wording: Do not backdate or reconstruct an entry as though it were contemporaneous.

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which supply update is supported?

A. Q-214 definitely failed manufacturing because a field is blank.
B. Q-214 remains under quality review; release timing is unconfirmed.
C. Q-214 is fully reviewed because laboratory results are available.
D. Q-214 is released because a customer needs it tomorrow.

2. What does the missing entry prove by itself?

A. That the check definitely did not occur
B. That every test result is invalid
C. That required recorded information is absent
D. That the check definitely occurred

3. A colleague needs the term for this meaning: The authorized instruction defining how a product is to be manufactured. Which term fits?

A. release authorization
B. effectiveness check
C. master production instruction
D. traceability

D Choose the response that fits / continued

4. Rebuild the exchange from "A gap in the batch record". Diego: "Someone suggested signing yesterday's date to tidy it up. I have stopped that and kept the original record unchanged." Helen: [missing response] Diego: "I can make the original records and relevant staff available. Should we keep the messages about this discrepancy as well?" Which response belongs in the gap?

A. I will. Once there is an actual decision, we will communicate it through the proper route. Until then, do not imply that release authorization exists.

B. Good. A later entry cannot be passed off as a contemporaneous record. Any permitted correction must accurately show its actual timing, basis, and required authorization.

C. Specify an effectiveness check tied to the intended result. Completing a session and showing that a recurring documentation weakness has been addressed are different things.

D. No. The batch record still has an unresolved entry, and quality review is open. Test results alone do not establish completed review or release.

Answers and explanations, page 148

> **One useful distinction**
>
> Available test results are not the same as a completed batch review or release. The missing entry needs investigation under the applicable process. A signature added later must not falsely represent when an activity occurred; any permitted correction needs an accurate, traceable basis.

E Conversation

A gap in the batch record

A gap in the batch record

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

contemporaneous record / release authorization / traceability / root cause / batch disposition / effectiveness check / out-of-specification result / documentation gap / deviation investigation / batch record

01 **Diego:** Supply wants to ship Q-214 tomorrow. The laboratory results are available, but a verification entry is missing. Can I at least say fully reviewed?

02 **Helen:** No. The [1] _____ still has an unresolved entry, and quality review is open. Test results alone do not establish completed review or release.

03 **Diego:** Do we know whether the check was missed or just not written down? The shift team remembers a busy handover.

04 **Helen:** Not yet. We have a [2] _____, not a supported explanation. Record the recollection as a source to investigate, not proof of what occurred.

05 **Diego:** Someone suggested signing yesterday's date to tidy it up. I have stopped that and kept the original record unchanged.

06 **Helen:** Good. A later entry cannot be passed off as a [3] _____. Any permitted correction must accurately show its actual timing, basis, and required authorization.

07 **Diego:** I can make the original records and relevant staff available. Should we keep the messages about this discrepancy as well?

08 **Helen:** Preserve the relevant evidence through the site process. We need [4] _____ between the activity, supporting information, and any documented correction.

09 **Diego:** Supply has called this an OOS. That sounds like a failed test, but I have not been shown a result outside a limit.

10 **Helen:** An [5] _____ is not the same as a missing entry. Use the documentation description unless the investigation establishes another finding.

Conversation continues on the next page.

A gap in the batch record

CONVERSATION / CONTINUED

11 **Diego:** Who is coordinating the next steps? Production can help, but I need a clear owner before I reply to planning.

12 **Helen:** Quality is coordinating the [6] _____ under our site procedure. We will identify the evidence needed and communicate the actual review status.

13 **Diego:** The proposed action is another training session. Would that be enough to close this before tomorrow's shipment?

14 **Helen:** Not without establishing the [7] _____ and appropriate action. Keep containment in place now; a convenient training proposal does not resolve the investigation or release the batch.

15 **Diego:** Then the action record needs more than a training date. How will we know the change addressed the problem?

16 **Helen:** Specify an [8] _____ tied to the intended result. Completing a session and showing that a recurring documentation weakness has been addressed are different things.

17 **Diego:** Planning will press for an estimate. I can promise a status update, but I cannot tell the customer that tomorrow's shipment is approved.

18 **Helen:** Correct. Authorized [9] _____ remains pending. Communicate that timing is unconfirmed; urgency does not transfer the decision to production or planning.

19 **Diego:** I will report the missing entry, open investigation, and unreleased status. Let me know the next update point without converting it into a release promise.

20 **Helen:** I will. Once there is an actual decision, we will communicate it through the proper route. Until then, do not imply that [10] _____ exists.

CHECK THE GAPS

1. batch record / **2.** documentation gap / **3.** contemporaneous record / **4.** traceability / **5.** out-of-specification result / **6.** deviation investigation / **7.** root cause / **8.** effectiveness check / **9.** batch disposition / **10.** release authorization

Why these answers fit, page 149

F Check accuracy and scope

A gap in the batch record

Return to the case

Batch Q-214 lacks a verification entry; whether the check occurred is unknown. The batch remains under quality review, not released. Production lead Diego asks quality specialist Helen whether available laboratory results justify saying fully reviewed. Site procedure requires documented investigation and authorized disposition. Neither may backdate an entry, assume the check occurred, or promise release to meet a shipment date.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. Which action preserves data integrity?

A. Add yesterday's date to make today's entry look timely.

B. Copy another batch's verification to fill the space.

C. Remove the blank page so the review looks complete.

D. Use the authorized correction process with an accurate, traceable account.

2. Which action plan best distinguishes completion from effectiveness?

A. Treat attendance at retraining as proof that the missing-entry problem cannot recur.

B. Link the action to the supported cause and specify evidence that it reduces recurrence.

C. Close the investigation when a training invitation is sent to the affected staff.

D. Release the batch when the team agrees that the record looks complete.

Answers and explanations, page 151

Listen for the distinction

Switch roles for the proposed correction. Stress actual timing, traceable evidence, immediate containment, and pending disposition; do not make tomorrow a release promise.

Describe the gap without deciding the cause

Missing documentation is a fact; assuming the activity happened or did not happen adds a conclusion. State the observed gap, current batch status, investigation owner, and next update. This is more useful to supply colleagues than either premature reassurance or an unsupported declaration that the product failed.

F Say it / transfer to a new case

A test report is not a release decision

A certificate of analysis is available, but the batch-disposition record is pending. The current status says not released. A customer has requested shipment tomorrow.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

results / released / authorization / pending

01 **Planner:** The certificate reports test [1] _____.

02 **Quality:** Batch disposition is still [2] _____.

03 **Planner:** The current status is not [3] _____.

04 **Quality:** The customer request does not supply release [4] _____.

CHECK THE GAPS

1. results / **2.** pending / **3.** released / **4.** authorization

Why these answers fit, page 151

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "A gap in the batch record". Diego: "The proposed action is another training session. Would that be enough to close this before tomorrow's shipment?" Helen: [missing response] Diego: "Then the action record needs more than a training date. How will we know the change addressed the problem?" Which response belongs in the gap?

A. Not without establishing the root cause and appropriate action. Keep containment in place now; a convenient training proposal does not resolve the investigation or release the batch.

B. Good. A later entry cannot be passed off as a contemporaneous record. Any permitted correction must accurately show its actual timing, basis, and required authorization.

C. I will. Once there is an actual decision, we will communicate it through the proper route. Until then, do not imply that release authorization exists.

D. Specify an effectiveness check tied to the intended result. Completing a session and showing that a recurring documentation weakness has been addressed are different things.

Answers and explanations, page 152

Review 3 / Meaning and context

Use language from the lessons you have completed. Choose one answer per question.

> **Case file**
>
> Revisit "Pharmacovigilance, Safety Signals, Risk-Benefit, and Label Updates" and "CMC, CGMP, Quality Events, Manufacturing, and Supply". Choose precise terminology, then reconstruct the missing responses using both neighboring turns. The final two questions also revisit earlier conversations.

1. A colleague needs the term for this meaning: An undesirable response for which a causal relationship is at least reasonably suspected in context. Which term fits?

A. case reconciliation

B. reference safety information

C. event onset

D. adverse reaction

2. A colleague needs the term for this meaning: A record made when the relevant activity occurs. Which term fits?

A. contemporaneous record

B. chemistry, manufacturing, and controls

C. certificate of analysis

D. preventive action

3. Rebuild the exchange from "A favorable secondary result takes over the headline". Luis: "I do not want to hide the secondary observation either. How should it appear in the readout?" Anika: [missing response] Luis: "The clinicians will ask whether two points matter to patients. We have not included any explanation of the symptom scale." Which response belongs in the gap?

A. Report the nominal p-value with its limitation. The observation remains available for discussion, but it does not have the confirmatory status implied by proved efficacy.

B. Not under this testing hierarchy. Confirmatory secondary testing required primary success. Changing the headline cannot change the rule after we have seen the data.

C. We need to check the estimand. It defines the treatment effect we intended to estimate, including population, outcome, treatments, event handling, and summary measure.

D. Yes. The primary endpoint did not meet its prespecified threshold: point one nine is above point zero five. We should lead with that result.

Review 3 / Meaning and context / continued

4. Rebuild the exchange from "A gap in the batch record". Diego: "Planning will press for an estimate. I can promise a status update, but I cannot tell the customer that tomorrow's shipment is approved." Helen: [missing response] Diego: "I will report the missing entry, open investigation, and unreleased status. Let me know the next update point without converting it into a release promise." Which response belongs in the gap?

A. An out-of-specification result is not the same as a missing entry. Use the documentation description unless the investigation establishes another finding.

B. Correct. Authorized batch disposition remains pending. Communicate that timing is unconfirmed; urgency does not transfer the decision to production or planning.

C. Preserve the relevant evidence through the site process. We need traceability between the activity, supporting information, and any documented correction.

D. Quality is coordinating the deviation investigation under our site procedure. We will identify the evidence needed and communicate the actual review status.

Answers and explanations, page 169

A Read the situation

LESSON 07 / LABELING, MEDICAL AFFAIRS, PROMOTION REVIEW, AND COMPLIANCE BOUNDARIES

Identify an unsupported promotional implication and return a precisely scoped claim through the required review route.

> **Stronger wording changes the claim**
>
> A fictional approved source supplied for an internal exercise describes a lower mean symptom score versus placebo at week 12 in a defined adult study population. A promotional draft instead says immediate relief for every patient and places risk information in an unreadable footnote. No evidence supports immediate onset or benefit in every patient. The complete revised item requires medical, legal, and regulatory review before external use. Medical reviewer Sofia and brand colleague Adam must restore the supported scope without treating familiar wording or an approaching deadline as final approval.

People: Sofia, Medical reviewer; Adam, Brand colleague.

Read the record

Choose one answer per question using the facts above.

1. Which part of the draft is unsupported by the supplied source?

A. Immediate relief for every patient
B. A comparison with placebo
C. Assessment at week 12
D. A defined adult study population

2. What remains necessary before external use?

A. Reuse of approval from any earlier version
B. Review and approval of the complete revised item through the stated process
C. Only a check that the headline uses familiar words
D. Only the brand colleague's confirmation of the deadline

3. Why is the risk footnote a concern?

A. Its unreadable presentation undermines meaningful communication of risk.
B. Any footnote is automatically prohibited in every context.
C. Risks cease to matter when the benefit is a mean difference.
D. Risk information belongs only in internal records.

Answers and explanations, page 153

B Vocabulary in use

CORE TERMS

approved labeling
The regulator-authorized product information in the relevant jurisdiction.

check approved labeling

prescribing information
The professional product information describing authorized use and relevant details.

consult the prescribing information

promotional claim
A statement or implication used to communicate a product benefit or attribute in promotion.

substantiate a promotional claim

claim substantiation
Evidence supporting the precise meaning of a claim.

verify claim substantiation

fair balance
Appropriate presentation of benefit and risk information in the relevant promotional context.

assess fair balance

risk disclosure
Communication of relevant risks and limitations.

improve risk disclosure

material limitation
A qualification important to understanding a claim accurately.

retain material limitations

intended audience
The people a communication is designed to reach.

identify the intended audience

indication boundary
The limits of the use or population described by the relevant authorization.

preserve the indication boundary

onset claim
A claim about when a treatment effect begins.

substantiate an onset claim

comparative claim
A claim comparing products, treatments, or outcomes.

review comparative claims

absolute claim
An unqualified assertion suggesting no exceptions.

challenge an absolute claim

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

approved labeling APPROVED LABELING

check approved labeling

Keep the meaning precise: The regulator-authorized product information in the relevant jurisdiction.

risk disclosure RISK DISCLOSURE

improve risk disclosure

Keep the meaning precise: Communication of relevant risks and limitations.

B Precision and professional shorthand

EXTEND YOUR RANGE

net impression

The overall meaning conveyed by words, visuals, and presentation together.

evaluate the net impression

qualifier

Language restricting the scope of a statement.

keep the qualifier visible

medical affairs

The function supporting scientific and medical exchange within its responsibilities.

consult medical affairs

medical information

The function providing appropriate responses to product-related medical inquiries.

route a medical-information inquiry

unsolicited request

An inquiry not prompted or solicited through the relevant promotional activity.

document an unsolicited request

off-label use

Use outside the authorized labeling in the relevant jurisdiction.

distinguish an off-label question

scientific exchange

Communication of scientific information in an appropriate context.

define the scientific-exchange context

medical-legal-regulatory review

Cross-functional review of materials under the applicable company process.

complete MLR review

reference pack

The supporting evidence linked to a communication's claims.

update the reference pack

redline

A version showing proposed text changes.

review the redline

approval expiry

The end of a material's permitted use period under the applicable process.

check approval expiry

final-form approval

Authorization of the complete version in its intended presentation.

obtain final-form approval

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

absolute claim ABSOLUTE CLAIM

challenge an absolute claim

Keep the meaning precise: An unqualified assertion suggesting no exceptions.

unsolicited request UNSOLICITED REQUEST

document an unsolicited request

Keep the meaning precise: An inquiry not prompted or solicited through the relevant promotional activity.

C Phrases that move the work forward

Identify the overreach

Immediate relief is not supported by the week-12 assessment.

Keep the measure

The finding is a difference in mean symptom score, not universal relief.

Challenge an absolute

Every patient implies a guarantee the source does not establish.

Make risk readable

Risk information must not be reduced to unreadable text.

Route the inquiry

The off-label question should follow the appropriate medical-information process.

Check the version

Approval of an earlier item does not establish approval of this revision.

Preserve the caveat

This qualifier belongs next to the claim it limits.

Close the action

I will identify the unsupported implication and the supported alternative.

Preserve the population

The supplied evidence concerns the defined adult study population.

Name the comparator

Retain versus placebo where it is part of the supported statement.

Review the whole item

The headline, visual, and footnote must be assessed together.

Tie the comment to evidence

Please align this sentence with the cited source and its limitations.

Separate contexts

A scientific response is not automatic permission to reuse a claim in promotion.

Request final review

Return the complete final-form item through the required review route.

Avoid deadline pressure

The publication date does not replace approval.

State current status

The item remains pending review and is not ready for external use.

Precision in real exchanges

Purpose: Name the comparator
Useful wording: Retain versus placebo where it is part of the supported statement.

Purpose: Challenge an absolute
Useful wording: Every patient implies a guarantee the source does not establish.

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which revision best preserves the supplied scope?

A. Immediate improvement in the average patient

B. Proven symptom relief for all adults who receive treatment

C. Lower mean symptom score versus placebo at week 12 in the defined adult study population

D. Relief in adults, with timing omitted because the study lasted twelve weeks

2. What is the best review comment on the headline?

A. It is acceptable once a source number is added.

B. It is only a stylistic change because relief and score are related.

C. It is acceptable if the universal claim is placed in a visual instead of text.

D. It adds an onset claim and an individual guarantee absent from the supplied evidence.

3. A colleague needs the term for this meaning: Evidence supporting the precise meaning of a claim. Which term fits?

A. claim substantiation

B. medical affairs

C. reference pack

D. indication boundary

D Choose the response that fits / continued

4. Rebuild the exchange from "Stronger wording changes the claim". Adam: "Immediate is also new. The source identifies a week-twelve assessment, but it does not say when any effect began." Sofia: [missing response] Adam: "Could the headline stay short with the conditions underneath? Show me which limits must be prominent so the designer can give them enough space." Which response belongs in the gap?

A. Correct. That absolute claim is unsupported. We should not turn a group result into an individual guarantee, even if the wording sounds accessible and avoids technical language.

B. Route it through medical information under the applicable process. The existence of a scientific question does not expand the approved indication or authorize a new promotional use.

C. Then remove the onset claim unless appropriate supporting evidence and review establish it. A timepoint of measurement cannot silently become a claim about the first moment of benefit.

D. Update the reference pack and check the exact scope. A source citation is useful only when it supports the statement actually made, including population, comparator, timing, and outcome.

Answers and explanations, page 153

> **One useful distinction**
>
> A group mean difference does not mean every patient benefits. A week-12 assessment does not establish immediate onset. Qualifiers about population, comparator, timing, and measure carry substantive meaning; removing them can create a different claim even when individual words remain familiar.

E Conversation

Stronger wording changes the claim

Stronger wording changes the claim

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

net impression / reference pack / material limitation / fair balance / medical-legal-regulatory review / onset claim / final-form approval / medical information / promotional claim / absolute claim

01 **Adam:** The creative team shortened the headline to immediate relief for every patient. They say it carries the same idea as the approved source, but more clearly.

02 **Sofia:** It changes the [1] _____. The source describes a lower mean symptom score versus placebo at week twelve in a defined adult population. Those qualifications are part of the meaning.

03 **Adam:** I see that every patient is stronger than an average difference. The study result does not tell us that all participants responded in the same way.

04 **Sofia:** Correct. That [2] _____ is unsupported. We should not turn a group result into an individual guarantee, even if the wording sounds accessible and avoids technical language.

05 **Adam:** Immediate is also new. The source identifies a week-twelve assessment, but it does not say when any effect began.

06 **Sofia:** Then remove the [3] _____ unless appropriate supporting evidence and review establish it. A timepoint of measurement cannot silently become a claim about the first moment of benefit.

E Conversation / continued

07 **Adam:** Could the headline stay short with the conditions underneath? Show me which limits must be prominent so the designer can give them enough space.

08 **Sofia:** A [4] _____ must remain meaningfully available in context. Moving essential conditions into unreadable text does not reliably repair the stronger message in the headline.

09 **Adam:** The risk information has the same problem. It is technically on the page, but at the current size it is difficult to read.

10 **Sofia:** Review the [5] _____ of the complete item. We need to assess how benefit and risk are communicated, not just tick a box saying that a block of risk text exists.

Conversation continues on the next page.

Stronger wording changes the claim

CONVERSATION / CONTINUED

11 **Adam:** The image shows a person returning immediately to normal activity. Even with a revised sentence, that could keep suggesting a rapid and universal benefit.

12 **Sofia:** Exactly. Assess the [6] _____ created by words and visuals together. Correcting one sentence is insufficient if the overall presentation still makes the unsupported implication.

13 **Adam:** I will send the revised wording with the evidence references. That should make it easier to see which source supports each part of the claim.

14 **Sofia:** Update the [7] _____ and check the exact scope. A source citation is useful only when it supports the statement actually made, including population, comparator, timing, and outcome.

15 **Adam:** A field colleague has also asked whether this could be used for a different, unapproved condition. That question should not go into this promotional revision.

16 **Sofia:** Route it through [8] _____ under the applicable process. The existence of a scientific question does not expand the approved indication or authorize a new promotional use.

17 **Adam:** Once the edits are complete, can the brand team publish because the original item was approved? The deadline is close, and most of the layout is unchanged.

18 **Sofia:** The revised item still needs [9] _____. Earlier approval does not automatically cover changed claims or presentation. The deadline does not replace the required decision.

19 **Adam:** I will return the full revised item with a redline. Keep it pending until that version has been reviewed; the earlier approval will not be our release basis.

20 **Sofia:** Good. The [10] _____ should see the actual version intended for use. We can make the message readable without deleting the limits that make it accurate.

CHECK THE GAPS

1. promotional claim / **2.** absolute claim / **3.** onset claim / **4.** material limitation / **5.** fair balance / **6.** net impression / **7.** reference pack / **8.** medical information / **9.** final-form approval / **10.** medical-legal-regulatory review

Why these answers fit, page 155

F Check accuracy and scope

Stronger wording changes the claim

Return to the case

A fictional approved source supplied for an internal exercise describes a lower mean symptom score versus placebo at week 12 in a defined adult study population. A promotional draft instead says immediate relief for every patient and places risk information in an unreadable footnote. No evidence supports immediate onset or benefit in every patient. The complete revised item requires medical, legal, and regulatory review before external use. Medical reviewer Sofia and brand colleague Adam must restore the supported scope without treating familiar wording or an approaching deadline as final approval.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. Which action addresses the footnote problem?

A. Remove the risk information to reduce visual clutter.

B. Assume the source document's risks need not appear in a reviewed communication.

C. Revise the risk presentation so it is readable and assess the complete item's balance.

D. Keep the text unchanged because it is technically present.

2. A colleague asks about an unapproved use. What is the appropriate response here?

A. Add the use to the promotional slide while review is pending.

B. Treat an internal discussion as a new approved indication.

C. Answer with the strongest benefit sentence from the draft.

D. Route the inquiry through the relevant medical-information process without repurposing it as promotion.

F Check accuracy and scope / continued

Answers and explanations, page 156

Listen for the distinction

Switch roles for the visual and risk review. Preserve population, comparator, timing, and readable risk information in the version sent for review.

Explain what changed in the meaning

A review comment is more useful when it identifies the unsupported implication and the evidence boundary, rather than merely saying noncompliant. Offer wording tied to the supplied source, keep risk information appropriately visible, and identify the decision still required before the material can be used.

F Say it / transfer to a new case

Earlier approval, changed meaning

An approved internal example says benefit was measured at week 12 in adults. A revision says rapid benefit in all ages. The revised item has not completed review.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

pending / version / onset / population

01 **Reviewer:** Rapid adds an unsupported [1] _____ claim.

02 **Author:** All ages expands the stated [2] _____.

03 **Reviewer:** Earlier approval does not cover this changed [3] _____.

04 **Author:** External use remains [4] _____.

CHECK THE GAPS

1. onset / **2.** population / **3.** version / **4.** pending

Why these answers fit, page 156

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "Stronger wording changes the claim". Adam: "I will send the revised wording with the evidence references. That should make it easier to see which source supports each part of the claim." Sofia: [missing response] Adam: "A field colleague has also asked whether this could be used for a different, unapproved condition. That question should not go into this promotional revision." Which response belongs in the gap?

A. Route it through medical information under the applicable process. The existence of a scientific question does not expand the approved indication or authorize a new promotional use.

B. Correct. That absolute claim is unsupported. We should not turn a group result into an individual guarantee, even if the wording sounds accessible and avoids technical language.

C. Update the reference pack and check the exact scope. A source citation is useful only when it supports the statement actually made, including population, comparator, timing, and outcome.

D. Then remove the onset claim unless appropriate supporting evidence and review establish it. A timepoint of measurement cannot silently become a claim about the first moment of benefit.

Answers and explanations, page 157

A Read the situation

LESSON 08 / MARKET ACCESS, RWE, LAUNCH READINESS, BIOSIMILARS, AND LIFECYCLE MANAGEMENT

Explain differences between study designs and keep evidence, economic assumptions, and access decisions distinct.

> **Two sources, different questions**
>
> An access deck combines a 12-week randomized placebo-controlled trial in 400 adults with a six-month observational analysis of routine-care claims from 5,000 adults. The trial measures symptom score; the claims analysis measures hospitalization. Baseline differences and treatment-selection factors remain incompletely assessed in the claims analysis. The draft calls the combined material one randomized proof of reduced hospitalizations and guaranteed coverage. Access lead Mina and evidence specialist Jules must separate the evidence streams and qualify both causal and payer claims. No payer decision has been received.

People: Mina, Market-access lead; Jules, Evidence specialist.

Read the record

Choose one answer per question using the facts above.

1. Which source measures hospitalization?

A. The 12-week trial's stated symptom-score endpoint
B. Both sources using an identical protocol
C. A completed payer coverage decision
D. The six-month observational claims analysis

2. Why cannot the combined deck be described as randomized proof of fewer hospitalizations?

A. Only the symptom-score trial is randomized; the hospitalization analysis is observational.
B. Both sources are randomized, but hospitalization is measured only at week twelve.
C. The claims comparison is randomized because it includes more adults than the trial.
D. Both sources use the same outcome, but six months is too short for randomization.

3. What is the status of coverage?

A. Identical to the investigator's efficacy assessment
B. No payer decision has been received.
C. Guaranteed by the observational sample size
D. Approved by the trial's completion

Answers and explanations, page 158

B Vocabulary in use

CORE TERMS

market access
Work addressing whether and how patients can obtain a product within a health system.

develop the market-access plan

coverage decision
A decision defining whether and under what conditions a payer funds a service or product.

track the coverage decision

prior authorization
A requirement for advance approval before specified coverage applies.

clarify prior-authorization criteria

value dossier
A structured presentation of clinical, economic, and other evidence for an access audience.

tailor the value dossier

real-world evidence
Clinical evidence generated by analyzing real-world data using an appropriate study design.

evaluate real-world evidence

observational study
A study in which the investigator does not assign the treatment under study.

describe the observational study

payer
An organization or body funding or reimbursing health care.

address payer evidence needs

formulary
A payer or system's list of medicines and associated access conditions.

review formulary status

health technology assessment
Structured evaluation of a health technology's effects and implications in context.

prepare for health technology assessment

real-world data
Routinely collected information about health or health-care delivery.

assess real-world data quality

claims data
Records created for billing or reimbursement of health-care services.

assess claims-data limitations

confounding
Distortion of an exposure-outcome comparison by other relevant factors.

address confounding

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

market access MARKET ACCESS

develop the market-access plan

Keep the meaning precise: Work addressing whether and how patients can obtain a product within a health system.

health technology assessment HEALTH TECHNOLOGY ASSESSMENT

prepare for health technology assessment

Keep the meaning precise: Structured evaluation of a health technology's effects and implications in context.

B Precision and professional shorthand

EXTEND YOUR RANGE

confounding by indication

Differences in treatment choice related to prognosis or clinical need that affect comparison.

assess confounding by indication

residual confounding

Confounding remaining after the applied adjustment.

acknowledge residual confounding

comparative effectiveness

Comparison of health outcomes from different care options in relevant settings.

evaluate comparative effectiveness

cost-effectiveness

Assessment of costs relative to health outcomes for compared options.

examine cost-effectiveness assumptions

launch readiness

Preparedness of the relevant functions for a planned market introduction.

verify launch readiness

selection bias

Distortion related to how participants or observations enter the analysis.

examine selection bias

external validity

How well findings apply beyond the studied setting or population.

assess external validity

budget impact

The expected change in spending for a specified budget holder over a defined period.

model budget impact

uptake assumption

An estimate of how many eligible people will use a product over time.

test the uptake assumption

biosimilar

A biological product shown to be highly similar to a reference product without clinically meaningful safety or effectiveness differences.

evaluate a biosimilar evidence package

B Precision and professional shorthand / continued

reference product

The authorized biological product used as the comparator for biosimilar development.

identify the reference product

lifecycle management

Planned development and stewardship of a product across its commercial and regulatory life.

align lifecycle management

Say it in context

Stress the highlighted expression. Keep the surrounding words connected; pause at the end of the phrase.

confounding CONFOUNDING

address confounding

Keep the meaning precise: Distortion of an exposure-outcome comparison by other relevant factors.

comparative effectiveness COMPARATIVE EFFECTIVENESS

evaluate comparative effectiveness

Keep the meaning precise: Comparison of health outcomes from different care options in relevant settings.

C Phrases that move the work forward

Separate the sources

The trial and claims analysis answer different questions.

State the design

The claims comparison is observational, not randomized.

Keep the outcome clear

The trial measures symptom score; the claims analysis measures hospitalization.

Preserve follow-up

The studies use different observation periods.

Qualify causality

Treatment-selection differences may affect the observed association.

Ask about adjustment

Which confounders were measured, and what uncertainty remains?

Avoid sample-size overreach

A larger dataset does not remove systematic bias by itself.

Retain useful context

Routine-care evidence may add context without replacing the trial's design.

Separate economic questions

Budget impact and cost-effectiveness are not interchangeable.

Expose model inputs

Uptake and eligible-population assumptions should remain visible.

Limit the coverage claim

No payer decision has been received.

Check the audience

Which evidence question is this payer asking us to address?

Distinguish similarity

Biosimilarity is a specific evidence and regulatory concept, not a loose claim of being similar.

Avoid automatic substitution claims

Substitution implications require the relevant product status and applicable rules.

Track launch dependencies

Clinical evidence does not complete every access and supply dependency.

Close the deck

Label each evidence stream, its limitations, and the decision it can inform.

Precision in real exchanges

Purpose: Preserve follow-up
Useful wording: The studies use different observation periods.

Purpose: Qualify causality
Useful wording: Treatment-selection differences may affect the observed association.

D Choose the response that fits

Choose one response for each situation. Preserve the facts, the decision boundary, and the purpose of the exchange.

1. Which summary is best supported?

A. The claims analysis is randomized because the trial is randomized.

B. The larger sample makes treatment-selection assessment unnecessary.

C. The trial assesses symptoms; the observational claims study examines hospitalizations with unresolved confounding questions.

D. Both studies independently prove the same hospitalization effect.

2. What should be checked before interpreting the claims difference causally?

A. Whether the trial used an unrelated symptom scale

B. Treatment selection, measured confounders, remaining bias, and the analysis design

C. Only whether the sample exceeds the trial's sample

D. Only whether the result supports the launch message

3. A colleague needs the term for this meaning: A payer or system's list of medicines and associated access conditions. Which term fits?

A. launch readiness

B. formulary

C. residual confounding

D. real-world evidence

D Choose the response that fits / continued

4. Rebuild the exchange from "Two sources, different questions". Mina: "Patients receiving one treatment could have had different clinical needs before follow-up began. Those needs might also affect their chances of hospitalization." Jules: [missing response] Mina: "The slide says adjustment removed all bias. Which characteristics were included, and what important differences could still remain unmeasured?" Which response belongs in the gap?

A. Then label the uptake assumption prominently. The model should not transform an estimate into a commitment simply because it appears in a launch forecast.

B. Yes. Discuss external validity without assuming that all routine-care records represent every intended patient. Data relevance and quality need examination alongside the study design.

C. That raises confounding by indication. We should ask which relevant factors were measured and addressed, while acknowledging that the current assessment is incomplete.

D. A coverage decision is distinct from the clinical findings and economic models. State the current status and any unresolved evidence requests instead of promising an outcome.

Answers and explanations, page 158

One useful distinction

Real-world describes data context, not guaranteed quality or one study design. Randomized studies can use routine data. This claims comparison, however, is observational, with treatment-selection issues still incompletely assessed.

E Conversation

Two sources, different questions

Two sources, different questions

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

launch-readiness / budget impact / external validity / coverage decision / evidence streams / confounding by indication / observational design / uptake assumption / Real-world data / Residual confounding

01 **Mina:** The access deck combines the trial and claims results under one headline: randomized proof of fewer hospitalizations and guaranteed coverage. That seems to merge several different claims.

02 **Jules:** Start by separating the [1] _____. The trial assessed symptom score over twelve weeks. The claims analysis assessed hospitalization over six months in a different, routine-care population.

03 **Mina:** The trial randomized four hundred adults, while the claims dataset contains five thousand. The larger number is attracting more attention in the discussion.

04 **Jules:** Size does not change the [2] _____. Treatment was not assigned by the claims study. We need to examine how people entered each treatment group and which differences might affect their outcomes.

05 **Mina:** Patients receiving one treatment could have had different clinical needs before follow-up began. Those needs might also affect their chances of hospitalization.

06 **Jules:** That raises [3] _____. We should ask which relevant factors were measured and addressed, while acknowledging that the current assessment is incomplete.

E Conversation / continued

07 **Mina:** The slide says adjustment removed all bias. Which characteristics were included, and what important differences could still remain unmeasured?

08 **Jules:** It cannot establish that from the information we have. [4] _____ may remain, particularly for factors not adequately captured. Explain the approach and limitations rather than describe adjustment as a universal cure.

09 **Mina:** We should still show what the routine-care analysis adds. It may help an access audience understand patterns outside the trial's narrower setting.

10 **Jules:** Yes. Discuss [5] _____ without assuming that all routine-care records represent every intended patient. Data relevance and quality need examination alongside the study design.

Conversation continues on the next page.

Two sources, different questions

CONVERSATION / CONTINUED

11 **Mina:** The phrase real-world evidence is being used as though it automatically means observational. That is too broad, even though this particular study is observational.

12 **Jules:** Correct. [6] _____ describes the data context. Different designs can use those data. In this case, we should state the actual observational design rather than let the label do the explaining.

13 **Mina:** The financial slide also combines budget impact with cost-effectiveness. One estimates payer spending, while the other relates costs to health outcomes.

14 **Jules:** Keep the [7] _____ model separate and state its horizon, eligible population, and uptake assumptions. A useful economic result depends on a clearly defined question and inputs.

15 **Mina:** The uptake assumption is twenty percent of an estimated eligible population. It is a planning input, not a confirmed number of people who will receive coverage.

16 **Jules:** Then label the [8] _____ prominently. The model should not transform an estimate into a commitment simply because it appears in a launch forecast.

17 **Mina:** We have no payer decision yet. The headline's guaranteed coverage wording could create expectations that neither the evidence team nor the access team can support.

18 **Jules:** A [9] _____ is distinct from the clinical findings and economic models. State the current status and any unresolved evidence requests instead of promising an outcome.

19 **Mina:** I will separate the evidence streams and remove guaranteed coverage. Please check that each headline still matches its own study design, population, and period.

20 **Jules:** That supports a realistic [10] _____ discussion. We can explain what is ready, what remains under review, and who owns each dependency without presenting a collection of evidence as a completed launch decision.

CHECK THE GAPS

1. evidence streams / **2.** observational design / **3.** confounding by indication / **4.** Residual confounding / **5.** external validity / **6.** Real-world data / **7.** budget impact / **8.** uptake assumption / **9.** coverage decision / **10.** launch-readiness

Why these answers fit, page 159

F Check accuracy and scope

Two sources, different questions

Return to the case

An access deck combines a 12-week randomized placebo-controlled trial in 400 adults with a six-month observational analysis of routine-care claims from 5,000 adults. The trial measures symptom score; the claims analysis measures hospitalization. Baseline differences and treatment-selection factors remain incompletely assessed in the claims analysis. The draft calls the combined material one randomized proof of reduced hospitalizations and guaranteed coverage. Access lead Mina and evidence specialist Jules must separate the evidence streams and qualify both causal and payer claims. No payer decision has been received.

Choose one answer per question. Use the case facts and keep any uncertainty or limits intact.

1. Which statement about economic evidence is accurate?

A. An uptake assumption is the same as an executed coverage decision.

B. Budget impact estimates spending for a defined budget holder; cost-effectiveness relates costs to health outcomes.

C. Budget impact proves that a product is clinically superior.

D. Cost-effectiveness guarantees that every payer will provide coverage.

2. Which launch update preserves the actual status?

A. The access evidence package is being revised; payer decisions remain pending.

B. Coverage is guaranteed because the evidence sources were combined.

C. Every launch dependency is complete once a trial ends.

D. The claims dataset authorizes unrestricted substitution of all biological products.

Answers and explanations, page 161

Listen for the distinction

Switch roles for the causal and payer claims. Preserve unresolved treatment-selection concerns and distinguish evidence preparation from a coverage decision.

Different evidence can complement without merging

Explain what each source answers before placing results side by side. A larger routine-care dataset may broaden context without becoming randomized evidence. Likewise, clinical results, economic models, and coverage decisions address different questions. A useful access conversation makes these boundaries visible without dismissing a source simply because it has limitations.

F Say it / transfer to a new case

An assumption is not a covered population

A fictional spending model assumes 10,000 eligible adults, 20% uptake, and $300 incremental annual spending per user. No offsets or other costs are included. No payer decision is available.

Complete the four gaps. With a partner, read alternate roles, then switch. Studying alone, read both roles and pause before each response.

WORD AND PHRASE BANK

assumption / $600,000 / 2,000 / undecided

01 **Analyst:** The model assumes [1] _____ users.

02 **Reviewer:** The modeled annual spending increase is [2] _____.

03 **Analyst:** The 20% figure is an uptake [3] _____.

04 **Reviewer:** Coverage remains [4] _____.

CHECK THE GAPS

1. 2,000 / **2.** $600,000 / **3.** assumption / **4.** undecided

Why these answers fit, page 161

Rebuild the exchange

Choose the missing response. Read the completed three-turn exchange aloud, changing roles on the second reading.

1. Rebuild the exchange from "Two sources, different questions". Mina: "The financial slide also combines budget impact with cost-effectiveness. One estimates payer spending, while the other relates costs to health outcomes." Jules: [missing response] Mina: "The uptake assumption is twenty percent of an estimated eligible population. It is a planning input, not a confirmed number of people who will receive coverage." Which response belongs in the gap?

A. Keep the budget impact model separate and state its horizon, eligible population, and uptake assumptions. A useful economic result depends on a clearly defined question and inputs.

B. Yes. Discuss external validity without assuming that all routine-care records represent every intended patient. Data relevance and quality need examination alongside the study design.

C. Start by separating the evidence streams. The trial assessed symptom score over twelve weeks. The claims analysis assessed hospitalization over six months in a different, routine-care population.

D. A coverage decision is distinct from the clinical findings and economic models. State the current status and any unresolved evidence requests instead of promising an outcome.

Answers and explanations, page 162

Review 4 / Meaning and context

Use language from the lessons you have completed. Choose one answer per question.

> **Case file**
>
> Revisit "Labeling, Medical Affairs, Promotion Review, and Compliance Boundaries" and "Market Access, RWE, Launch Readiness, Biosimilars, and Lifecycle Management". Choose precise terminology, then reconstruct the missing responses using both neighboring turns. The final two questions also revisit earlier conversations.

1. A colleague needs the term for this meaning: A statement or implication used to communicate a product benefit or attribute in promotion. Which term fits?

A. medical-legal-regulatory review

B. promotional claim

C. intended audience

D. qualifier

2. A colleague needs the term for this meaning: Routinely collected information about health or health-care delivery. Which term fits?

A. confounding by indication

B. market access

C. cost-effectiveness

D. real-world data

3. Rebuild the exchange from "A gap in the batch record". Diego: "Supply has called this an OOS. That sounds like a failed test, but I have not been shown a result outside a limit." Helen: [missing response] Diego: "Who is coordinating the next steps? Production can help, but I need a clear owner before I reply to planning." Which response belongs in the gap?

A. Quality is coordinating the deviation investigation under our site procedure. We will identify the evidence needed and communicate the actual review status.

B. An out-of-specification result is not the same as a missing entry. Use the documentation description unless the investigation establishes another finding.

C. Correct. Authorized batch disposition remains pending. Communicate that timing is unconfirmed; urgency does not transfer the decision to production or planning.

D. Preserve the relevant evidence through the site process. We need traceability between the activity, supporting information, and any documented correction.

Review 4 / Meaning and context / continued

4. Rebuild the exchange from "Two sources, different questions". Mina: "We have no payer decision yet. The headline's guaranteed coverage wording could create expectations that neither the evidence team nor the access team can support." Jules: [missing response] Mina: "I will separate the evidence streams and remove guaranteed coverage. Please check that each headline still matches its own study design, population, and period." Which response belongs in the gap?

A. Start by separating the evidence streams. The trial assessed symptom score over twelve weeks. The claims analysis assessed hospitalization over six months in a different, routine-care population.

B. A coverage decision is distinct from the clinical findings and economic models. State the current status and any unresolved evidence requests instead of promising an outcome.

C. Keep the budget impact model separate and state its horizon, eligible population, and uptake assumptions. A useful economic result depends on a clearly defined question and inputs.

D. Yes. Discuss external validity without assuming that all routine-care records represent every intended patient. Data relevance and quality need examination alongside the study design.

Answers and explanations, page 171

Language for the next case

The shipment is cold again, but its history still matters

quarantine status

Quarantine status prevents use or distribution pending the authorized quality decision; current temperature alone does not settle suitability.

Elena: Keep it in the appropriate quarantine status under our procedure. What does the original logger record show?

raw data

Raw data are the original recorded measurements and associated information needed to assess the excursion reliably.

Elena: Preserve the raw data and the logger details. A screenshot alone may omit information needed for the assessment.

affected scope

Affected scope identifies which product and quantities may be involved, rather than assuming every shipment or no shipment was exposed.

Elena: Good. That establishes the affected scope for review. Do not infer that one logger necessarily represents every container.

temperature excursion

A temperature excursion is exposure outside the specified range; visual appearance does not determine its effect on the product.

Elena: Record that observation, but appearance does not establish product quality after a temperature excursion.

stability data

Stability data support assessment of this product under relevant conditions; another product's decision cannot be transferred automatically.

Elena: No. The assessment needs the relevant stability data, history, and approved process for this product.

disposition

Disposition is the authorized decision about how the product may be handled after the evidence has been reviewed.

Elena: I will assign the review and document the disposition once the evidence supports it.

Read the phrases aloud. Then complete the conversation on the following pages using the word bank.

Conversation

The shipment is cold again, but its history still matters

A logger for a fictional refrigerated medicine shipment shows a period above its specified transport range. The shipment is now within range. Duration, peak temperature, affected lots, and product-specific stability support need qualified review. Distribution has not been authorized.

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

quarantine status / temperature excursion / affected scope / stability data / disposition / raw data

01 **Ravi:** The logger alarmed during transit, but the delivery is back within range. Can the warehouse receive it into available stock?

02 **Elena:** Keep it in the appropriate [1] _____ under our procedure. What does the original logger record show?

03 **Ravi:** A period above the upper limit. I have the graph but have not checked the time zone or downloaded the complete file.

04 **Elena:** Preserve the [2] _____ and the logger details. A screenshot alone may omit information needed for the assessment.

05 **Ravi:** The carrier says the alarm lasted only briefly. Should I use their estimate in the incident summary?

06 **Elena:** Attribute it as the carrier's statement, not the verified duration. We need the recorded interval, peak, and time basis checked.

07 **Ravi:** I will also confirm the lot numbers and which shipping containers the logger accompanied.

08 **Elena:** Good. That establishes the [3] _____ for review. Do not infer that one logger necessarily represents every container.

09 **Ravi:** The products look normal. There is no visible damage to the cartons.

10 **Elena:** Record that observation, but appearance does not establish product quality after a [4] _____.

Conversation continues on the next page.

The shipment is cold again, but its history still matters

CONVERSATION / CONTINUED

11 **Ravi:** Could we use the allowance applied to another medicine last month? That shipment was eventually released.

12 **Elena:** No. The assessment needs the relevant [5] _____, history, and approved process for this product.

13 **Ravi:** The receiving site is asking whether the delivery can be used today. I need a sentence that is clear without sounding evasive.

14 **Elena:** Say: "The shipment is held for quality assessment; use is not authorized. We will provide the next status update at the agreed time."

15 **Ravi:** Should I arrange replacement stock now, or wait for the assessment?

16 **Elena:** Ask supply to assess the options in parallel. Do not make the quality decision depend on whether a replacement is convenient.

17 **Ravi:** I will send the complete record and container mapping, and keep the current storage conditions controlled under the procedure.

18 **Elena:** I will assign the review and document the [6] _____ once the evidence supports it.

19 **Ravi:** The summary will distinguish the verified readings from the carrier's account. I will not describe the shipment as released.

20 **Elena:** Exactly. Returning to the specified range ends the current temperature event; it does not erase the exposure history.

CHECK THE GAPS

1. quarantine status / **2.** raw data / **3.** affected scope / **4.** temperature excursion / **5.** stability data / **6.** disposition

Why these answers fit, page 174

A missing logger file is treated as a normal result

A logger file cannot yet be opened. Cartons are intact and the receiving temperature is within range. Transport history has not been verified.

Complete the exchange using each phrase once. Read both roles, then switch roles and repeat with the bank covered.

WORD AND PHRASE BANK

unverified / record / result / review

01 **Coordinator:** The receiving reading is within range, but the journey history is [1] _____.

02 **Quality:** Preserve the device and pursue recovery of the original [2] _____.

03 **Coordinator:** I will not translate missing data into a passing [3] _____.

04 **Quality:** Keep the product in its required controlled status pending authorized [4] _____.

CHECK THE GAPS

1. unverified / **2.** record / **3.** result / **4.** review

Why these answers fit, page 174

> **Keep the distinction audible**
>
> Report a temperature excursion without making an unsupported product-quality decision.

Language for the next case

A passing retest does not make the first result disappear

out-of-specification result

An out-of-specification result falls outside the established acceptance criterion and requires documented investigation, not silent replacement.

Alex: We must retain both. The original out-of-specification result still requires a supported investigation conclusion.

assignable cause

An assignable cause is a cause established by evidence; an analyst's suspicion alone does not establish one.

Alex: Then record it as a hypothesis. We cannot designate an assignable cause because it would explain away an inconvenient result.

audit trail

The audit trail records relevant actions and changes so the investigation can reconstruct what actually occurred.

Alex: And preserve the audit trail. The original files and their history need to remain available.

retesting

Retesting repeats analysis under a justified investigation plan; it must not become repeated testing until a favorable result appears.

Alex: Any further retesting needs a documented rationale and authorized plan, not an open-ended search for a pass.

Resampling

Resampling takes another sample; it is different from retesting and needs its own justified basis in the investigation.

Alex: Yes. Resampling is a separate action with its own justification. Changing the sample can change what evidence we have.

investigation conclusion

The investigation conclusion must explain the supported findings and treatment of all relevant results; it remains pending here.

Alex: Include the investigation conclusion as pending, with the owner and next review step.

Read the phrases aloud. Then complete the conversation on the following pages using the word bank.

Conversation

A passing retest does not make the first result disappear

A fictional assay result falls outside its approved acceptance criterion. A later test passes. No assignable laboratory error has been established, and the investigation is still open. The laboratory manager and quality reviewer must decide what to document and investigate, not release the batch.

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

assignable cause / Resampling / retesting / audit trail / investigation conclusion / out-of-specification result

01 **Noemi:** The later assay passes. Production wants the summary updated to show the passing result instead of the original failure.

02 **Alex:** We must retain both. The original [1] _____ still requires a supported investigation conclusion.

03 **Noemi:** The analyst suspects a preparation error, but has not found evidence of one.

04 **Alex:** Then record it as a hypothesis. We cannot designate an [2] _____ because it would explain away an inconvenient result.

05 **Noemi:** I will review the preparation records, instrument information, calculations, and the original sample sequence with the analyst.

06 **Alex:** And preserve the [3] _____. The original files and their history need to remain available.

07 **Noemi:** Would averaging the passing and failing values give us a representative number for the batch?

08 **Alex:** Not as a way to cancel the failing result. Follow the approved method and investigation procedure; do not invent an averaging rule after seeing the outcomes.

09 **Noemi:** The next question is whether more testing would clarify the suspected preparation issue.

10 **Alex:** Any further [4] _____ needs a documented rationale and authorized plan, not an open-ended search for a pass.

Conversation continues on the next page.

A passing retest does not make the first result disappear

CONVERSATION / CONTINUED

11 **Noemi:** We should also establish whether the retained sample can answer the question or whether someone is proposing a new sample.

12 **Alex:** Yes. [5] _____ is a separate action with its own justification. Changing the sample can change what evidence we have.

13 **Noemi:** If the laboratory review finds no supported error, the investigation cannot simply stop with "retest passed."

14 **Alex:** Correct. The review must address the potential manufacturing issue and the full body of evidence through the applicable process.

15 **Noemi:** I will notify production of the open investigation and request the relevant batch records without declaring a manufacturing defect.

16 **Alex:** That keeps the scope open appropriately. We need evidence, not an early conclusion about which department is responsible.

17 **Noemi:** For today's report, I will show both results, the hypothesis, completed checks, and the questions still unanswered.

18 **Alex:** Include the [6] _____ as pending, with the owner and next review step.

19 **Noemi:** And no batch-release statement, even if the shipment date is close.

20 **Alex:** Exactly. A passing later test is evidence to evaluate, not permission to delete the first result or bypass disposition.

CHECK THE GAPS

1. out-of-specification result / **2.** assignable cause / **3.** audit trail / **4.** retesting / **5.** Resampling / **6.** investigation conclusion

Why these answers fit, page 176

Testing continues until the desired number appears

After a failing result, a colleague proposes repeated testing without a defined plan and retaining only the first passing result.

Complete the exchange using each phrase once. Read both roles, then switch roles and repeat with the bank covered.

WORD AND PHRASE BANK

one / cause / plan / decision

01 **Reviewer:** We need an authorized investigation [1] _____ before further testing.

02 **Analyst:** I will preserve every original result, not just the favorable [2] _____.

03 **Reviewer:** A possible preparation error is still a hypothesis, not an established [3] _____.

04 **Analyst:** The batch status remains subject to the authorized quality [4] _____.

CHECK THE GAPS

1. plan / **2.** one / **3.** cause / **4.** decision

Why these answers fit, page 176

Keep the distinction audible

Discuss an out-of-specification result while protecting the complete investigation record.

Language for the next case

The packaging artwork is current, but the line has an older version

artwork revision

Artwork revision identifies the specific approved labeling version; similar appearance does not establish that two revisions are equivalent.

Hassan: Keep the start on hold and secure the suspect materials under the procedure. First confirm the artwork revision against the controlled records.

material traceability

Material traceability links the issued packaging components to their identity, quantities, and handling records.

Hassan: That is an important boundary. Preserve the material traceability so we can identify exactly what may be affected.

line clearance

Line clearance verifies that the line is ready for the intended operation and free of relevant materials from an incompatible or previous operation.

Hassan: Not by itself. The correct materials, required review, and line clearance must be confirmed before authorized restart.

segregation

Segregation keeps materials with different or restricted status from being mixed or used inadvertently.

Hassan: Use the required segregation and status controls. Record their location rather than leave a loose note beside an unmarked stack.

change control

Change control evaluates, approves, and documents a proposed change rather than allowing an undocumented substitution.

Hassan: Correct. Route it through change control and retain the reason, approvals, and applicable version history.

restart authorization

Restart authorization is the formal permission to resume after the required conditions are met, not merely the arrival of replacement cartons.

Hassan: And document the restart authorization so the final decision is traceable to the actual completed checks.

Read the phrases aloud. Then complete the conversation on the following pages using the word bank.

Conversation

The packaging artwork is current, but the line has an older version

Before packaging starts, an operator finds that the issued carton artwork revision differs from the revision required by the approved batch instruction. No units have been packaged in this run. The discrepancy must be controlled through the site quality process; no one may silently substitute files.

Complete the numbered gaps. Use each bank entry once; some entries contain several words.

WORD AND PHRASE BANK

change control / material traceability / segregation / artwork revision / line clearance / restart authorization

01 **Dina:** The batch instruction calls for revision C, but the issued cartons show revision B. We caught it before starting the run.

02 **Hassan:** Keep the start on hold and secure the suspect materials under the procedure. First confirm the [1] _____ against the controlled records.

03 **Dina:** The visible difference is small. Someone suggested crossing out the instruction's revision letter and using what is already at the line.

04 **Hassan:** Do not change the instruction informally. We need to establish whether the issue is the instruction, material issuance, or something else.

05 **Dina:** I will record both identifiers and the quantities issued. No units from this run have been packaged.

06 **Hassan:** That is an important boundary. Preserve the [2] _____ so we can identify exactly what may be affected.

07 **Dina:** Should we check whether the same carton revision was issued to another line?

08 **Hassan:** Yes, through the investigation. Expand the scope using records rather than assume this is the only occurrence or that all other runs are affected.

09 **Dina:** The planner can deliver revision C this afternoon. Is that enough to restart?

10 **Hassan:** Not by itself. The correct materials, required review, and [3] _____ must be confirmed before authorized restart.

Conversation continues on the next page.

The packaging artwork is current, but the line has an older version

CONVERSATION / CONTINUED

11 **Dina:** I want to make sure the older cartons cannot be picked up again while we wait.

12 **Hassan:** Use the required [4] _____ and status controls. Record their location rather than leave a loose note beside an unmarked stack.

13 **Dina:** Could the supplier tell us whether the revisions are interchangeable?

14 **Hassan:** They can provide relevant information. Equivalence and permitted use still require our authorized review; the supplier's reassurance is not our release decision.

15 **Dina:** If the instruction itself needs correcting, we should not overwrite the original file.

16 **Hassan:** Correct. Route it through [5] _____ and retain the reason, approvals, and applicable version history.

17 **Dina:** For the shift handover, I will state that the run has not started and the material mismatch is under review.

18 **Hassan:** Add the hold location, open actions, and responsible contacts. The incoming shift should not interpret a replacement delivery as permission to begin.

19 **Dina:** Once the issue is resolved, I will reconcile the issued, returned, and used quantities through the batch record.

20 **Hassan:** And document the [6] _____ so the final decision is traceable to the actual completed checks.

CHECK THE GAPS

1. artwork revision / **2.** material traceability / **3.** line clearance / **4.** segregation / **5.** change control / **6.** restart authorization

Why these answers fit, page 178

Correct cartons arrive during a shift change

Replacement cartons arrive while the line remains on hold. Required review and clearance checks are incomplete.

Complete the exchange using each phrase once. Read both roles, then switch roles and repeat with the bank covered.

WORD AND PHRASE BANK

owners / confirmed / hold / identified

01 **Supervisor:** Delivery of the cartons does not cancel the current [1] _____.

02 **Quality:** The handover must identify the remaining checks and their [2] _____.

03 **Supervisor:** I will keep the two material versions physically controlled and clearly [3] _____.

04 **Quality:** Resume only after the required conditions and authorization are [4] _____.

CHECK THE GAPS

1. hold / **2.** owners / **3.** identified / **4.** confirmed

Why these answers fit, page 178

> **Keep the distinction audible**
>
> Coordinate a labeling discrepancy with precise version, scope, and release language.

01 / Answers and explanations

A / Read the situation

A1. Where was the observed result obtained?

A. In a cell-based assay across three experimental runs

The supplied evidence is laboratory evidence, with no patient study or approved label in the case.

Return to the exercise, page 4

A2. What does the target product profile represent here?

A. Desired future characteristics to guide development

The proposed profile sets aspirations; it does not establish that those characteristics have been demonstrated.

Return to the exercise, page 4

A3. Which claim exceeds the evidence?

D. PX-17 improves patient mobility.

No patients were studied, so the laboratory result cannot establish a patient mobility benefit.

Return to the exercise, page 4

D / Response choices

D1. Which replacement is best supported?

C. PX-17 changed the marker in the tested cell assay; patient benefit remains unestablished.

The replacement preserves the observed laboratory finding while explicitly limiting the clinical inference.

Return to the exercise, page 9

D2. What does testing one concentration fail to characterize?

A. How the effect changes across different concentrations

A response across concentrations requires additional concentration conditions; the supplied experiment used only one.

Return to the exercise, page 9

01 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: The biological process through which an intervention produces an effect. Which term fits?

C. mechanism of action

A Not here. assay limitation: A restriction affecting what a test can establish. The usual combination is "disclose assay limitations".

B Not here. pharmacokinetics: Study of the time course of drug exposure through absorption, distribution, metabolism, and elimination. The usual combination is "characterize pharmacokinetics".

C Correct. mechanism of action: The biological process through which an intervention produces an effect. The usual combination is "investigate the mechanism of action".

D Not here. surrogate endpoint: A substitute outcome used to predict a clinical outcome, with context-dependent support. The usual combination is "justify a surrogate endpoint".

Return to the exercise, page 9

D4. Rebuild the exchange from "A laboratory signal becomes a patient-benefit claim". Owen: "The commercial planning page also lists improved mobility. That is in the target profile, so perhaps someone copied it into the result summary." Leila: [missing response] Owen: "Can we call this proof of concept? The phrase is familiar, but different colleagues may hear very different claims when they read it." Which response belongs in the gap?

A. The target product profile is aspirational at this stage. We should label the desired outcome separately from the finding so readers do not confuse the development goal with achieved performance.

A Correct. This is Leila's response in turn 6. It connects Owen's preceding point with Owen's follow-up. The target product profile sets desired future characteristics rather than recording benefits already demonstrated.

B Not here. This line belongs to turn 16, after Owen says: "Let us make the next decision concrete. What missing information matters most before the program can support a stronger development proposal?" It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 12, after Owen says: "What can we say about the three runs? I do not want to discard useful repetition merely because the result is still early." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 20, after Owen says: "I will replace the mobility headline with the assay finding. Send me the limits and proposed next question so the committee sees what this result supports." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 9

E / Conversation

1. cell-based assay

Turn 2. A cell-based assay is laboratory evidence, not a study establishing outcomes in patients.

01 / Answers and explanations / continued

2. biomarker

Turn 4. The biomarker is the measured biological characteristic; its change does not itself establish improved patient function.

3. target product profile

Turn 6. The target product profile sets desired future characteristics rather than recording benefits already demonstrated.

4. clinical benefit

Turn 8. Clinical benefit concerns meaningful patient outcomes; this experiment did not measure any patient outcome.

5. dose-response relationship

Turn 10. Dose-response describes effects across doses; the single tested concentration cannot establish that relationship.

6. repeatability

Turn 12. Repeatability concerns agreement under the same specified measurement conditions; it does not establish performance across laboratories or in patients.

7. tolerability

Turn 14. Tolerability concerns enduring unwanted effects in a defined context; this laboratory experiment did not establish human tolerability.

8. evidence gap

Turn 16. The evidence gap is missing information connecting the assay finding to a more meaningful development conclusion.

9. go/no-go criterion

Turn 18. A go/no-go criterion identifies evidence needed for a development decision rather than treating all additional data as progress.

10. development milestone

Turn 20. A development milestone is a defined progress or review point, not automatic authorization for human studies.

Return to the conversation, page 11

01 / Answers and explanations / continued

F / Accuracy and scope

F1. Which additional question most directly tests the proposed marker-to-mobility claim?

A. What evidence links changes in this marker to patient mobility in the intended context?

The missing bridge is evidence connecting the measured marker with the intended patient outcome. Repetition, an added population label, and a planning aspiration do not establish that connection.

Return to the exercise, page 15

F2. What can the portfolio discussion decide on the supplied authority?

C. A proposed next investigation for appropriate review, not automatic clinical-study authorization

The brief excludes clinical-study authorization and supplies no basis for treatment or marketing claims.

Return to the exercise, page 15

F / Guided speaking

1. goal

Scientist: Once-daily use is currently a goal.

Turn 1. The planning profile states an intended characteristic, while no human dosing evidence is supplied.

2. target engagement

Reviewer: The measured laboratory finding concerns target engagement.

Turn 2. The test measured interaction with the intended target, not patient symptoms.

3. unestablished

Scientist: Human dosing remains unestablished.

Turn 3. The brief explicitly says there are no human dosing data.

4. improvement

Reviewer: We cannot claim symptom improvement.

Turn 4. Patient symptoms were not measured, so an improvement claim would exceed the evidence.

Return to the conversation, page 16

01 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "A laboratory signal becomes a patient-benefit claim". Owen: "Someone has added well tolerated to the summary. There were no human participants, so I cannot see what that statement is based on." Leila: [missing response] Owen: "Let us make the next decision concrete. What missing information matters most before the program can support a stronger development proposal?" Which response belongs in the gap?

C. Human tolerability is unestablished. We can report the actual observations, but we should not translate the absence of a particular laboratory finding into a general human safety claim.

A Not here. This line belongs to turn 18, after Owen says: "The committee will ask what would count as enough progress to continue. A broad instruction to produce more data will not help them compare programs." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 4, after Owen says: "Then the headline has moved beyond the evidence. The team wanted a clear link to the disease, but the wording makes that link sound demonstrated." It addresses that point, not the missing turn in the supplied exchange.

C Correct. This is Leila's response in turn 14. It connects Owen's preceding point with Owen's follow-up. Tolerability concerns enduring unwanted effects in a defined context; this laboratory experiment did not establish human tolerability.

D Not here. This line belongs to turn 12, after Owen says: "What can we say about the three runs? I do not want to discard useful repetition merely because the result is still early." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 16

02 / Answers and explanations

A / Read the situation

A1. What is wrong with the current question?

C. It asks broadly about the entire plan instead of the specific unresolved proposal.

The real issue concerns population and endpoint rationale, while the draft requests undifferentiated agreement with the entire plan.

Return to the exercise, page 17

A2. What is the status of two supporting reports?

B. Unfinished

The brief explicitly says two supporting reports remain unfinished.

Return to the exercise, page 17

A3. Which statement is accurate now?

B. No agency response has been received.

Preparation is ongoing, and the case supplies no response or study authorization.

Return to the exercise, page 17

D / Response choices

D1. Which agency question is focused on the immediate unresolved proposal?

C. Does the proposed population and endpoint rationale in section three address the intended study objective?

The focused question identifies the population, endpoint rationale, supporting section, and study objective. It does not seek blanket endorsement, alter document status, or imply a future marketing decision.

Return to the exercise, page 21

D2. Which package statement is accurate?

D. Two reports remain unfinished and their status is identified.

The statement preserves the actual document status and allows reviewers to understand the information gap.

Return to the exercise, page 21

02 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: A US application seeking approval to market a new drug. Which term fits?

B. new drug application

A Not here. sponsor position: The development sponsor's proposed approach and rationale. The usual combination is "state the sponsor position".

B Correct. new drug application: A US application seeking approval to market a new drug. The usual combination is "plan the NDA content".

C Not here. cross-reference: A pointer connecting a statement to supporting material. The usual combination is "verify the cross-reference".

D Not here. clinical hold: A regulatory order delaying or suspending a clinical investigation in the US context. The usual combination is "address a clinical hold".

Return to the exercise, page 21

D4. Rebuild the exchange from "A meeting question that asks for blanket agreement". Evan: "The supporting summaries are available, although two underlying reports are unfinished. I do not want the package to imply that those reports are final." Priya: [missing response] Evan: "Section three names a narrower population than the summary. Which one is our proposal? We need that settled before the package goes out." Which response belongs in the gap?

B. Identify that information gap and explain its relevance. We should not conceal unfinished work or claim that a summary has the status of a completed report.

A Not here. This line belongs to turn 20, after Evan says: "I will revise the question and flag the unfinished reports. Please check the executive summary too; it must not suggest that we already have agency agreement." It addresses that point, not the missing turn in the supplied exchange.

B Correct. This is Priya's response in turn 6. It connects Evan's preceding point with Evan's follow-up. The information gap is the unfinished supporting material, whose status must remain visible in the package.

C Not here. This line belongs to turn 2, after Evan says: "Our lead question asks whether the whole development plan is acceptable. It is concise, but I suspect it will not give us the advice we need." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 4, after Evan says: "The team has a preferred approach. Should we present alternatives equally, or make clear which approach we are proposing and why we favor it?" It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 21

E / Conversation

1. focused question

Turn 2. A focused question identifies the specific unresolved issue, here population and endpoint rationale.

02 / Answers and explanations / continued

2. sponsor position

Turn 4. The sponsor position is the company's proposed approach, which must not be confused with agency agreement.

3. information gap

Turn 6. The information gap is the unfinished supporting material, whose status must remain visible in the package.

4. briefing document

Turn 8. The briefing document should present a coherent proposal, evidence, and questions rather than conflicting population descriptions.

5. cross-reference

Turn 10. A cross-reference must lead to material that actually supports the statement, including after document revisions.

6. investigational new drug application

Turn 12. An investigational new drug application concerns clinical investigation in the US; a meeting package is not itself that authorization.

7. regulatory pathway

Turn 14. The regulatory pathway is the applicable route for this product, which qualified regulatory colleagues must confirm.

8. conditional advice

Turn 16. Conditional advice applies within stated circumstances; removing those conditions changes the meaning of the response.

9. meeting minutes

Turn 18. Meeting minutes record the discussion and outcomes; discrepancies should be resolved through the applicable process.

10. approval status

Turn 20. Approval status must reflect the actual authorization record, not expectations created by meeting preparation or positive discussion.

Return to the conversation, page 23

02 / Answers and explanations / continued

F / Accuracy and scope

F1. Which distinction is correct in US terminology?

D. An IND supports clinical investigation; an NDA seeks marketing approval for a new drug.

These submissions serve different purposes, and permission or advice at one stage does not guarantee later approval.

Return to the exercise, page 27

F2. How should limited advice be summarized?

D. Preserve its subject, assumptions, conditions, and unresolved issues.

An accurate record retains the boundaries of the actual response instead of changing its meaning.

Return to the exercise, page 27

F / Guided speaking

1. conditional

Regulatory lead: The endpoint advice is conditional.

Turn 1. The note limits support to the population actually described, so it is not unconditional agreement.

2. unresolved

Clinical lead: Follow-up duration remains unresolved.

Turn 2. Further justification is requested, so the note does not settle that issue.

3. population

Regulatory lead: Preserve the stated population.

Turn 3. The population defines the scope within which the endpoint advice applies.

4. authorization

Clinical lead: This is not marketing authorization.

Turn 4. The briefing explicitly excludes marketing authorization, regardless of the favorable endpoint comment.

Return to the conversation, page 28

02 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "A meeting question that asks for blanket agreement". Evan: "The long-range plan also mentions an NDA or BLA. We should explain the applicable route without suggesting that either application is already ready." Priya: [missing response] Evan: "After the meeting, the executive team will want a short update. How do we keep it concise without losing qualifications that affect the advice?" Which response belongs in the gap?

C. Exactly. The regulatory pathway depends on the product and context. Regulatory affairs should confirm the route, while the current discussion remains focused on the specific development question.

A Not here. This line belongs to turn 2, after Evan says: "Our lead question asks whether the whole development plan is acceptable. It is concise, but I suspect it will not give us the advice we need." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 8, after Evan says: "Section three names a narrower population than the summary. Which one is our proposal? We need that settled before the package goes out." It addresses that point, not the missing turn in the supplied exchange.

C Correct. This is Priya's response in turn 14. It connects Evan's preceding point with Evan's follow-up. The regulatory pathway is the applicable route for this product, which qualified regulatory colleagues must confirm.

D Not here. This line belongs to turn 10, after Evan says: "I will also check the references. One footnote points to an earlier table, and the current table no longer contains the information cited." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 28

03 / Answers and explanations

A / Read the situation

A1. What is the documented conflict?

D. Three-day versus seven-day visit windows in two sources

The brief supplies conflicting window descriptions and no confirmed controlling version.

Return to the exercise, page 31

A2. What has not been confirmed?

C. The controlling version and its implementation status at the site

Both document authority and site applicability remain unresolved in the supplied facts.

Return to the exercise, page 31

A3. Which response avoids an unauthorized assumption?

B. Escalate the exact discrepancy for documented clarification through the applicable process.

The case requires authorized clarification, not a convenient choice, inferred authority, or silent alteration of records.

Return to the exercise, page 31

D / Response choices

D1. Which message best resolves the three-day versus seven-day discrepancy?

B. Please identify the approved site-effective protocol version and section governing this visit window.

The governing version and site applicability are missing. Neither recency, scheduling convenience, nor an invented midpoint identifies the actual approved requirement.

Return to the exercise, page 36

D2. What does a newer file date prove by itself?

C. Only that the file carries a later date, not that it governs the site

File recency alone does not establish approval, effective status, training, or authority to change records.

Return to the exercise, page 36

03 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: The planned timing of study procedures and measurements. Which term fits?

A. schedule of assessments

A Correct. schedule of assessments: The planned timing of study procedures and measurements. The usual combination is "reconcile the schedule of assessments".

B Not here. randomization: Allocation to study groups using a chance-based method. The usual combination is "protect the randomization process".

C Not here. clinical research associate: A professional supporting study oversight and monitoring within assigned responsibilities. The usual combination is "contact the clinical research associate".

D Not here. monitoring finding: An issue identified during study monitoring. The usual combination is "document a monitoring finding".

Return to the exercise, page 36

D4. Rebuild the exchange from "Two visit windows and no confirmed controlling version". Maya: "I will quote both wordings in the request. Calling it a scheduling question without the numbers would make the reviewer reconstruct the discrepancy." Ben: [missing response] Maya: "Nobody is waiting for an urgent care decision. This concerns next week's appointments, but I want to resolve it before we send instructions to participants." Which response belongs in the gap?

D. Include the relevant schedule of assessments references as well. The team needs enough context to identify whether the quick-reference material reflects an authorized change or an error.

A Not here. This line belongs to turn 2, after Maya says: "I am scheduling next week's visits and have two different windows. The protocol attachment says plus or minus three days; the quick-reference sheet says seven." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 12, after Maya says: "The quick-reference sheet is useful at the desk, but it cannot create a separate set of rules. We should reconcile it after the answer is confirmed." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 16, after Maya says: "One colleague suggested changing the old dates in the tracker to remove the apparent conflict. That would conceal what happened rather than resolve it." It addresses that point, not the missing turn in the supplied exchange.

D Correct. This is Ben's response in turn 6. It connects Maya's preceding point with Maya's follow-up. The schedule of assessments supplies timing details that must be reconciled with the governing protocol.

Return to the exercise, page 36

03 / Answers and explanations / continued

E / Conversation

1. controlling version

Turn 2. The controlling version is the authorized document applicable to this site, not whichever file is easier to use.

2. effective date

Turn 4. The effective date establishes when a document applies; a later email alone does not establish implementation status.

3. schedule of assessments

Turn 6. The schedule of assessments supplies timing details that must be reconciled with the governing protocol.

4. principal investigator

Turn 8. The principal investigator leads the study at the site and is part of the appropriate clarification route.

5. protocol amendment

Turn 10. A protocol amendment changes the study document and requires applicable authorization and implementation, not merely receipt by email.

6. training record

Turn 12. The training record helps establish whether the site team received the required instruction for an implemented change.

7. protocol deviation

Turn 14. A protocol deviation is a departure from the protocol; classification requires knowing which requirement actually applied.

8. audit trail

Turn 16. The audit trail preserves the history of recorded changes; silently altering dates would conceal rather than resolve the discrepancy.

9. read-back

Turn 18. A read-back repeats the key instruction and source, allowing misunderstanding to be detected before the team acts.

03 / Answers and explanations / continued

10. essential records

Turn 20. Essential records support reconstruction and evaluation of trial conduct, including the documented clarification relevant to this discrepancy.

Return to the conversation, page 38

F / Accuracy and scope

F1. How should a necessary record correction be handled?

A. Through the applicable traceable correction process, preserving the original history

Traceable correction preserves what was recorded and why it changed, instead of concealing the original entry.

Return to the exercise, page 42

F2. What should follow an authorized clarification?

C. Confirm understanding and assess affected appointments or records through the appropriate process.

The clarified instruction must reach the relevant people, and any impact needs assessment rather than an unsupported conclusion.

Return to the exercise, page 42

F / Guided speaking

1. received

Coordinator: Version 5 has been received.

Turn 1. Email receipt is established, but the briefing supplies no confirmation of authorized implementation.

2. unconfirmed

Monitor: Its site applicability remains unconfirmed.

Turn 2. The available record still names version 4, so applicability must be clarified.

3. confirmation

Coordinator: I will request implementation confirmation.

Turn 3. The missing information concerns whether and when the newer version applies at the site.

03 / Answers and explanations / continued

4. recency

Monitor: Do not infer authority from file recency.

Turn 4. A newer file alone does not establish the approval and effective status required to govern the work.

Return to the conversation, page 43

F / Rebuild the exchange

1. Rebuild the exchange from "Two visit windows and no confirmed controlling version". Maya: "Should we review earlier visits as well? I need the version that applied then, not just today's confirmed window, before labeling any visit outside it." Ben: [missing response] Maya: "One colleague suggested changing the old dates in the tracker to remove the apparent conflict. That would conceal what happened rather than resolve it." Which response belongs in the gap?

A. Exactly. Assess any potential protocol deviation through the appropriate process after establishing the facts. Do not assume that today's clarification automatically governs every historical visit.

A Correct. This is Ben's response in turn 14. It connects Maya's preceding point with Maya's follow-up. A protocol deviation is a departure from the protocol; classification requires knowing which requirement actually applied.

B Not here. This line belongs to turn 6, after Maya says: "I will quote both wordings in the request. Calling it a scheduling question without the numbers would make the reviewer reconstruct the discrepancy." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 2, after Maya says: "I am scheduling next week's visits and have two different windows. The protocol attachment says plus or minus three days; the quick-reference sheet says seven." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 4, after Maya says: "The seven-day sheet arrived in a newer email. The coordinator who forwarded it assumed that made it the current instruction for every site." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 43

04 / Answers and explanations

A / Read the situation

A1. Did the primary analysis meet the specified significance threshold?

A. No; p=0.19 is above 0.05.

The supplied p-value exceeds the prespecified threshold, and the interval actually includes both negative and positive values.

Return to the exercise, page 44

A2. What is the secondary endpoint's status under this plan?

B. A nominal result that cannot support the stated confirmatory claim after primary failure

The specified testing sequence requires primary success before a confirmatory secondary claim.

Return to the exercise, page 44

A3. What does the negative point estimate indicate?

C. The estimated difference favors treatment on this lower-is-better scale.

Treatment minus control is negative on a scale where lower is better, but uncertainty and the test result still matter.

Return to the exercise, page 44

D / Response choices

D1. Which topline preserves the testing plan and uncertainty?

A. Primary threshold not met; the favorable secondary result remains nominal under the hierarchy.

The primary p=0.19 exceeds 0.05, and the hierarchy requires primary success for a confirmatory secondary claim. A favorable estimate is not primary success, while a missed threshold does not establish an exactly absent effect.

Return to the exercise, page 49

D2. What does the interval from -5 to +1 show?

A. It includes values favoring treatment, zero, and values favoring control on this scale.

The interval crosses zero and spans both directions of the treatment-minus-control difference.

Return to the exercise, page 49

04 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: The precise treatment-effect target defined for a clinical question. Which term fits?

A. estimand

A Correct. estimand: The precise treatment-effect target defined for a clinical question. The usual combination is "specify the estimand".

B Not here. subgroup analysis: Examination of results within defined subsets of participants. The usual combination is "qualify a subgroup analysis".

C Not here. statistical analysis plan: The detailed prespecified methods for study analyses. The usual combination is "follow the statistical analysis plan".

D Not here. testing hierarchy: A prespecified order and conditions for confirmatory tests. The usual combination is "respect the testing hierarchy".

Return to the exercise, page 49

D4. Rebuild the exchange from "A favorable secondary result takes over the headline". Luis: "The interval is minus five to plus one. Someone suggested calling that no effect because it crosses zero." Anika: [missing response] Luis: "What about the secondary result? Its p-value is smaller. Could that become the main efficacy message instead?" Which response belongs in the gap?

C. The confidence interval includes zero and values in both directions. It does not establish an effect of exactly zero or prove that every patient had the same response.

A Not here. This line belongs to turn 20, after Luis says: "I will revise the headline, retain the estimate and interval, and label the secondary finding nominal. Please check the population wording before circulation." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 2, after Luis says: "The headline says the trial proved efficacy. It leads with the secondary p-value of point zero one. Is that too strong?" It addresses that point, not the missing turn in the supplied exchange.

C Correct. This is Anika's response in turn 6. It connects Luis's preceding point with Luis's follow-up. The confidence interval crosses zero and includes differences in both directions, showing uncertainty around the estimate.

D Not here. This line belongs to turn 10, after Luis says: "I do not want to hide the secondary observation either. How should it appear in the readout?" It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 49

E / Conversation

1. primary endpoint

Turn 2. The primary endpoint addresses the principal prespecified objective and cannot be replaced by a favorable secondary finding.

04 / Answers and explanations / continued

2. point estimate

Turn 4. The point estimate is the estimated two-point difference favoring treatment; it must be reported with uncertainty.

3. confidence interval

Turn 6. The confidence interval crosses zero and includes differences in both directions, showing uncertainty around the estimate.

4. testing hierarchy

Turn 8. The testing hierarchy conditions the confirmatory secondary claim on primary success, which the supplied result did not achieve.

5. nominal p-value

Turn 10. The nominal p-value can be reported with its limitation, but lacks the confirmatory status required by this testing plan.

6. Clinical importance

Turn 12. Clinical importance concerns the meaning of the effect for patients, which a p-value alone cannot establish.

7. estimand

Turn 14. The estimand defines the treatment effect of interest, including how relevant intercurrent events are handled.

8. intercurrent event

Turn 16. An intercurrent event occurs after treatment starts and affects the interpretation or existence of relevant measurements.

9. sensitivity analyses

Turn 18. Sensitivity analyses examine robustness to assumptions for the target question, rather than searching only for a favorable result.

10. topline readout

Turn 20. The topline readout should summarize the key results with their uncertainty and actual testing status.

Return to the conversation, page 51

04 / Answers and explanations / continued

F / Accuracy and scope

F1. Why does clinical importance need separate discussion?

C. The practical meaning of a score difference is not established by a p-value alone.

Statistical evidence alone does not determine whether an effect matters to patients in the relevant clinical context.

Return to the exercise, page 54

F2. Which statement about later subgroup work is appropriate?

B. Label it according to its actual prespecification and multiplicity status.

Transparency about timing and testing status prevents exploratory work from being misrepresented as confirmatory evidence.

Return to the exercise, page 54

F / Guided speaking

1. 20%

Analyst: The control event rate is 20%.

Turn 1. Dividing 200 events by 1,000 participants gives 0.20, or 20%.

2. 15%

Reviewer: The treatment event rate is 15%.

Turn 2. Dividing 150 events by 1,000 participants gives 0.15, or 15%.

3. five percentage points

Analyst: The absolute reduction is five percentage points.

Turn 3. Subtract 15% from 20%; the difference between the rates is five percentage points.

4. 25%

Reviewer: Relative to control, the reduction is 25%.

Turn 4. The five-percentage-point reduction divided by the 20% control rate equals 25%; this does not establish statistical significance.

Return to the conversation, page 55

04 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "A favorable secondary result takes over the headline". Luis: "There is another inconsistency. One slide says all assigned patients; another talks only about people who stayed on treatment. Are those equivalent?" Anika: [missing response] Luis: "So stopping treatment is not automatically a reason to remove someone from the question or to assume every later observation is missing?" Which response belongs in the gap?

C. We need to check the estimand. It defines the treatment effect we intended to estimate, including population, outcome, treatments, event handling, and summary measure.

A Not here. This line belongs to turn 8, after Luis says: "What about the secondary result? Its p-value is smaller. Could that become the main efficacy message instead?" It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 2, after Luis says: "The headline says the trial proved efficacy. It leads with the secondary p-value of point zero one. Is that too strong?" It addresses that point, not the missing turn in the supplied exchange.

C Correct. This is Anika's response in turn 14. It connects Luis's preceding point with Luis's follow-up. The estimand defines the treatment effect of interest, including how relevant intercurrent events are handled.

D Not here. This line belongs to turn 4, after Luis says: "But the treatment-minus-control difference is minus two points. Lower symptoms are better, so the direction is favorable. Can we still say that?" It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 55

05 / Answers and explanations

A / Read the situation

A1. What has the clinician actually reported?

A. Dizziness described as severe after product use

The message reports dizziness and its described intensity, while key assessment details remain missing.

Return to the exercise, page 58

A2. Can severe automatically be treated as serious in the regulatory sense?

D. No; intensity and seriousness criteria are different concepts.

Severe describes intensity, whereas seriousness depends on applicable outcome or medical criteria that are not confirmed here.

Return to the exercise, page 58

A3. What does the stated company procedure require?

D. Prompt internal routing of available information with documented follow-up

The procedure requires routing despite incompleteness and does not authorize diagnosis or treatment advice.

Return to the exercise, page 58

D / Response choices

D1. Which intake summary accurately separates the three safety concepts?

D. Severe dizziness reported after use; seriousness unconfirmed and causality unassessed.

Severe records the clinician's intensity description. Missing seriousness information cannot establish nonseriousness, and occurrence after use does not establish causation or a recovered outcome.

Return to the exercise, page 64

D2. What should happen while missing details are sought?

C. Route available information promptly under the stated procedure and document follow-up.

The stated process requires timely internal routing and follow-up, not waiting for complete information or providing treatment advice.

Return to the exercise, page 64

05 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: Classification based on applicable outcomes or medical criteria, not intensity alone. Which term fits?

A. seriousness

A Correct. seriousness: Classification based on applicable outcomes or medical criteria, not intensity alone. The usual combination is "assess seriousness".

B Not here. temporal association: A relationship in timing without necessarily proving causation. The usual combination is "distinguish temporal association from causation".

C Not here. safety signal: Information suggesting a possible new or changed risk that warrants further evaluation. The usual combination is "evaluate a safety signal".

D Not here. duplicate report: Another report concerning a case already recorded. The usual combination is "investigate a possible duplicate report".

Return to the exercise, page 64

D4. Rebuild the exchange from "An incomplete safety report". Nora: "The event occurred after product use, but the message gives no exact dates. I cannot tell how close the timing was or whether other explanations were considered." Rafael: [missing response] Nora: "We are missing dose, outcome, and other medicines. I will list those gaps so they remain visible in the structured record." Which response belongs in the gap?

B. Describe a reported temporal association, not proven causation. The safety assessment will consider the available evidence. Do not add a causal conclusion that the source did not establish.

A Not here. This line belongs to turn 16, after Nora says: "The commercial team may ask whether this is a new safety signal. I do not think this handoff alone answers that broader question." It addresses that point, not the missing turn in the supplied exchange.

B Correct. This is Rafael's response in turn 6. It connects Nora's preceding point with Nora's follow-up. Temporal association concerns timing and does not by itself prove that the product caused the event.

C Not here. This line belongs to turn 18, after Nora says: "I also want the internal update to avoid implying that a label change has been decided. We are much earlier in the information process." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 10, after Nora says: "The message says severe dizziness. For the structured follow-up, should I standardize it or keep the clinician's exact phrase alongside the coded fields?" It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 64

E / Conversation

1. case intake

Turn 2. Case intake records the available report so assessment and follow-up can proceed under the required process.

05 / Answers and explanations / continued

2. severity

Turn 4. Severity describes the reported intensity, while seriousness depends on separate applicable criteria.

3. temporal association

Turn 6. Temporal association concerns timing and does not by itself prove that the product caused the event.

4. concomitant medications

Turn 8. Concomitant medications are other medicines used in the relevant period and are missing from this report.

5. verbatim term

Turn 10. The verbatim term preserves the original reporter's wording, even when standardized coding is added later.

6. follow-up request

Turn 12. A follow-up request seeks missing facts neutrally rather than suggesting the answer or demanding proof of causation.

7. case reconciliation

Turn 14. Case reconciliation checks related records and handoffs so the same case is consistently accounted for across teams.

8. safety signal

Turn 16. A safety signal suggests a possible new or changed risk requiring evaluation, not a confirmed causal conclusion from this handoff.

9. benefit-risk assessment

Turn 18. Benefit-risk assessment considers favorable and unfavorable effects in context; this incomplete report does not determine a labeling decision.

10. receipt confirmation

Turn 20. Receipt confirmation closes the handoff by establishing that the safety team received the information.

Return to the conversation, page 66

05 / Answers and explanations / continued

F / Accuracy and scope

F1. Which follow-up asks for a missing fact without leading the reporter?

D. Was the patient hospitalized, and what was the outcome?

The neutral question seeks hospitalization and outcome facts that are absent without suggesting the desired answer.

Return to the exercise, page 70

F2. What does a temporal association establish?

A. That the event and product use have a stated timing relationship, not necessarily causation

Temporal association concerns timing; causal conclusions and labeling decisions require the relevant assessment.

Return to the exercise, page 70

F / Guided speaking

1. unknown

Associate: Hospitalization status is unknown.

Turn 1. The report contains no hospitalization information, so neither occurrence nor absence can be asserted.

2. causation

Reviewer: The timing does not establish causation.

Turn 2. Occurrence after use supplies a temporal relationship, not proof that the product caused the rash.

3. now

Associate: I will route the available report now.

Turn 3. The stated process requires routing the information now rather than waiting for every missing detail.

4. follow-up

Reviewer: Record the missing details for follow-up.

Turn 4. Additional information is needed, and the reporting process must preserve the outstanding questions.

Return to the conversation, page 71

05 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "An incomplete safety report". Nora: "Who owns that contact? I do not want both teams to approach the reporter independently and create two incomplete versions of the same conversation." Rafael: [missing response] Nora: "The commercial team may ask whether this is a new safety signal. I do not think this handoff alone answers that broader question." Which response belongs in the gap?

A. We will assign an owner and use case reconciliation to keep the records aligned. If another report arrives, assess whether it is connected rather than assuming similar wording means a duplicate.

A Correct. This is Rafael's response in turn 14. It connects Nora's preceding point with Nora's follow-up. Case reconciliation checks related records and handoffs so the same case is consistently accounted for across teams.

B Not here. This line belongs to turn 6, after Nora says: "The event occurred after product use, but the message gives no exact dates. I cannot tell how close the timing was or whether other explanations were considered." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 18, after Nora says: "I also want the internal update to avoid implying that a label change has been decided. We are much earlier in the information process." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 2, after Nora says: "I received a clinician's message about dizziness after LM-4 use. The clinician called it severe, but several details are missing. I want to route it correctly." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 71

06 / Answers and explanations

A / Read the situation

A1. What is established about batch Q-214?

B. A verification entry is missing and quality review remains open.

The brief establishes a documentation gap and open review, not completion, release, or a confirmed manufacturing failure.

Return to the exercise, page 72

A2. Why is fully reviewed inaccurate?

C. The required investigation and authorized disposition are not complete.

Laboratory results do not resolve the missing verification or complete the specified quality process.

Return to the exercise, page 72

A3. Which action is excluded by the case?

A. Backdating an entry to make the record appear complete

The brief expressly prohibits backdating and requires a documented investigation rather than retrospective fabrication.

Return to the exercise, page 72

D / Response choices

D1. Which supply update is supported?

B. Q-214 remains under quality review; release timing is unconfirmed.

The update states the actual open status without assuming either release or a confirmed product defect.

Return to the exercise, page 77

D2. What does the missing entry prove by itself?

C. That required recorded information is absent

A missing entry establishes a documentation gap; the underlying event requires investigation.

Return to the exercise, page 77

06 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: The authorized instruction defining how a product is to be manufactured. Which term fits?

C. master production instruction

A Not here. release authorization: Formal permission for a batch to proceed to the defined use or distribution. The usual combination is "verify release authorization".

B Not here. effectiveness check: A review of whether an action achieved its intended result. The usual combination is "plan an effectiveness check".

C Correct. master production instruction: The authorized instruction defining how a product is to be manufactured. The usual combination is "follow master production instructions".

D Not here. traceability: The ability to follow an activity, material, or result through its records. The usual combination is "preserve traceability".

Return to the exercise, page 77

D4. Rebuild the exchange from "A gap in the batch record". Diego: "Someone suggested signing yesterday's date to tidy it up. I have stopped that and kept the original record unchanged." Helen: [missing response] Diego: "I can make the original records and relevant staff available. Should we keep the messages about this discrepancy as well?" Which response belongs in the gap?

B. Good. A later entry cannot be passed off as a contemporaneous record. Any permitted correction must accurately show its actual timing, basis, and required authorization.

A Not here. This line belongs to turn 20, after Diego says: "I will report the missing entry, open investigation, and unreleased status. Let me know the next update point without converting it into a release promise." It addresses that point, not the missing turn in the supplied exchange.

B Correct. This is Helen's response in turn 6. It connects Diego's preceding point with Diego's follow-up. A contemporaneous record is made when the activity occurs; a later entry must not falsely appear to have been made earlier.

C Not here. This line belongs to turn 16, after Diego says: "Then the action record needs more than a training date. How will we know the change addressed the problem?" It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 2, after Diego says: "Supply wants to ship Q-214 tomorrow. The laboratory results are available, but a verification entry is missing. Can I at least say fully reviewed?" It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 77

E / Conversation

1. batch record

Turn 2. The batch record contains a missing verification entry, so the required review is not complete.

06 / Answers and explanations / continued

2. documentation gap

Turn 4. The documentation gap is the established missing information; its underlying cause has not yet been determined.

3. contemporaneous record

Turn 6. A contemporaneous record is made when the activity occurs; a later entry must not falsely appear to have been made earlier.

4. traceability

Turn 8. Traceability preserves the links between activities, evidence, and changes so the discrepancy can be reconstructed.

5. out-of-specification result

Turn 10. An out-of-specification result is a test result outside acceptance specifications, not merely missing documentation.

6. deviation investigation

Turn 12. The deviation investigation examines the discrepancy through the applicable documented process and identifies supported conclusions.

7. root cause

Turn 14. The root cause must be supported by investigation rather than selected because it is a convenient explanation.

8. effectiveness check

Turn 16. An effectiveness check evaluates whether the action achieved its intended result, rather than merely confirming task completion.

9. batch disposition

Turn 18. Batch disposition is the authorized decision on batch use or handling, which remains pending here.

10. release authorization

Turn 20. Release authorization is formal permission to proceed; it cannot be inferred from available tests or a desired shipment date.

Return to the conversation, page 79

06 / Answers and explanations / continued

F / Accuracy and scope

F1. Which action preserves data integrity?

D. Use the authorized correction process with an accurate, traceable account.

An authorized, traceable correction preserves the true record rather than inventing evidence or concealing the gap.

Return to the exercise, page 81

F2. Which action plan best distinguishes completion from effectiveness?

B. Link the action to the supported cause and specify evidence that it reduces recurrence.

A completed administrative task does not by itself show the intended preventive effect. The action needs a supported cause and a suitable effectiveness check; neither replaces authorized batch disposition.

Return to the exercise, page 81

F / Guided speaking

1. results

Planner: The certificate reports test results.

Turn 1. A certificate of analysis reports specified testing information, not every aspect of the release decision.

2. pending

Quality: Batch disposition is still pending.

Turn 2. The briefing states that the authorized disposition record has not been completed.

3. released

Planner: The current status is not released.

Turn 3. The supplied status expressly says not released, despite the available certificate.

4. authorization

Quality: The customer request does not supply release authorization.

Turn 4. A delivery request cannot replace the applicable authorized decision on whether the batch may ship.

Return to the conversation, page 82

06 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "A gap in the batch record". Diego: "The proposed action is another training session. Would that be enough to close this before tomorrow's shipment?" Helen: [missing response] Diego: "Then the action record needs more than a training date. How will we know the change addressed the problem?" Which response belongs in the gap?

A. Not without establishing the root cause and appropriate action. Keep containment in place now; a convenient training proposal does not resolve the investigation or release the batch.

A Correct. This is Helen's response in turn 14. It connects Diego's preceding point with Diego's follow-up. The root cause must be supported by investigation rather than selected because it is a convenient explanation.

B Not here. This line belongs to turn 6, after Diego says: "Someone suggested signing yesterday's date to tidy it up. I have stopped that and kept the original record unchanged." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 20, after Diego says: "I will report the missing entry, open investigation, and unreleased status. Let me know the next update point without converting it into a release promise." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 16, after Diego says: "Then the action record needs more than a training date. How will we know the change addressed the problem?" It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 82

07 / Answers and explanations

A / Read the situation

A1. Which part of the draft is unsupported by the supplied source?

A. Immediate relief for every patient

The supplied claim concerns a mean score at week 12, not immediate onset or a benefit for every individual.

Return to the exercise, page 85

A2. What remains necessary before external use?

B. Review and approval of the complete revised item through the stated process

The case requires review of the complete revised item, not just a phrase or an earlier version.

Return to the exercise, page 85

A3. Why is the risk footnote a concern?

A. Its unreadable presentation undermines meaningful communication of risk.

The problem is the unreadable presentation, not a universal prohibition on footnotes or the existence of a benefit claim.

Return to the exercise, page 85

D / Response choices

D1. Which revision best preserves the supplied scope?

C. Lower mean symptom score versus placebo at week 12 in the defined adult study population

The correct wording retains the measured outcome, comparator, timepoint, and defined population without adding a universal or onset claim.

Return to the exercise, page 89

D2. What is the best review comment on the headline?

D. It adds an onset claim and an individual guarantee absent from the supplied evidence.

The draft changes the substantive meaning; a citation or change of presentation does not establish the missing evidence.

Return to the exercise, page 89

07 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: Evidence supporting the precise meaning of a claim. Which term fits?

A. claim substantiation

A Correct. claim substantiation: Evidence supporting the precise meaning of a claim. The usual combination is "verify claim substantiation".

B Not here. medical affairs: The function supporting scientific and medical exchange within its responsibilities. The usual combination is "consult medical affairs".

C Not here. reference pack: The supporting evidence linked to a communication's claims. The usual combination is "update the reference pack".

D Not here. indication boundary: The limits of the use or population described by the relevant authorization. The usual combination is "preserve the indication boundary".

Return to the exercise, page 89

D4. Rebuild the exchange from "Stronger wording changes the claim". Adam: "Immediate is also new. The source identifies a week-twelve assessment, but it does not say when any effect began." Sofia: [missing response] Adam: "Could the headline stay short with the conditions underneath? Show me which limits must be prominent so the designer can give them enough space." Which response belongs in the gap?

C. Then remove the onset claim unless appropriate supporting evidence and review establish it. A timepoint of measurement cannot silently become a claim about the first moment of benefit.

A Not here. This line belongs to turn 4, after Adam says: "I see that every patient is stronger than an average difference. The study result does not tell us that all participants responded in the same way." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 16, after Adam says: "A field colleague has also asked whether this could be used for a different, unapproved condition. That question should not go into this promotional revision." It addresses that point, not the missing turn in the supplied exchange.

C Correct. This is Sofia's response in turn 6. It connects Adam's preceding point with Adam's follow-up. An onset claim concerns when an effect begins; a week-12 assessment does not establish immediate relief.

D Not here. This line belongs to turn 14, after Adam says: "I will send the revised wording with the evidence references. That should make it easier to see which source supports each part of the claim." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 89

07 / Answers and explanations / continued

E / Conversation

1. promotional claim

Turn 2. The promotional claim now adds immediate onset and universal benefit, neither of which the supplied evidence supports.

2. absolute claim

Turn 4. An absolute claim suggests no exceptions; a group mean result does not establish benefit for every patient.

3. onset claim

Turn 6. An onset claim concerns when an effect begins; a week-12 assessment does not establish immediate relief.

4. material limitation

Turn 8. A material limitation affects the accurate interpretation of the claim and should not disappear into unreadable text.

5. fair balance

Turn 10. Fair balance concerns appropriate benefit-risk presentation; merely including unreadable risk text does not resolve the issue.

6. net impression

Turn 12. Net impression is the combined meaning of wording, imagery, and layout, not just the literal accuracy of one sentence.

7. reference pack

Turn 14. The reference pack links claims to their supporting evidence and allows reviewers to check the precise scope.

8. medical information

Turn 16. Medical information provides the appropriate route for the inquiry; the question does not authorize an expanded promotional claim.

9. final-form approval

Turn 18. Final-form approval applies to the complete version intended for use; approval of an earlier item does not cover every revision.

07 / Answers and explanations / continued

10. medical-legal-regulatory review

Turn 20. Medical-legal-regulatory review assesses the revised communication through the required cross-functional process before external use.

Return to the conversation, page 91

F / Accuracy and scope

F1. Which action addresses the footnote problem?

C. Revise the risk presentation so it is readable and assess the complete item's balance.

Meaningful risk communication requires attention to presentation and context, not merely the physical presence of tiny text.

Return to the exercise, page 94

F2. A colleague asks about an unapproved use. What is the appropriate response here?

D. Route the inquiry through the relevant medical-information process without repurposing it as promotion.

The communication context and approved scope matter; the inquiry does not authorize an expanded promotional claim.

Return to the exercise, page 94

F / Guided speaking

1. onset

Reviewer: Rapid adds an unsupported onset claim.

Turn 1. The source supplies a week-12 measurement, not evidence about how quickly benefit begins.

2. population

Author: All ages expands the stated population.

Turn 2. The original example concerns adults, while the revision extends the claim beyond that group.

3. version

Reviewer: Earlier approval does not cover this changed version.

Turn 3. A substantive revision must be assessed under the applicable process rather than inheriting approval automatically.

07 / Answers and explanations / continued

4. pending

Author: External use remains pending.

Turn 4. The revised item has not completed review, so the briefing supplies no authorization for external use.

Return to the conversation, page 96

F / Rebuild the exchange

1. Rebuild the exchange from "Stronger wording changes the claim". Adam: "I will send the revised wording with the evidence references. That should make it easier to see which source supports each part of the claim." Sofia: [missing response] Adam: "A field colleague has also asked whether this could be used for a different, unapproved condition. That question should not go into this promotional revision." Which response belongs in the gap?

C. Update the reference pack and check the exact scope. A source citation is useful only when it supports the statement actually made, including population, comparator, timing, and outcome.

A Not here. This line belongs to turn 16, after Adam says: "A field colleague has also asked whether this could be used for a different, unapproved condition. That question should not go into this promotional revision." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 4, after Adam says: "I see that every patient is stronger than an average difference. The study result does not tell us that all participants responded in the same way." It addresses that point, not the missing turn in the supplied exchange.

C Correct. This is Sofia's response in turn 14. It connects Adam's preceding point with Adam's follow-up. The reference pack links claims to their supporting evidence and allows reviewers to check the precise scope.

D Not here. This line belongs to turn 6, after Adam says: "Immediate is also new. The source identifies a week-twelve assessment, but it does not say when any effect began." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 96

08 / Answers and explanations

A / Read the situation

A1. Which source measures hospitalization?

D. The six-month observational claims analysis

The brief assigns hospitalization to the claims analysis and symptom score to the trial.

Return to the exercise, page 97

A2. Why cannot the combined deck be described as randomized proof of fewer hospitalizations?

A. Only the symptom-score trial is randomized; the hospitalization analysis is observational.

The trial studies symptoms for twelve weeks, while routine-care claims examine hospitalization for six months without randomized treatment assignment. Sample size and combining sources do not change either design.

Return to the exercise, page 97

A3. What is the status of coverage?

B. No payer decision has been received.

The briefing explicitly says no payer decision exists, so coverage cannot be reported as guaranteed.

Return to the exercise, page 97

D / Response choices

D1. Which summary is best supported?

C. The trial assesses symptoms; the observational claims study examines hospitalizations with unresolved confounding questions.

The accurate summary preserves each source's design, outcome, and unresolved limitation.

Return to the exercise, page 102

D2. What should be checked before interpreting the claims difference causally?

B. Treatment selection, measured confounders, remaining bias, and the analysis design

Causal interpretation depends on the design and assumptions, not merely sample size or a favorable result.

Return to the exercise, page 102

08 / Answers and explanations / continued

D3. A colleague needs the term for this meaning: A payer or system's list of medicines and associated access conditions. Which term fits?

B. formulary

A Not here. launch readiness: Preparedness of the relevant functions for a planned market introduction. The usual combination is "verify launch readiness".

B Correct. formulary: A payer or system's list of medicines and associated access conditions. The usual combination is "review formulary status".

C Not here. residual confounding: Confounding remaining after the applied adjustment. The usual combination is "acknowledge residual confounding".

D Not here. real-world evidence: Clinical evidence generated by analyzing real-world data using an appropriate study design. The usual combination is "evaluate real-world evidence".

Return to the exercise, page 102

D4. Rebuild the exchange from "Two sources, different questions". Mina: "Patients receiving one treatment could have had different clinical needs before follow-up began. Those needs might also affect their chances of hospitalization." Jules: [missing response] Mina: "The slide says adjustment removed all bias. Which characteristics were included, and what important differences could still remain unmeasured?" Which response belongs in the gap?

C. That raises confounding by indication. We should ask which relevant factors were measured and addressed, while acknowledging that the current assessment is incomplete.

A Not here. This line belongs to turn 16, after Mina says: "The uptake assumption is twenty percent of an estimated eligible population. It is a planning input, not a confirmed number of people who will receive coverage." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 10, after Mina says: "We should still show what the routine-care analysis adds. It may help an access audience understand patterns outside the trial's narrower setting." It addresses that point, not the missing turn in the supplied exchange.

C Correct. This is Jules's response in turn 6. It connects Mina's preceding point with Mina's follow-up. Confounding by indication arises when treatment choice relates to clinical need or prognosis that also affects outcomes.

D Not here. This line belongs to turn 18, after Mina says: "We have no payer decision yet. The headline's guaranteed coverage wording could create expectations that neither the evidence team nor the access team can support." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 102

E / Conversation

1. evidence streams

Turn 2. The evidence streams differ in study design, outcome, and follow-up, so their findings cannot be presented as one randomized result.

08 / Answers and explanations / continued

2. observational design

Turn 4. The observational design did not assign treatment randomly; a larger sample does not change that fact.

3. confounding by indication

Turn 6. Confounding by indication arises when treatment choice relates to clinical need or prognosis that also affects outcomes.

4. Residual confounding

Turn 8. Residual confounding is distortion remaining after adjustment, which cannot be ruled out merely by listing adjusted variables.

5. external validity

Turn 10. External validity concerns applicability beyond the studied setting; routine-care context may help but does not automatically guarantee generalizability.

6. Real-world data

Turn 12. Real-world data describes routinely collected health information; the study design using those data must be specified separately.

7. budget impact

Turn 14. Budget impact estimates spending consequences for a specified budget holder over a defined period, not value per health outcome.

8. uptake assumption

Turn 16. The uptake assumption is a model input about use, not a confirmed patient count or payer commitment.

9. coverage decision

Turn 18. A coverage decision is the payer's funding determination, which has not been received in this case.

10. launch-readiness

Turn 20. Launch-readiness assessment checks multiple dependencies; evidence preparation alone does not establish coverage, supply, or complete operational readiness.

Return to the conversation, page 104

08 / Answers and explanations / continued

F / Accuracy and scope

F1. Which statement about economic evidence is accurate?

B. Budget impact estimates spending for a defined budget holder; cost-effectiveness relates costs to health outcomes.

These economic analyses address different questions and do not themselves determine clinical superiority or coverage.

Return to the exercise, page 107

F2. Which launch update preserves the actual status?

A. The access evidence package is being revised; payer decisions remain pending.

The brief states that no payer decision has been received and the deck still needs correction.

Return to the exercise, page 107

F / Guided speaking

1. 2,000

Analyst: The model assumes 2,000 users.

Turn 1. Twenty percent of 10,000 eligible adults equals 2,000 modeled users, not a confirmed covered population.

2. $600,000

Reviewer: The modeled annual spending increase is $600,000.

Turn 2. Multiply 2,000 modeled users by $300 each, with no other costs or offsets included.

3. assumption

Analyst: The 20% figure is an uptake assumption.

Turn 3. The briefing identifies the figure as a planning input rather than an observed or guaranteed use rate.

4. undecided

Reviewer: Coverage remains undecided.

Turn 4. No payer decision has been supplied, so the model cannot establish that coverage has been granted.

Return to the conversation, page 108

08 / Answers and explanations / continued

F / Rebuild the exchange

1. Rebuild the exchange from "Two sources, different questions". Mina: "The financial slide also combines budget impact with cost-effectiveness. One estimates payer spending, while the other relates costs to health outcomes." Jules: [missing response] Mina: "The uptake assumption is twenty percent of an estimated eligible population. It is a planning input, not a confirmed number of people who will receive coverage." Which response belongs in the gap?

A. Keep the budget impact model separate and state its horizon, eligible population, and uptake assumptions. A useful economic result depends on a clearly defined question and inputs.

A Correct. This is Jules's response in turn 14. It connects Mina's preceding point with Mina's follow-up. Budget impact estimates spending consequences for a specified budget holder over a defined period, not value per health outcome.

B Not here. This line belongs to turn 10, after Mina says: "We should still show what the routine-care analysis adds. It may help an access audience understand patterns outside the trial's narrower setting." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 2, after Mina says: "The access deck combines the trial and claims results under one headline: randomized proof of fewer hospitalizations and guaranteed coverage. That seems to merge several different claims." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 18, after Mina says: "We have no payer decision yet. The headline's guaranteed coverage wording could create expectations that neither the evidence team nor the access team can support." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 108

Review 1 / Explanations

Cumulative review

1. A colleague needs the term for this meaning: A selected investigational substance being evaluated for further development. Which term fits?

A. development candidate

A Correct. development candidate: A selected investigational substance being evaluated for further development. The usual combination is "nominate a development candidate".

B Not here. repeatability: Agreement between repeated results under the same specified measurement conditions. The usual combination is "assess repeatability".

C Not here. exposure-response relationship: How an effect varies with drug exposure in the body. The usual combination is "examine exposure-response relationships".

D Not here. biomarker: A measured characteristic indicating a biological process or response. The usual combination is "evaluate a candidate biomarker".

Return to the exercise, page 29

2. A colleague needs the term for this meaning: A structured account of background, evidence, proposals, and questions. Which term fits?

A. briefing document

A Correct. briefing document: A structured account of background, evidence, proposals, and questions. The usual combination is "finalize the briefing document".

B Not here. written response: A documented reply to a submitted question or issue. The usual combination is "review the written response".

C Not here. regulatory strategy: A plan for addressing applicable regulatory requirements and interactions. The usual combination is "align the regulatory strategy".

D Not here. submission readiness: The state of being sufficiently prepared for an intended submission. The usual combination is "assess submission readiness".

Return to the exercise, page 29

Review 1 / Explanations / continued

3. Rebuild the exchange from "A laboratory signal becomes a patient-benefit claim". Owen: "Why is there a rising dose-response arrow if we tested only one concentration? The graphic seems to be claiming something the experiment did not measure." Leila: [missing response] Owen: "What can we say about the three runs? I do not want to discard useful repetition merely because the result is still early." Which response belongs in the gap?

D. Remove that implication. A dose-response relationship has not been characterized here. The visual should not make a stronger statement than the accompanying words or the experiment itself.

A Not here. This line belongs to turn 12, after Owen says: "What can we say about the three runs? I do not want to discard useful repetition merely because the result is still early." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 4, after Owen says: "Then the headline has moved beyond the evidence. The team wanted a clear link to the disease, but the wording makes that link sound demonstrated." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 8, after Owen says: "Can we call this proof of concept? The phrase is familiar, but different colleagues may hear very different claims when they read it." It addresses that point, not the missing turn in the supplied exchange.

D Correct. This is Leila's response in turn 10. It connects Owen's preceding point with Owen's follow-up. Dose-response describes effects across doses; the single tested concentration cannot establish that relationship.

Return to the exercise, page 29

Review 1 / Explanations / continued

4. Rebuild the exchange from "A meeting question that asks for blanket agreement". Evan: "We should compare our notes with the official record through the applicable process. A colleague's enthusiastic recollection should not become the only account." Priya: [missing response] Evan: "I will revise the question and flag the unfinished reports. Please check the executive summary too; it must not suggest that we already have agency agreement." Which response belongs in the gap?

D. Yes, reconcile the meeting minutes and track unresolved differences. Assign owners to follow-up items and commitments rather than treating a successful conversation as completion of all required work.

A Not here. This line belongs to turn 8, after Evan says: "Section three names a narrower population than the summary. Which one is our proposal? We need that settled before the package goes out." It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 10, after Evan says: "I will also check the references. One footnote points to an earlier table, and the current table no longer contains the information cited." It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 16, after Evan says: "After the meeting, the executive team will want a short update. How do we keep it concise without losing qualifications that affect the advice?" It addresses that point, not the missing turn in the supplied exchange.

D Correct. This is Priya's response in turn 18. It connects Evan's preceding point with Evan's follow-up. Meeting minutes record the discussion and outcomes; discrepancies should be resolved through the applicable process.

Return to the exercise, page 29

Review 2 / Explanations

Cumulative review

1. A colleague needs the term for this meaning: The allowed timing range for a specified study visit or assessment. Which term fits?

A. visit window

A Correct. visit window: The allowed timing range for a specified study visit or assessment. The usual combination is "confirm the visit window".

B Not here. eligibility criterion: A condition determining whether someone may enter a study. The usual combination is "verify eligibility criteria".

C Not here. principal investigator: The investigator responsible for leading the study at a site. The usual combination is "consult the principal investigator".

D Not here. protocol deviation: A departure from the study protocol. The usual combination is "assess a potential protocol deviation".

Return to the exercise, page 56

2. A colleague needs the term for this meaning: The defined group included in an analysis. Which term fits?

C. analysis population

A Not here. primary endpoint: The outcome designated to address the study's principal objective. The usual combination is "report the primary endpoint".

B Not here. missing data: Required or relevant observations not available for analysis. The usual combination is "assess missing data".

C Correct. analysis population: The defined group included in an analysis. The usual combination is "specify the analysis population".

D Not here. significance threshold: The prespecified criterion used for a statistical test decision. The usual combination is "apply the significance threshold".

Return to the exercise, page 56

Review 2 / Explanations / continued

3. Rebuild the exchange from "A meeting question that asks for blanket agreement". Evan: "I will also check the references. One footnote points to an earlier table, and the current table no longer contains the information cited." Priya: [missing response] Evan: "Some colleagues are calling the meeting package our IND approval package. That seems to mix the meeting purpose with the status of a regulatory submission." Which response belongs in the gap?

B. Verify each cross-reference after the content is settled. A broken evidence trail forces reviewers to guess which material we intended and can weaken an otherwise useful discussion.

A Not here. This line belongs to turn 14, after Evan says: "The long-range plan also mentions an NDA or BLA. We should explain the applicable route without suggesting that either application is already ready." It addresses that point, not the missing turn in the supplied exchange.

B Correct. This is Priya's response in turn 10. It connects Evan's preceding point with Evan's follow-up. A cross-reference must lead to material that actually supports the statement, including after document revisions.

C Not here. This line belongs to turn 2, after Evan says: "Our lead question asks whether the whole development plan is acceptable. It is concise, but I suspect it will not give us the advice we need." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 8, after Evan says: "Section three names a narrower population than the summary. Which one is our proposal? We need that settled before the package goes out." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 56

Review 2 / Explanations / continued

4. Rebuild the exchange from "A favorable secondary result takes over the headline". Luis: "Would alternative assumptions help show how robust the result is? I do not want the team shopping for the most favorable answer." Anika: [missing response] Luis: "I will revise the headline, retain the estimate and interval, and label the secondary finding nominal. Please check the population wording before circulation." Which response belongs in the gap?

C. Use the planned sensitivity analyses for the same target question. Label exploratory work separately, especially when it changes the population or treatment-effect question.

A Not here. This line belongs to turn 8, after Luis says: "What about the secondary result? Its p-value is smaller. Could that become the main efficacy message instead?" It addresses that point, not the missing turn in the supplied exchange.

B Not here. This line belongs to turn 2, after Luis says: "The headline says the trial proved efficacy. It leads with the secondary p-value of point zero one. Is that too strong?" It addresses that point, not the missing turn in the supplied exchange.

C Correct. This is Anika's response in turn 18. It connects Luis's preceding point with Luis's follow-up. Sensitivity analyses examine robustness to assumptions for the target question, rather than searching only for a favorable result.

D Not here. This line belongs to turn 4, after Luis says: "But the treatment-minus-control difference is minus two points. Lower symptoms are better, so the direction is favorable. Can we still say that?" It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 56

Review 3 / Explanations

Cumulative review

1. A colleague needs the term for this meaning: An undesirable response for which a causal relationship is at least reasonably suspected in context. Which term fits?

D. adverse reaction

A Not here. case reconciliation: Checking that related records are consistent and accounted for across sources. The usual combination is "perform case reconciliation".

B Not here. reference safety information: The designated information used for defined safety assessments in context. The usual combination is "identify reference safety information".

C Not here. event onset: The time an event began. The usual combination is "confirm event onset".

D Correct. adverse reaction: An undesirable response for which a causal relationship is at least reasonably suspected in context. The usual combination is "assess a suspected adverse reaction".

Return to the exercise, page 83

2. A colleague needs the term for this meaning: A record made when the relevant activity occurs. Which term fits?

A. contemporaneous record

A Correct. contemporaneous record: A record made when the relevant activity occurs. The usual combination is "maintain contemporaneous records".

B Not here. chemistry, manufacturing, and controls: Product and process information addressing composition, manufacture, and quality control. The usual combination is "assess CMC readiness".

C Not here. certificate of analysis: A document reporting specified test results for a material or batch. The usual combination is "verify the certificate of analysis".

D Not here. preventive action: Action intended to prevent a potential problem from occurring. The usual combination is "define preventive action".

Return to the exercise, page 83

Review 3 / Explanations / continued

3. Rebuild the exchange from "A favorable secondary result takes over the headline". Luis: "I do not want to hide the secondary observation either. How should it appear in the readout?" Anika: [missing response] Luis: "The clinicians will ask whether two points matter to patients. We have not included any explanation of the symptom scale." Which response belongs in the gap?

A. Report the nominal p-value with its limitation. The observation remains available for discussion, but it does not have the confirmatory status implied by proved efficacy.

A Correct. This is Anika's response in turn 10. It connects Luis's preceding point with Luis's follow-up. The nominal p-value can be reported with its limitation, but lacks the confirmatory status required by this testing plan.

B Not here. This line belongs to turn 8, after Luis says: "What about the secondary result? Its p-value is smaller. Could that become the main efficacy message instead?" It addresses that point, not the missing turn in the supplied exchange.

C Not here. This line belongs to turn 14, after Luis says: "There is another inconsistency. One slide says all assigned patients; another talks only about people who stayed on treatment. Are those equivalent?" It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 2, after Luis says: "The headline says the trial proved efficacy. It leads with the secondary p-value of point zero one. Is that too strong?" It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 83

4. Rebuild the exchange from "A gap in the batch record". Diego: "Planning will press for an estimate. I can promise a status update, but I cannot tell the customer that tomorrow's shipment is approved." Helen: [missing response] Diego: "I will report the missing entry, open investigation, and unreleased status. Let me know the next update point without converting it into a release promise." Which response belongs in the gap?

B. Correct. Authorized batch disposition remains pending. Communicate that timing is unconfirmed; urgency does not transfer the decision to production or planning.

A Not here. This line belongs to turn 10, after Diego says: "Supply has called this an OOS. That sounds like a failed test, but I have not been shown a result outside a limit." It addresses that point, not the missing turn in the supplied exchange.

B Correct. This is Helen's response in turn 18. It connects Diego's preceding point with Diego's follow-up. Batch disposition is the authorized decision on batch use or handling, which remains pending here.

C Not here. This line belongs to turn 8, after Diego says: "I can make the original records and relevant staff available. Should we keep the messages about this discrepancy as well?" It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 12, after Diego says: "Who is coordinating the next steps? Production can help, but I need a clear owner before I reply to planning." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 83

Review 4 / Explanations

Cumulative review

1. A colleague needs the term for this meaning: A statement or implication used to communicate a product benefit or attribute in promotion. Which term fits?

B. promotional claim

A Not here. medical-legal-regulatory review: Cross-functional review of materials under the applicable company process. The usual combination is "complete MLR review".

B Correct. promotional claim: A statement or implication used to communicate a product benefit or attribute in promotion. The usual combination is "substantiate a promotional claim".

C Not here. intended audience: The people a communication is designed to reach. The usual combination is "identify the intended audience".

D Not here. qualifier: Language restricting the scope of a statement. The usual combination is "keep the qualifier visible".

Return to the exercise, page 109

2. A colleague needs the term for this meaning: Routinely collected information about health or health-care delivery. Which term fits?

D. real-world data

A Not here. confounding by indication: Differences in treatment choice related to prognosis or clinical need that affect comparison. The usual combination is "assess confounding by indication".

B Not here. market access: Work addressing whether and how patients can obtain a product within a health system. The usual combination is "develop the market-access plan".

C Not here. cost-effectiveness: Assessment of costs relative to health outcomes for compared options. The usual combination is "examine cost-effectiveness assumptions".

D Correct. real-world data: Routinely collected information about health or health-care delivery. The usual combination is "assess real-world data quality".

Return to the exercise, page 109

Review 4 / Explanations / continued

3. Rebuild the exchange from "A gap in the batch record". Diego: "Supply has called this an OOS. That sounds like a failed test, but I have not been shown a result outside a limit." Helen: [missing response] Diego: "Who is coordinating the next steps? Production can help, but I need a clear owner before I reply to planning." Which response belongs in the gap?

B. An out-of-specification result is not the same as a missing entry. Use the documentation description unless the investigation establishes another finding.

A Not here. This line belongs to turn 12, after Diego says: "Who is coordinating the next steps? Production can help, but I need a clear owner before I reply to planning." It addresses that point, not the missing turn in the supplied exchange.

B Correct. This is Helen's response in turn 10. It connects Diego's preceding point with Diego's follow-up. An out-of-specification result is a test result outside acceptance specifications, not merely missing documentation.

C Not here. This line belongs to turn 18, after Diego says: "Planning will press for an estimate. I can promise a status update, but I cannot tell the customer that tomorrow's shipment is approved." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 8, after Diego says: "I can make the original records and relevant staff available. Should we keep the messages about this discrepancy as well?" It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 109

Review 4 / Explanations / continued

4. Rebuild the exchange from "Two sources, different questions". Mina: "We have no payer decision yet. The headline's guaranteed coverage wording could create expectations that neither the evidence team nor the access team can support." Jules: [missing response] Mina: "I will separate the evidence streams and remove guaranteed coverage. Please check that each headline still matches its own study design, population, and period." Which response belongs in the gap?

B. A coverage decision is distinct from the clinical findings and economic models. State the current status and any unresolved evidence requests instead of promising an outcome.

A Not here. This line belongs to turn 2, after Mina says: "The access deck combines the trial and claims results under one headline: randomized proof of fewer hospitalizations and guaranteed coverage. That seems to merge several different claims." It addresses that point, not the missing turn in the supplied exchange.

B Correct. This is Jules's response in turn 18. It connects Mina's preceding point with Mina's follow-up. A coverage decision is the payer's funding determination, which has not been received in this case.

C Not here. This line belongs to turn 14, after Mina says: "The financial slide also combines budget impact with cost-effectiveness. One estimates payer spending, while the other relates costs to health outcomes." It addresses that point, not the missing turn in the supplied exchange.

D Not here. This line belongs to turn 10, after Mina says: "We should still show what the routine-care analysis adds. It may help an access audience understand patterns outside the trial's narrower setting." It addresses that point, not the missing turn in the supplied exchange.

Return to the exercise, page 109

Additional case 1 / Explanations

Extended conversation

1. quarantine status

Turn 2. Quarantine status prevents use or distribution pending the authorized quality decision; current temperature alone does not settle suitability.

2. raw data

Turn 4. Raw data are the original recorded measurements and associated information needed to assess the excursion reliably.

3. affected scope

Turn 8. Affected scope identifies which product and quantities may be involved, rather than assuming every shipment or no shipment was exposed.

4. temperature excursion

Turn 10. A temperature excursion is exposure outside the specified range; visual appearance does not determine its effect on the product.

5. stability data

Turn 12. Stability data support assessment of this product under relevant conditions; another product's decision cannot be transferred automatically.

6. disposition

Turn 18. Disposition is the authorized decision about how the product may be handled after the evidence has been reviewed.

Return to the conversation, page 112

Transfer exchange

1. unverified

Coordinator: The receiving reading is within range, but the journey history is unverified.

Turn 1. The current reading cannot establish the temperatures experienced earlier in transit.

Additional case 1 / Explanations / continued

2. record

Quality: Preserve the device and pursue recovery of the original record.

Turn 2. The original logger record is the evidence needed to assess the transport history.

3. result

Coordinator: I will not translate missing data into a passing result.

Turn 3. No passing assessment can be inferred merely because the data cannot be read.

4. review

Quality: Keep the product in its required controlled status pending authorized review.

Turn 4. The unresolved history requires the appropriate quality review before a use decision.

Return to the conversation, page 114

Additional case 2 / Explanations

Extended conversation

1. out-of-specification result

Turn 2. An out-of-specification result falls outside the established acceptance criterion and requires documented investigation, not silent replacement.

2. assignable cause

Turn 4. An assignable cause is a cause established by evidence; an analyst's suspicion alone does not establish one.

3. audit trail

Turn 6. The audit trail records relevant actions and changes so the investigation can reconstruct what actually occurred.

4. retesting

Turn 10. Retesting repeats analysis under a justified investigation plan; it must not become repeated testing until a favorable result appears.

5. Resampling

Turn 12. Resampling takes another sample; it is different from retesting and needs its own justified basis in the investigation.

6. investigation conclusion

Turn 18. The investigation conclusion must explain the supported findings and treatment of all relevant results; it remains pending here.

Return to the conversation, page 116

Transfer exchange

1. plan

Reviewer: We need an authorized investigation plan before further testing.

Turn 1. The proposed repeated testing lacks a defined, justified investigation approach.

Additional case 2 / Explanations / continued

2. one

Analyst: I will preserve every original result, not just the favorable one.

Turn 2. Selecting only a favorable result would conceal relevant evidence from the investigation.

3. cause

Reviewer: A possible preparation error is still a hypothesis, not an established cause.

Turn 3. No evidence establishing the suspected preparation error has been supplied.

4. decision

Analyst: The batch status remains subject to the authorized quality decision.

Turn 4. Testing alone does not transfer batch-disposition authority to the analyst.

Return to the conversation, page 118

Additional case 3 / Explanations

Extended conversation

1. artwork revision

Turn 2. Artwork revision identifies the specific approved labeling version; similar appearance does not establish that two revisions are equivalent.

2. material traceability

Turn 6. Material traceability links the issued packaging components to their identity, quantities, and handling records.

3. line clearance

Turn 10. Line clearance verifies that the line is ready for the intended operation and free of relevant materials from an incompatible or previous operation.

4. segregation

Turn 12. Segregation keeps materials with different or restricted status from being mixed or used inadvertently.

5. change control

Turn 16. Change control evaluates, approves, and documents a proposed change rather than allowing an undocumented substitution.

6. restart authorization

Turn 20. Restart authorization is the formal permission to resume after the required conditions are met, not merely the arrival of replacement cartons.

Return to the conversation, page 120

Transfer exchange

1. hold

Supervisor: Delivery of the cartons does not cancel the current hold.

Turn 1. The hold remains because the required review and clearance checks are incomplete.

Additional case 3 / Explanations / continued

2. owners

Quality: The handover must identify the remaining checks and their owners.

Turn 2. Named owners make responsibility for the unfinished checks explicit across the shift change.

3. identified

Supervisor: I will keep the two material versions physically controlled and clearly identified.

Turn 3. Clear identification and control help prevent unintended use of the wrong packaging version.

4. confirmed

Quality: Resume only after the required conditions and authorization are confirmed.

Turn 4. Replacement delivery alone does not establish that the required restart conditions are satisfied.

Return to the conversation, page 122

Find a phrase by purpose

Drug Development Strategy, Unmet Need, TPP, and Evidence Logic

Locate the evidence: The observation comes from a cell-based assay, not a patient study.

Narrow the claim: We observed a marker change under the tested conditions.

Protect the distinction: Clinical benefit has not been established.

Full phrase bank, page 8

Regulatory Pathways, IND Readiness, NDA/BLA Strategy, and Agency Interaction

Focus the question: We seek advice on the proposed population and endpoint rationale.

State the sponsor view: Our proposed approach is set out in section three.

Supply the reason: The rationale rests on the following evidence and assumptions.

Full phrase bank, page 20

Clinical Trial Design, Protocols, GCP, and Operations

State the discrepancy: The protocol attachment says three days; the quick-reference sheet says seven.

Request the authority: Which approved version is effective for this site?

Identify the source: I will include the document identifiers and exact sections.

Full phrase bank, page 35

Endpoints, Estimands, Statistics, Data Readouts, and Clinical Meaning

Lead with the primary: The primary analysis did not meet the prespecified significance threshold.

State the direction: The point estimate favors treatment on this lower-is-better scale.

Preserve uncertainty: The confidence interval includes no difference.

Full phrase bank, page 48

Find a phrase by purpose / continued

Pharmacovigilance, Safety Signals, Risk-Benefit, and Label Updates

Open the handoff: I have an incomplete report that needs safety intake.

Preserve the wording: The clinician described the dizziness as severe.

Separate the classification: Seriousness has not yet been confirmed.

Full phrase bank, page 63

CMC, CGMP, Quality Events, Manufacturing, and Supply

State the observed gap: The verification entry is missing from the batch record.

Avoid an assumed cause: We have not established whether the check occurred.

Report the status: The batch remains under quality review and is not released.

Full phrase bank, page 76

Labeling, Medical Affairs, Promotion Review, and Compliance Boundaries

Identify the overreach: Immediate relief is not supported by the week-12 assessment.

Preserve the population: The supplied evidence concerns the defined adult study population.

Keep the measure: The finding is a difference in mean symptom score, not universal relief.

Full phrase bank, page 88

Market Access, RWE, Launch Readiness, Biosimilars, and Lifecycle Management

Separate the sources: The trial and claims analysis answer different questions.

State the design: The claims comparison is observational, not randomized.

Keep the outcome clear: The trial measures symptom score; the claims analysis measures hospitalization.

Full phrase bank, page 101

Vocabulary index

absolute claim
Lesson 07, p. 86

adverse event
Lesson 05, p. 59

adverse reaction
Lesson 05, p. 59

affected scope
Case 1, p. 111

agency interaction
Lesson 02, p. 18

aggregate safety review
Lesson 05, p. 62

analysis population
Lesson 04, p. 45

approval expiry
Lesson 07, p. 87

approval status
Lesson 02, p. 19

approved labeling
Lesson 07, p. 86

artwork revision
Case 3, p. 119

assay limitation
Lesson 01, p. 6

assignable cause
Case 2, p. 115

audit trail
Lesson 03, p. 33, Case 2, p. 115

batch disposition
Lesson 06, p. 74

batch record
Lesson 06, p. 73

benefit-risk assessment
Lesson 05, p. 62

biological plausibility
Lesson 01, p. 5

biologics license application
Lesson 02, p. 18

biomarker
Lesson 01, p. 5

biosimilar
Lesson 08, p. 99

blinding
Lesson 03, p. 33

briefing document
Lesson 02, p. 18

budget impact
Lesson 08, p. 99

CAPA
Lesson 06, p. 74

case intake
Lesson 05, p. 59

case reconciliation
Lesson 05, p. 61

causality assessment
Lesson 05, p. 59

cell-based assay
Lesson 01, p. 5

certificate of analysis
Lesson 06, p. 74

change control
Lesson 06, p. 74, Case 3, p. 119

chemistry, manufacturing, and controls
Lesson 06, p. 73

claim substantiation
Lesson 07, p. 86

claims data
Lesson 08, p. 98

clinical benefit
Lesson 01, p. 5

clinical hold
Lesson 02, p. 19

clinical importance
Lesson 04, p. 46

clinical research associate
Lesson 03, p. 32

comparative claim
Lesson 07, p. 86

comparative effectiveness
Lesson 08, p. 99

concomitant medication
Lesson 05, p. 61

conditional advice
Lesson 02, p. 19

confidence interval
Lesson 04, p. 45

confounding
Lesson 08, p. 98

Vocabulary index / continued

confounding by indication
Lesson 08, p. 99

contemporaneous record
Lesson 06, p. 73

controlling version
Lesson 03, p. 32

corrective action
Lesson 06, p. 73

cost-effectiveness
Lesson 08, p. 99

coverage decision
Lesson 08, p. 98

cross-reference
Lesson 02, p. 19

current good manufacturing practice
Lesson 06, p. 73

data cutoff
Lesson 04, p. 47

data integrity
Lesson 06, p. 73

data query
Lesson 03, p. 33

dechallenge
Lesson 05, p. 61

development candidate
Lesson 01, p. 5

development milestone
Lesson 01, p. 7

deviation investigation
Lesson 06, p. 73

disposition
Case 1, p. 111

documentation gap
Lesson 06, p. 73

dose-response relationship
Lesson 01, p. 6

dossier
Lesson 02, p. 19

duplicate report
Lesson 05, p. 61

effect size
Lesson 04, p. 46

effective date
Lesson 03, p. 32

effectiveness check
Lesson 06, p. 74

electronic case report form
Lesson 03, p. 33

eligibility criterion
Lesson 03, p. 33

essential record
Lesson 03, p. 34

estimand
Lesson 04, p. 45

event onset
Lesson 05, p. 61

evidence gap
Lesson 01, p. 7

expectedness
Lesson 05, p. 59

exploratory endpoint
Lesson 04, p. 45

exposure-response relationship
Lesson 01, p. 6

external validity
Lesson 08, p. 99

fair balance
Lesson 07, p. 86

final-form approval
Lesson 07, p. 87

focused question
Lesson 02, p. 18

follow-up request
Lesson 05, p. 61

formulary
Lesson 08, p. 98

go/no-go criterion
Lesson 01, p. 6

good clinical practice
Lesson 03, p. 32

health technology assessment
Lesson 08, p. 98

hypothetical strategy
Lesson 04, p. 45

indication
Lesson 02, p. 19

indication boundary
Lesson 07, p. 86

Vocabulary index / continued

individual case safety report
Lesson 05, p. 60

information gap
Lesson 02, p. 18

informed consent
Lesson 03, p. 32

institutional review board
Lesson 03, p. 33

intended audience
Lesson 07, p. 86

intercurrent event
Lesson 04, p. 45

investigation conclusion
Case 2, p. 115

investigational new drug application
Lesson 02, p. 18

launch readiness
Lesson 08, p. 99

lifecycle management
Lesson 08, p. 100

line clearance
Case 3, p. 119

market access
Lesson 08, p. 98

master production instruction
Lesson 06, p. 73

material limitation
Lesson 07, p. 86

material traceability
Case 3, p. 119

mechanism of action
Lesson 01, p. 5

MedDRA
Lesson 05, p. 61

medical affairs
Lesson 07, p. 87

medical information
Lesson 07, p. 87

medical-legal-regulatory review
Lesson 07, p. 87

meeting minutes
Lesson 02, p. 19

meeting package
Lesson 02, p. 18

missing data
Lesson 04, p. 46

monitoring finding
Lesson 03, p. 33

multiplicity
Lesson 04, p. 46

net impression
Lesson 07, p. 87

new drug application
Lesson 02, p. 18

nominal p-value
Lesson 04, p. 46

nonclinical study
Lesson 01, p. 6

observational study
Lesson 08, p. 98

off-label use
Lesson 07, p. 87

onset claim
Lesson 07, p. 86

out-of-specification result
Lesson 06, p. 74, Case 2, p. 115

out-of-trend result
Lesson 06, p. 74

p-value
Lesson 04, p. 45

payer
Lesson 08, p. 98

pharmacodynamics
Lesson 01, p. 6

pharmacokinetics
Lesson 01, p. 6

pharmacovigilance
Lesson 05, p. 59

point estimate
Lesson 04, p. 45

post hoc analysis
Lesson 04, p. 46

prescribing information
Lesson 07, p. 86

preventive action
Lesson 06, p. 74

primary endpoint
Lesson 04, p. 45

Vocabulary index / continued

principal investigator
Lesson 03, p. 32

promotional claim
Lesson 07, p. 86

protocol
Lesson 03, p. 32

protocol deviation
Lesson 03, p. 33

quarantine status
Case 1, p. 111

raw data
Case 1, p. 111

real-world evidence
Lesson 08, p. 98

redline
Lesson 07, p. 87

reference product
Lesson 08, p. 100

regulatory affairs
Lesson 02, p. 19

regulatory pathway
Lesson 02, p. 18

release authorization
Lesson 06, p. 74

reporter
Lesson 05, p. 60

residual confounding
Lesson 08, p. 99

retesting
Case 2, p. 115

root cause
Lesson 06, p. 73

schedule of assessments
Lesson 03, p. 32

secondary endpoint
Lesson 04, p. 45

selection bias
Lesson 08, p. 99

seriousness
Lesson 05, p. 59

significance threshold
Lesson 04, p. 46

source record
Lesson 03, p. 33

prior authorization
Lesson 08, p. 98

proof of concept
Lesson 01, p. 5

protocol amendment
Lesson 03, p. 32

qualifier
Lesson 07, p. 87

randomization
Lesson 03, p. 33

real-world data
Lesson 08, p. 98

rechallenge
Lesson 05, p. 61

reference pack
Lesson 07, p. 87

reference safety information
Lesson 05, p. 59

regulatory commitment
Lesson 02, p. 19

regulatory strategy
Lesson 02, p. 18

repeatability
Lesson 01, p. 6

Resampling
Case 2, p. 115

restart authorization
Case 3, p. 119

risk disclosure
Lesson 07, p. 86

safety signal
Lesson 05, p. 59

scientific exchange
Lesson 07, p. 87

segregation
Case 3, p. 119

sensitivity analysis
Lesson 04, p. 46

severity
Lesson 05, p. 59

site activation
Lesson 03, p. 32

specification
Lesson 06, p. 74

Vocabulary index / continued

sponsor position
Lesson 02, p. 18

stability program
Lesson 06, p. 75

study coordinator
Lesson 03, p. 32

submission readiness
Lesson 02, p. 19

supporting rationale
Lesson 02, p. 18

target engagement
Lesson 01, p. 5

temperature excursion
Case 1, p. 111

testing hierarchy
Lesson 04, p. 46

tolerability
Lesson 01, p. 6

traceability
Lesson 06, p. 73

translational evidence
Lesson 01, p. 5

unmet medical need
Lesson 01, p. 5

uptake assumption
Lesson 08, p. 99

verbatim term
Lesson 05, p. 61

version control
Lesson 02, p. 19

written response
Lesson 02, p. 19

stability data
Case 1, p. 111

statistical analysis plan
Lesson 04, p. 45

subgroup analysis
Lesson 04, p. 46

supply constraint
Lesson 06, p. 75

surrogate endpoint
Lesson 01, p. 5

target product profile
Lesson 01, p. 5

temporal association
Lesson 05, p. 61

therapeutic window
Lesson 01, p. 6

topline readout
Lesson 04, p. 47

training record
Lesson 03, p. 34

treatment-policy strategy
Lesson 04, p. 45

unsolicited request
Lesson 07, p. 87

value dossier
Lesson 08, p. 98

verification entry
Lesson 06, p. 73

visit window
Lesson 03, p. 32

References and publication notes

All cases, organizations, figures, and conversations are fictional teaching material, not transcripts. Communication practice does not replace qualified advice or current workplace procedures.

[1] US Food and Drug Administration. The Drug Development Process.

Background on development stages and post-market monitoring. All products, figures, and conversations in this book are invented. Source checked 10 October 2026.

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[2] FDA / ICH. E9(R1): Estimands and Sensitivity Analysis in Clinical Trials.

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https://www.fda.gov/about-fda/cder-offices-and-divisions/office-prescription-drug-promotion-opdp

Companion course

The website offers the eight-lesson course and related practice. This book can be used independently. No audio access or external account is needed for its exercises.

https://englishladder.com/efsp-pharmaceutical.html

State the evidence. Qualify the claim. Close the communication gap.

Follow eight cross-functional conversations from early evidence to development decisions, safety, quality, and access.

Drug Development Strategy, Unmet Need, TPP, and Evidence Logic

A biomarker change is not automatically clinical benefit. Its meaning depends on the marker, context, and supporting evidence. Here, one concentration and three laboratory runs do not establish human exposure, tolerability, dose response, or improved mobility.

Regulatory Pathways, IND Readiness, NDA/BLA Strategy, and Agency Interaction

IND means investigational new drug application; NDA means new drug application; BLA means biologics license application in US usage. These terms name different regulatory submissions. An IND is not a marketing approval, and preparing a meeting package does not authorize a study.

Clinical Trial Design, Protocols, GCP, and Operations

A later file date does not establish that a document is approved and effective for a particular site. Confirm identity, version, authorization, and applicability. A quick-reference sheet should not silently replace the controlling protocol or become an unofficial amendment.

Endpoints, Estimands, Statistics, Data Readouts, and Clinical Meaning

The estimate favors treatment, but the interval includes zero and the primary p-value exceeds 0.05. This is not proof of no effect. It is also not a successful primary test. The secondary finding retains its actual status under the prespecified plan.

State the evidence. Qualify the claim. Close the communication gap. / continued

Phil Stilwell

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